

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement Low Leakage 1.2V MOSFET SPICE Model Document

**Version 1.1 Phase 1
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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2_p1	2010/06/30	Kelly Pao	1. Original 2. Copy from G-05-LOGIC/MIXED_MODE11-MMC/L110E-SPICE and revise the Idlin corner.
0.2_p2	2010/08/11	Shih-Hsin Yeh	1. Copy Igate model from L110E model.
0.3_p1	2010/10/22	Kelly Pao	1. Removed the m in igate area formula.(area='w*0.9*l*0.9*m*1e4*tcig' --> area='w*0.9*l*0.9*1e4*tcig') 2. Updated the lmin
0.4_p1	2010/10/28	Yealing Lin	1. Add Twell NMOS model (copy from Twin well NMOS). 2. Modify Igate model: (a) m=m to m=1. (b) area='w*0.9*l*0.9*1e4*tcig'to area='w*0.9*(l*0.9+1.2e-8)*1e4*tcig'. (c) 'rsh*nrd/m' to 'rsh*nrd'. (d) 'rsh*nrs/m' to 'rsh*nrd'.
0.5_p1	2010/12/24	Yealing Lin	1. Re-center the model to align EDR target.
1.0_p1	2012/09/14	Laney Fan	1.Version up to 1.0 based on 6 pcs Si data.
1.1_p1	2013/09/10	Chih hung Chuang	Modify the PMOS noise parameters.

1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-Logic/Mixed_Mode11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.2V n-ch MOSFET

1.2V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of this compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that could be used. One is binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is global model, which means one scalable parameter set is provided and it could cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for its many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Model usage

The compact model is provided in various formats. It's guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2012.6
Cadence Spectre version 10.1.1.111.isr8 32 bit
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib" CaseName
, where CaseName could be TT, FF, SS, FNFP, or SNFP.

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
- Add the following statement in the input file to the simulator.
include "/ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, SS, FNFP, or SNFP.

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib.eldo" CaseName
, where CaseName could be TT, FF, SS, FNFP, or SNFP.
For 1.2V n-ch MOSFET: n_12_111110e_ig
For 1.2V p-ch MOSFET: p_12_111110e_ig

For 1.2V n-ch MOSFET: n_12bpw_1l110e_ig (for Twell model)

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range:(After 90% shrink)

For 1.2V n-ch MOSFET:

$$0.108 \mu\text{m} \leq L_{\text{DES}} \leq 45 \mu\text{m}$$

$$0.144 \mu\text{m} \leq W_{\text{DES}} \leq 90 \mu\text{m}$$

For 1.2V p-ch MOSFET:

$$0.108 \mu\text{m} \leq L_{\text{DES}} \leq 45 \mu\text{m}$$

$$0.144 \mu\text{m} \leq W_{\text{DES}} \leq 90 \mu\text{m}$$

Adopt multi-finger structures if $W_{\text{DES}} > 9\text{mm}$.

$$0.306\mu\text{m} \leq S_A, S_B \leq 1.98\mu\text{m}, S_D > 0.36\mu\text{m}$$

Note: $S_A, S_B = 1.98\mu\text{m}$ is the reference dimension of the STI effect.

Voltage range:

For 1.2V n-ch MOSFET:

$$0\text{V} \leq V_{\text{GS}} \leq 1.2\text{V} (*1.1)$$

$$0\text{V} \leq V_{\text{DS}} \leq 1.2\text{V} (*1.1)$$

$$-1.2\text{V} \leq V_{\text{BS}} \leq 0\text{V}$$

For 1.2V p-ch MOSFET:

$$0\text{V} \geq V_{\text{GS}} \geq -1.2\text{V} (*1.1)$$

$$0\text{V} \geq V_{\text{DS}} \geq -1.2\text{V} (*1.1)$$

$$1.2\text{V} \geq V_{\text{BS}} \geq 0\text{V}$$

Temperature range:

For both 1.2V n-ch and p-ch MOSFET:

$$-50\text{ }^{\circ}\text{C} \sim +125\text{ }^{\circ}\text{C}$$

UMC L110E is UMC L130E 90% direct shrunk option. The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 12 nm) 9/0.12
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Slotting inserting	metal line width>5.556um	metal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 1.2 V n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

er information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGSML	QGSML	QGSML	QGSML
Wafer Number	#12	#12	#12	#12

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 2-1 Electrical Parameters for NMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
-----------	-----------	------	--------	----------------	-----------------	----------------

Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.453	0.454	0.459	0.459
I _{ON}	V _{DS} = 1.2 V	μA/ μm	6.901	6.95	6.85	6.84

Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.577	0.565	0.580	0.577
V _{TSAT}	V _{DS} = 1.2 V	V	0.510	0.502	0.513	0.510
Body Effect V _T shift	V _{DS} = 1.2 V V _{BS} = 0 ~ -1.2 V	V	--	0.173	0.174	0.174
DIBL V _T shift	V _{DS} = 0.05~1.2 V	V	0.067	0.063	0.067	0.067
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	365	364.80	365.52	365.00
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	4e-12	2.7E-12	2.3E-12	2.5E-12

Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.42	0.419	0.423	0.423
V _{TSAT}	V _{DS} = 1.2 V	V	--	0.406	0.415	0.415
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	7.24	7.27	7.11	7.11

Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.516	0.515	0.517	0.515
V _{TSAT}	V _{DS} = 1.2 V	V	0.445	0.445	0.454	0.453
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	496.4	497.22	496.56	494.39
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	1.39e-11	8.6E-12	8.2E-12	8.6E-12

Capacitance

C _J	V _J = 0 V	F/m ²	6.92e-4	7.13E-04	7.20E-04	7.20E-04
C _{JSW}	V _J = 0 V	F/m	1.01e-10	9.98E-11	9.94E-11	9.94E-11
C _{JSWG}	V _J = 0 V	F/m	4.3e-10	4.21E-10	4.11E-10	4.11E-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 2-2 Geometry of NMOS devices measured for DC parameter extraction

W/L	10	1	0.8	0.5	0.24	0.18	0.15	0.13	0.12
10	X	X	X	X	X	X	X	X	X
1	X				X				X
0.8					X				X
0.72	X								
0.65					X				X
0.43	X		X	X	X	X	X	X	X
0.35									X
0.25	X				X				X
0.16	X				X				X
0.14	X		X	X	X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-3 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.05	1.2	1.15
V_{GS} (V)	-0.5	1.35	0.025
V_{BS} (V)	0	-1.2	-0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	1.35	0.025
V_{GS} (V)	0.6	1.2	0.15
V_{BS} (V)	0	-1.2	-1.2

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots denote the dimensions after shrink, which are $L_{DES} * 0.9 + 12\text{nm}$ and $W_{DES} * 0.9$, respectively.

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

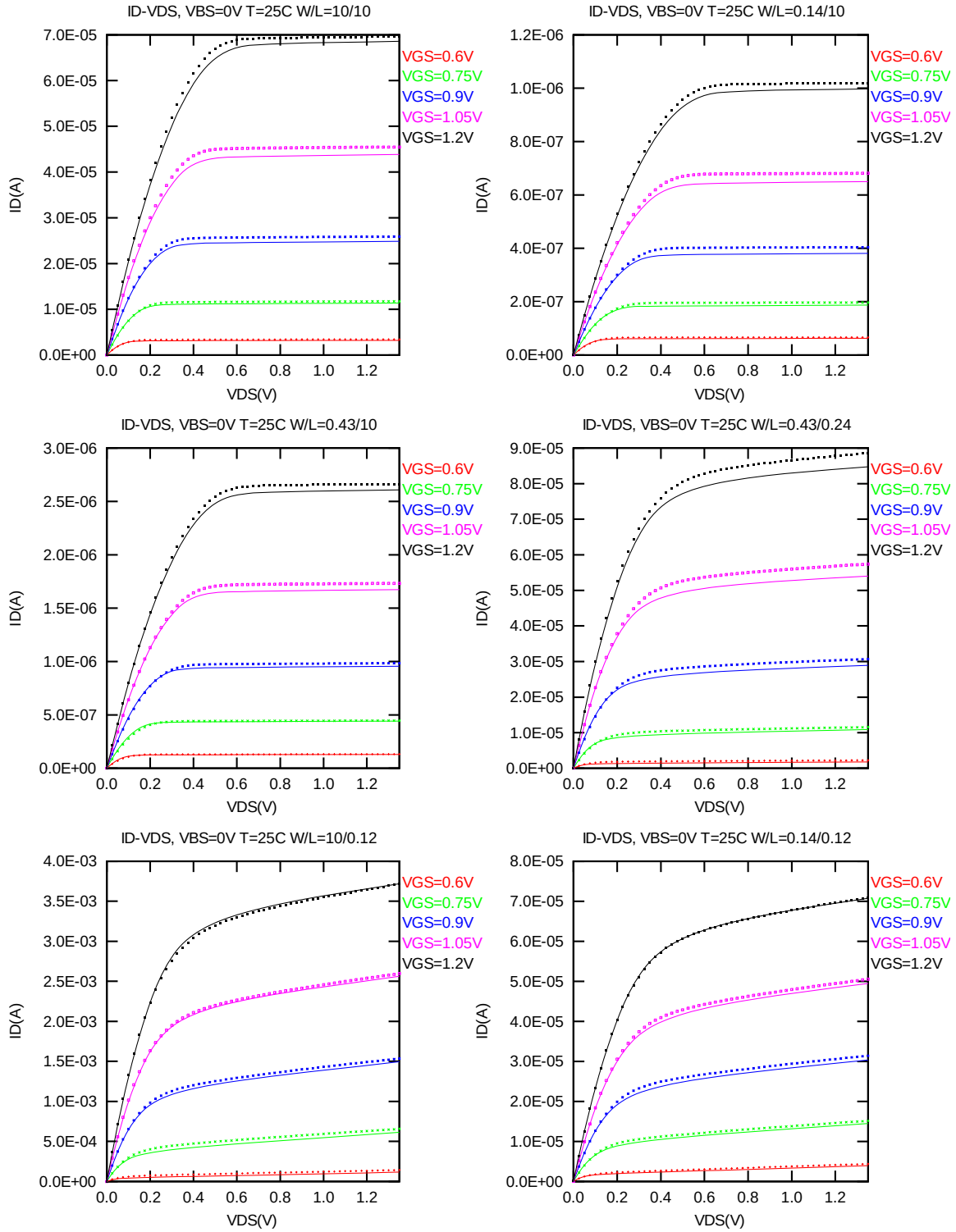


Figure 2-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for NMOS

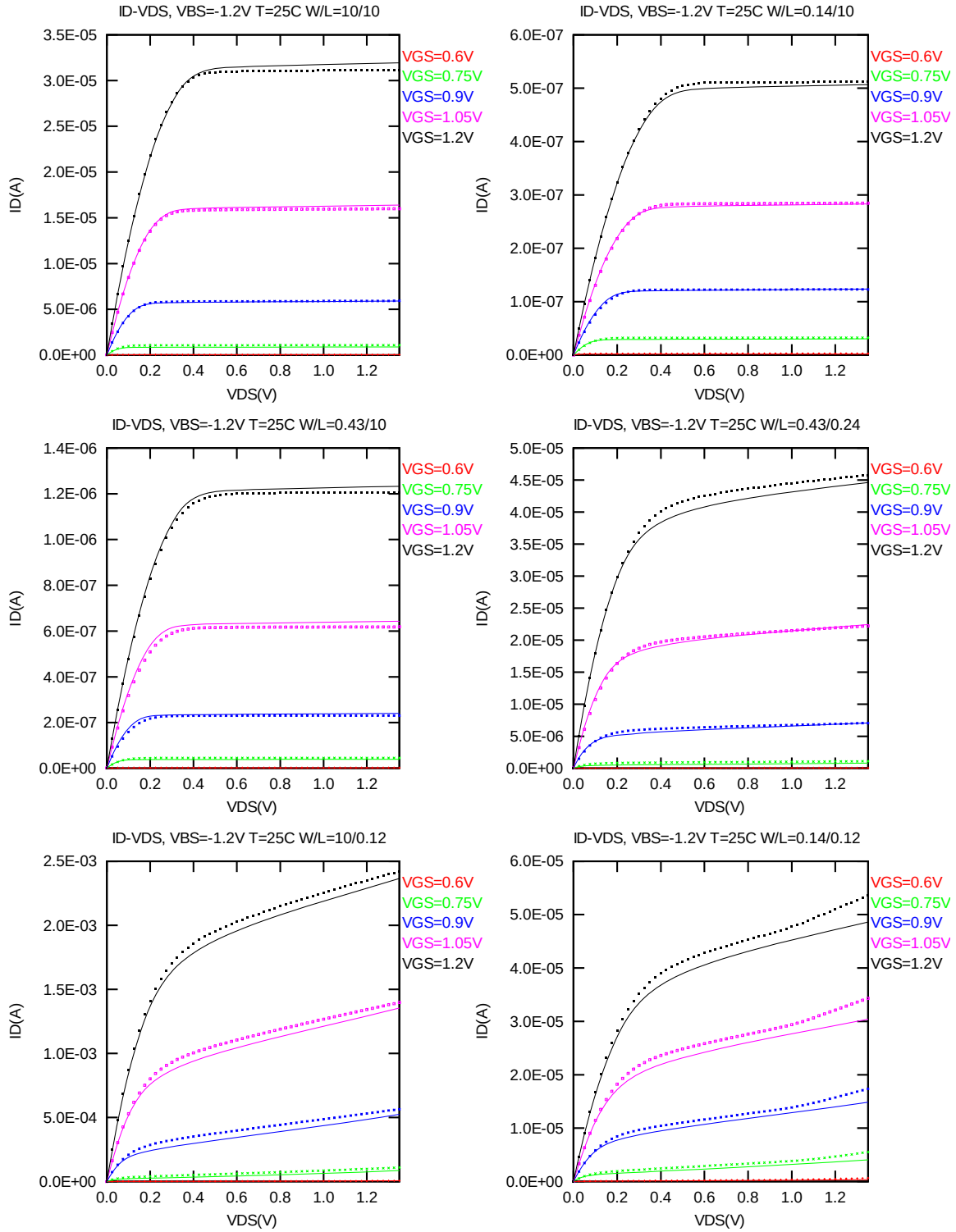


Figure 2-2 I_D vs. V_{DS} at $V_{BS} = -1.2V$ for NMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

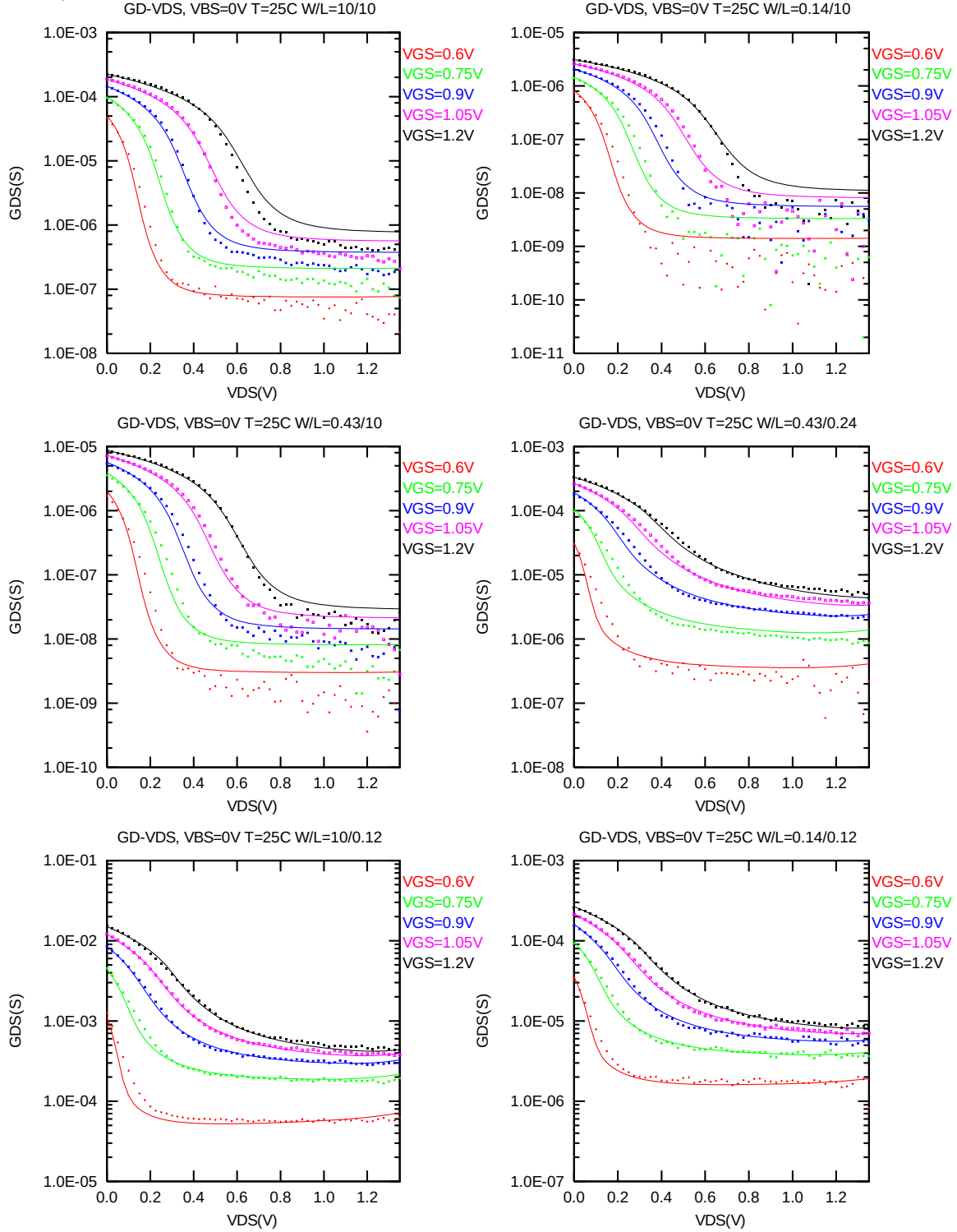


Figure 2-3 G_{DS} vs. V_{DS} at $V_{BS} = 0$ V for NMOS

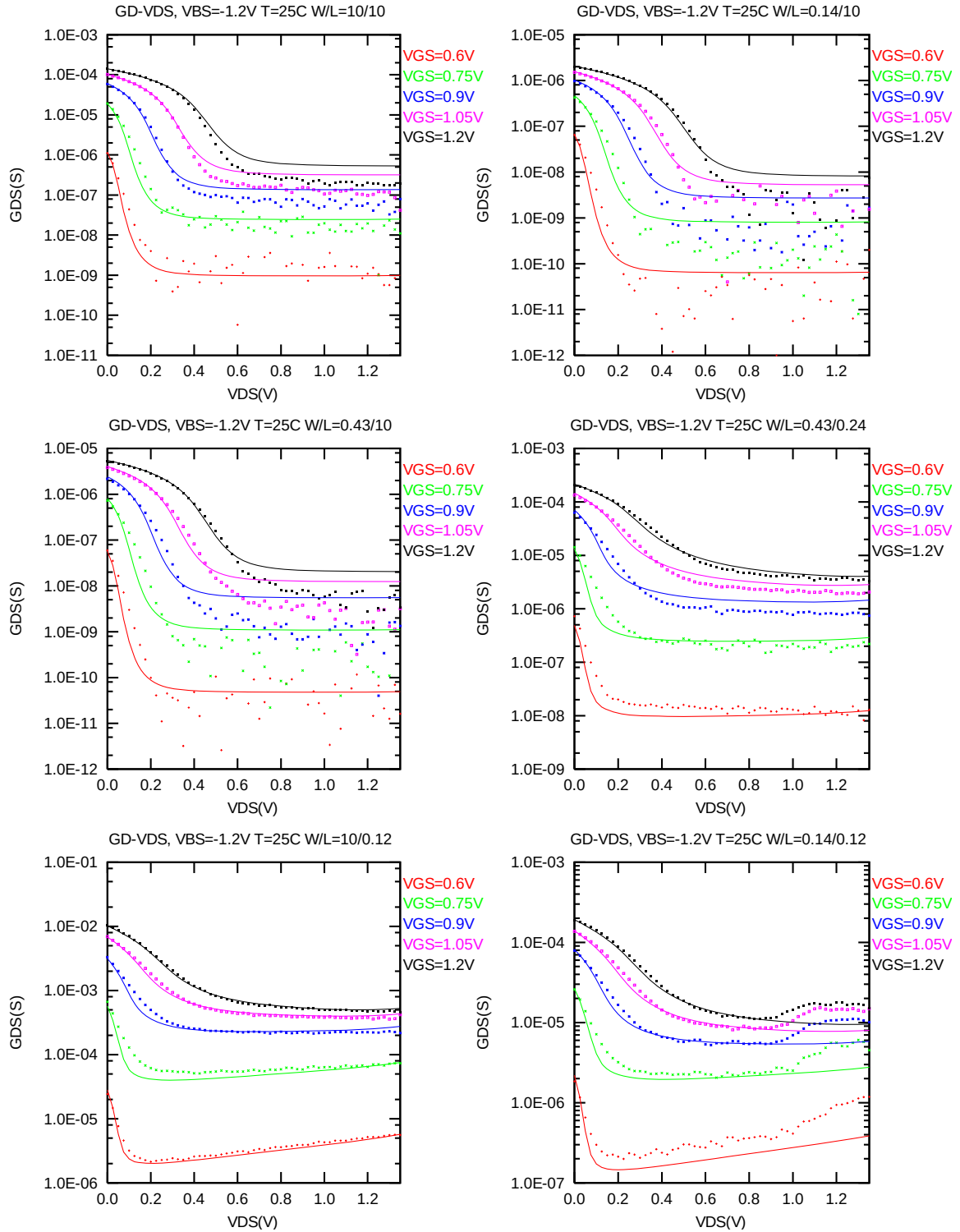


Figure 2-4 G_{DS} vs. V_{DS} at $V_{BS} = -1.2$ V for NMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

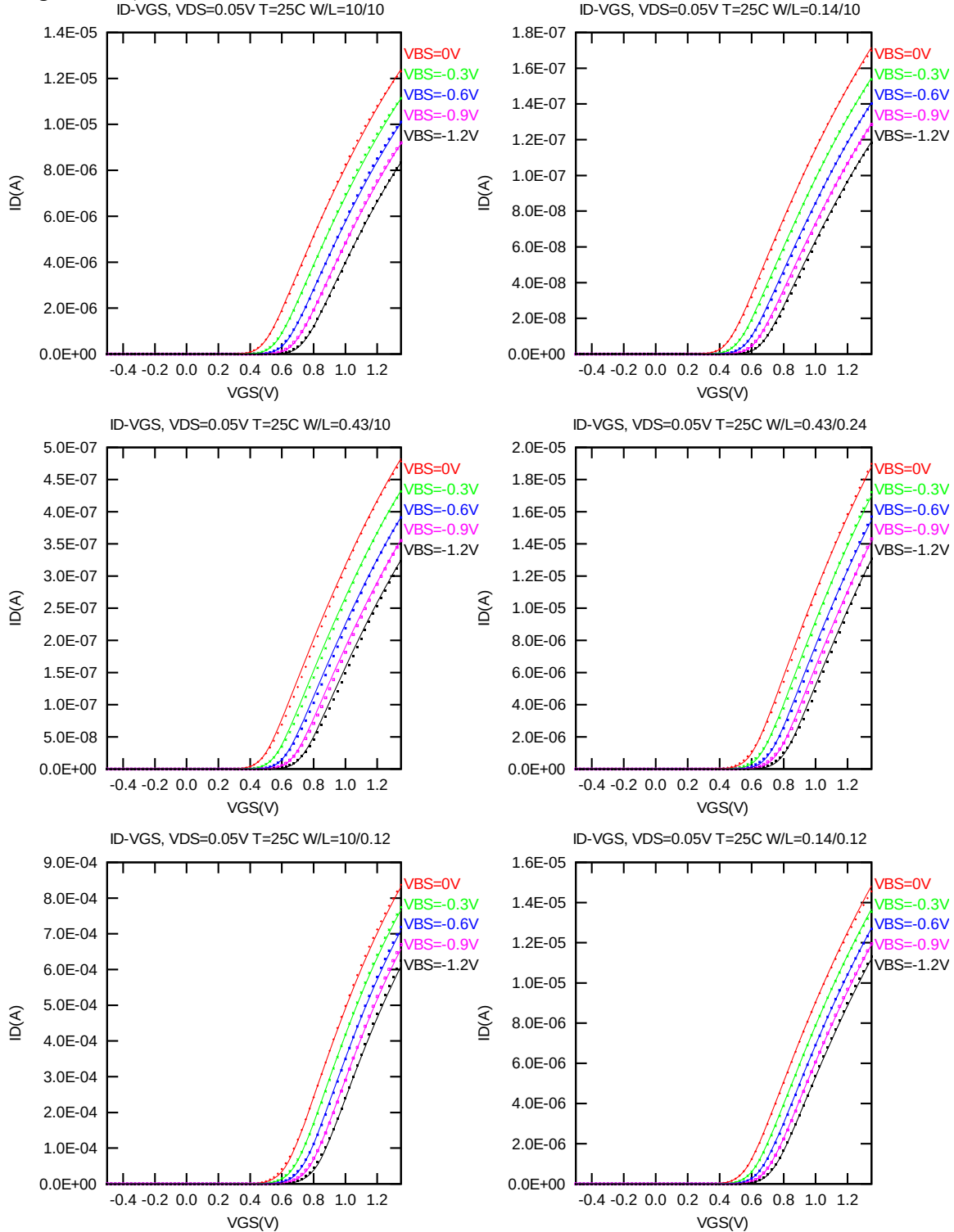


Figure 2-5 I_D vs. V_{GS} at $V_{DS} = 0.05V$ for NMOS

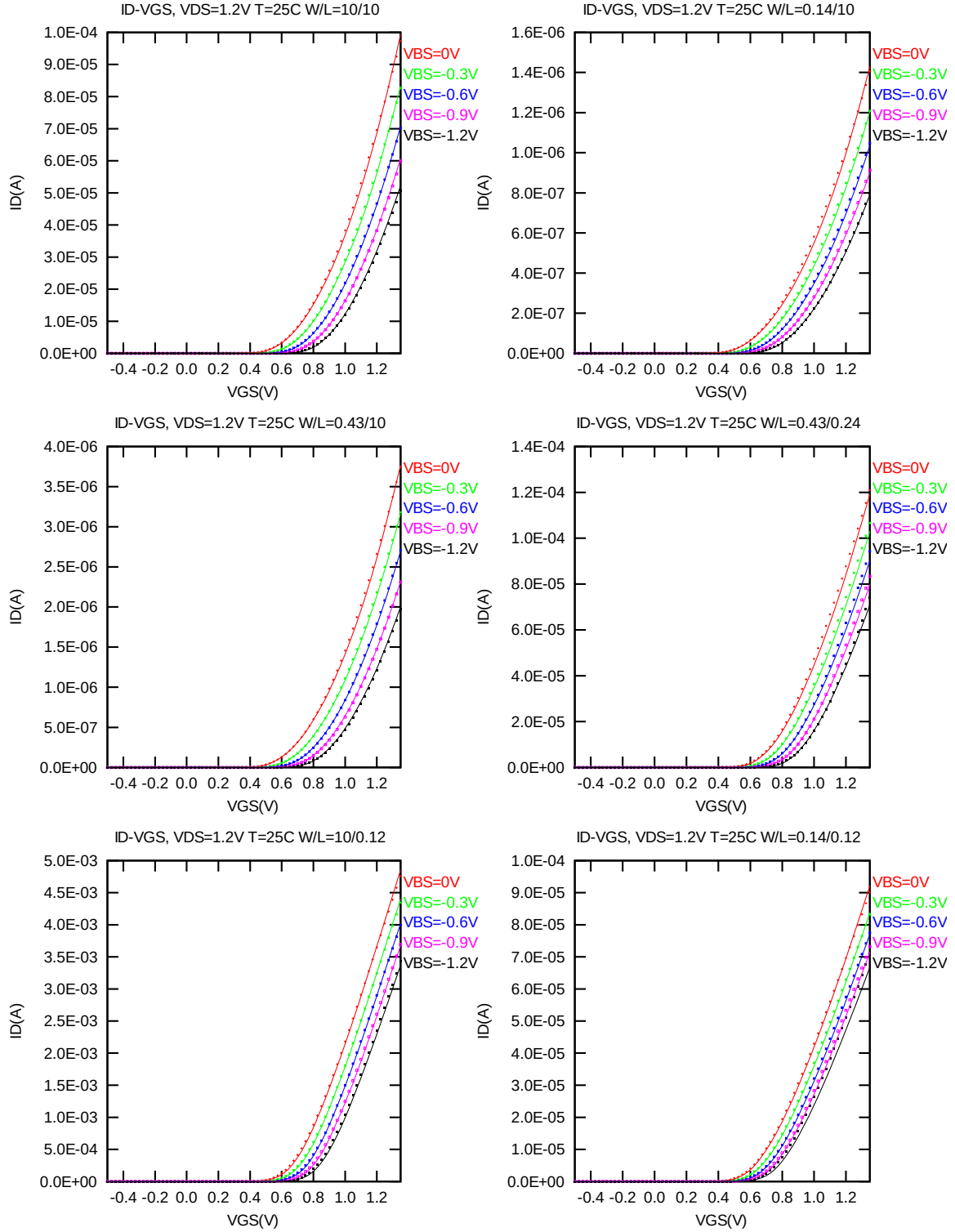


Figure 2-6 I_D vs. V_{GS} at $V_{DS} = 1.2$ V for NMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 °C. The simulation result is with $g_{mndc}=1e-13$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

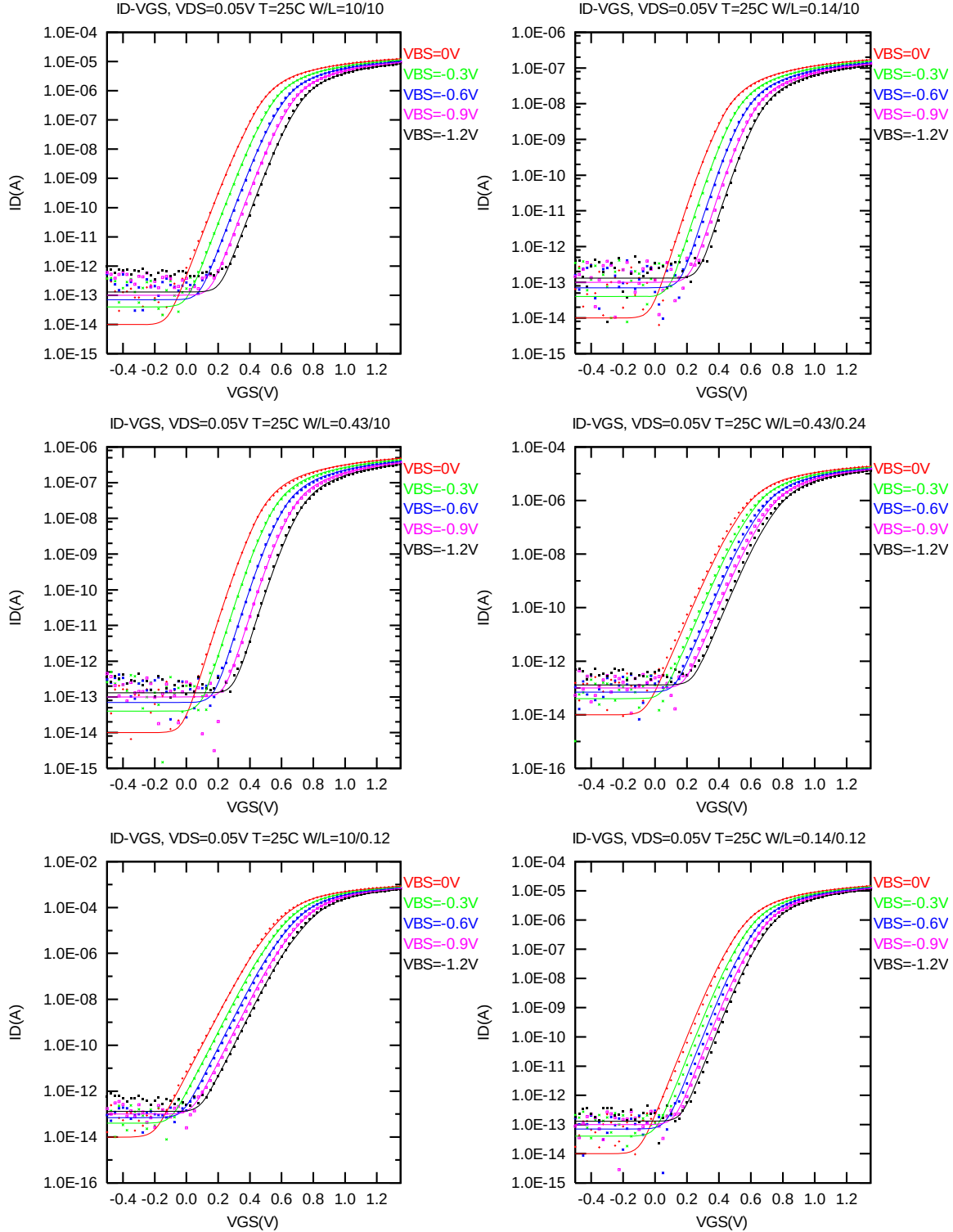


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS} = 0.05V$ for NMOS

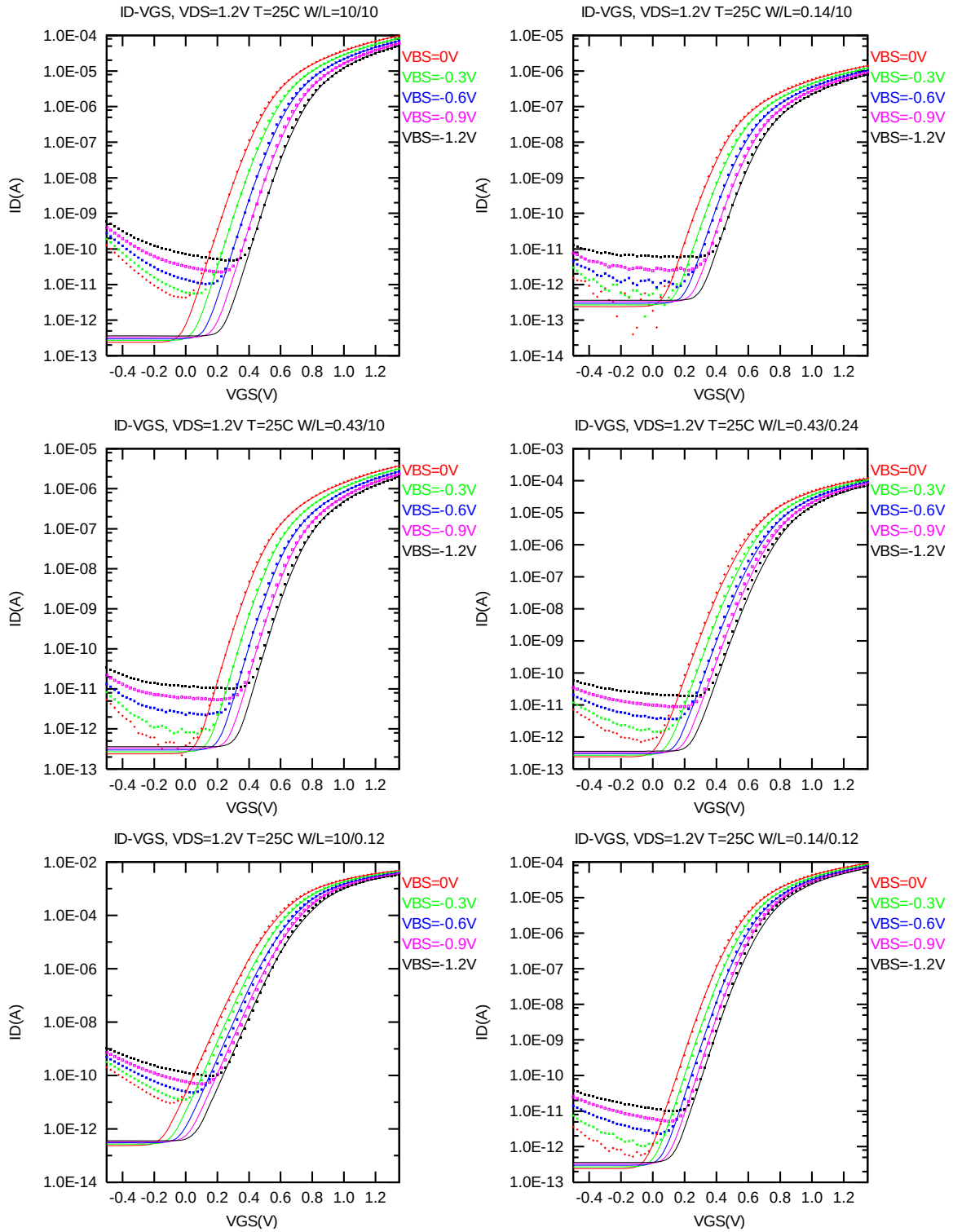


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS} = 1.2V$ for NMOS

G_M vs. V_{GS} Plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

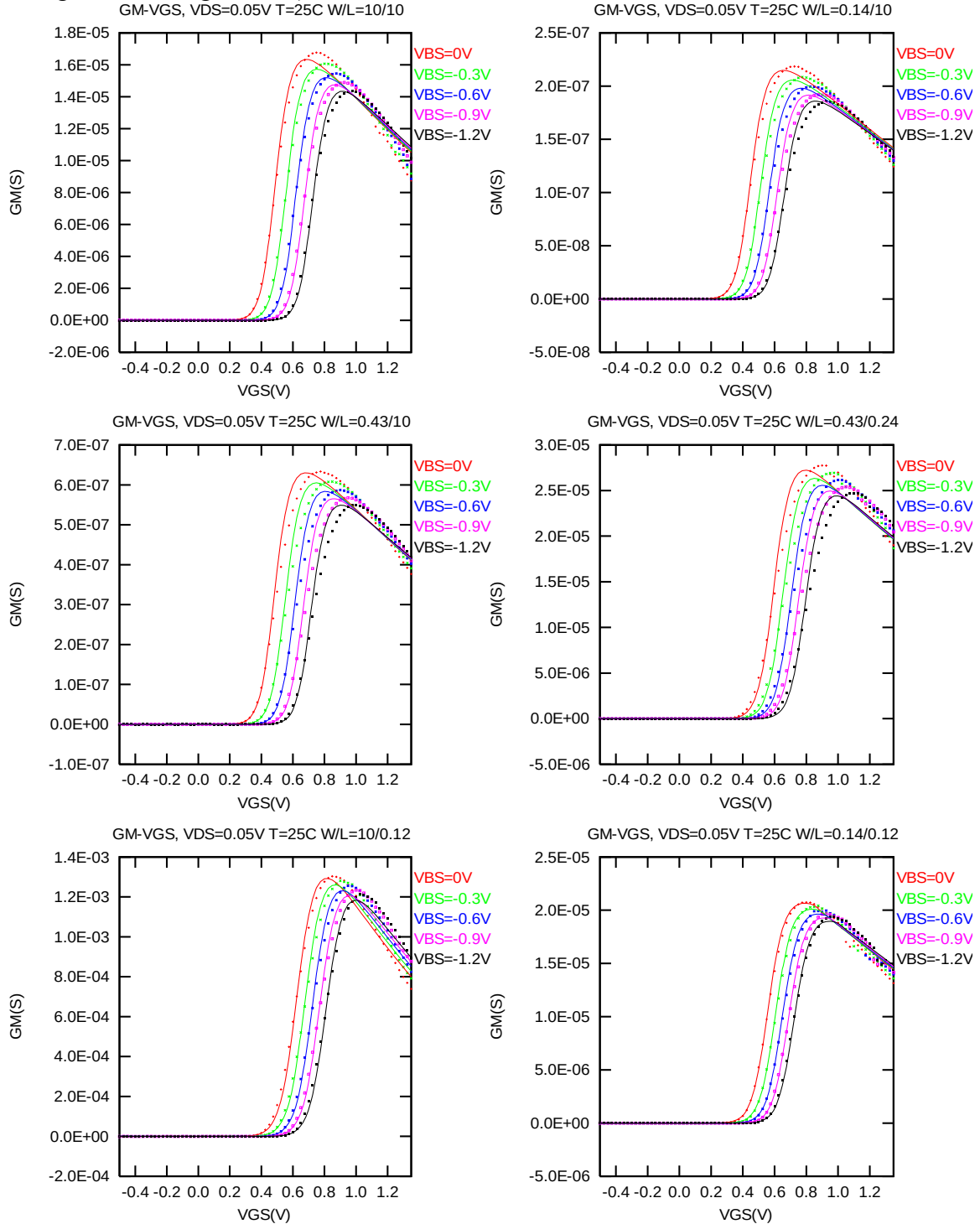


Figure 2-9 G_M vs. V_{GS} at $V_{DS} = 0.05V$ for NMOS

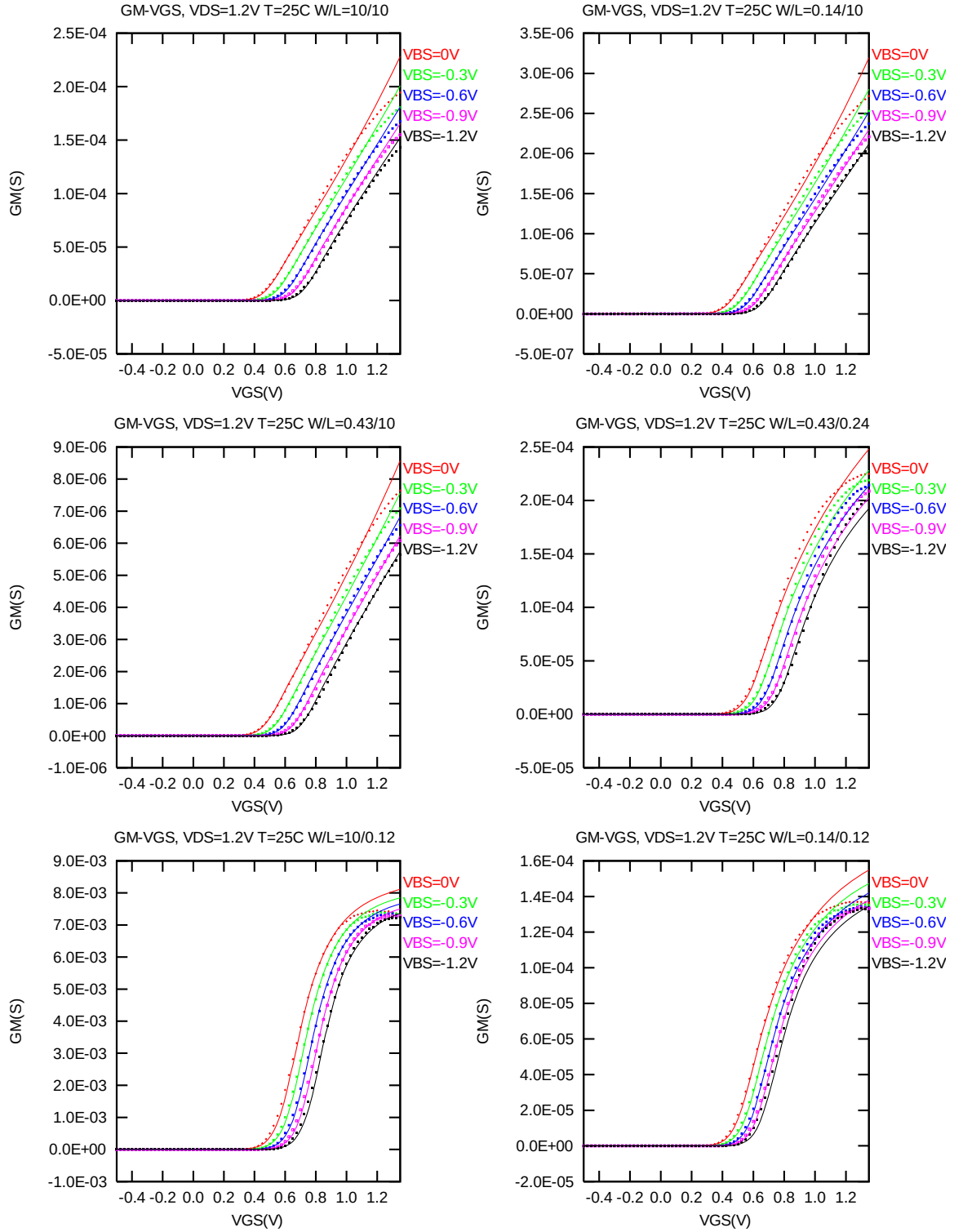


Figure 2-10 G_M vs. V_{GS} at $V_{DS} = 1.2\text{ V}$ for NMOS

V_T vs. L_{DES} & V_T vs. W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$ at 25 °C. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

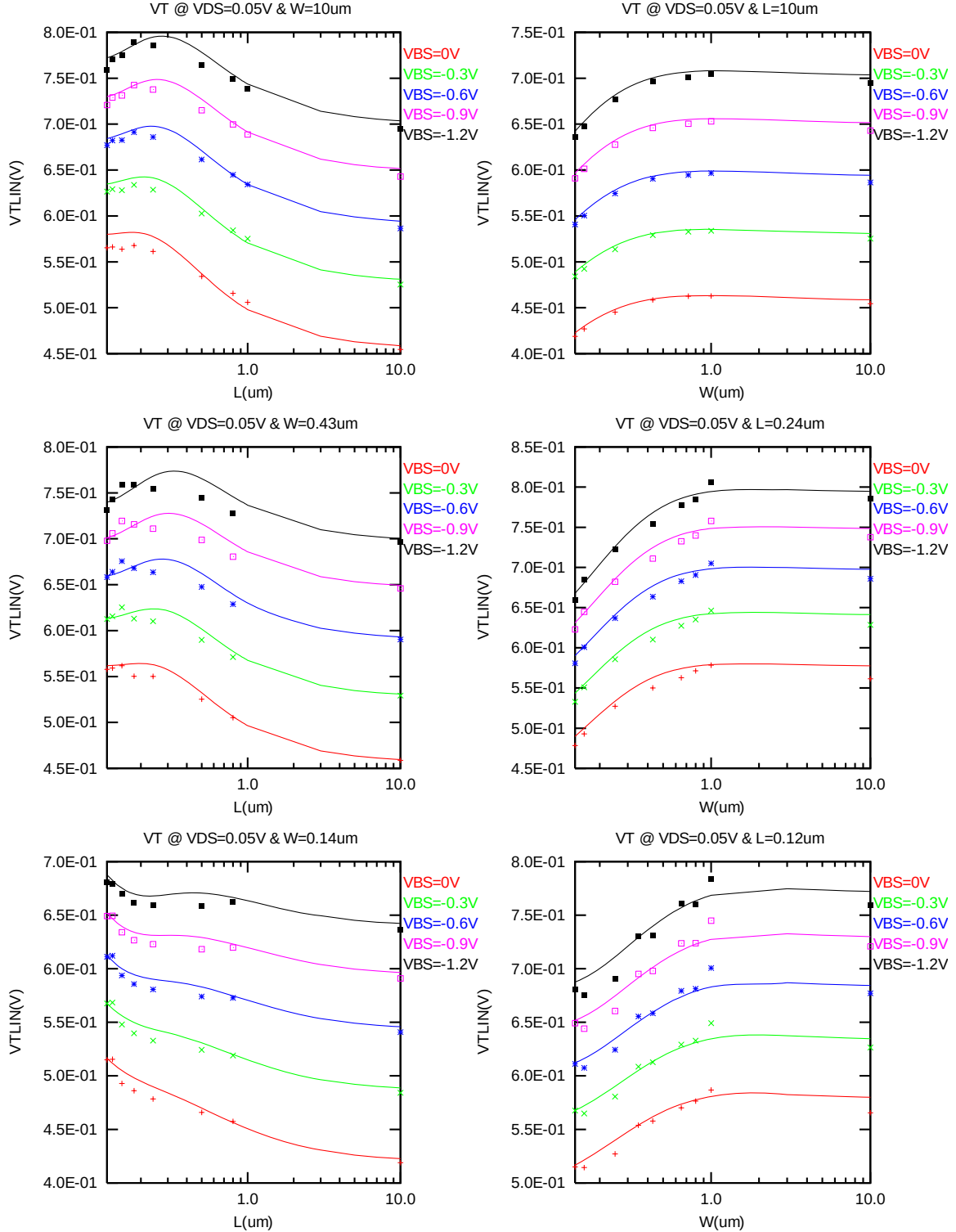


Figure 2-11 V_T vs. channel length & channel width for low drain bias ($V_{DS} = 0.05 V$) for NMOS

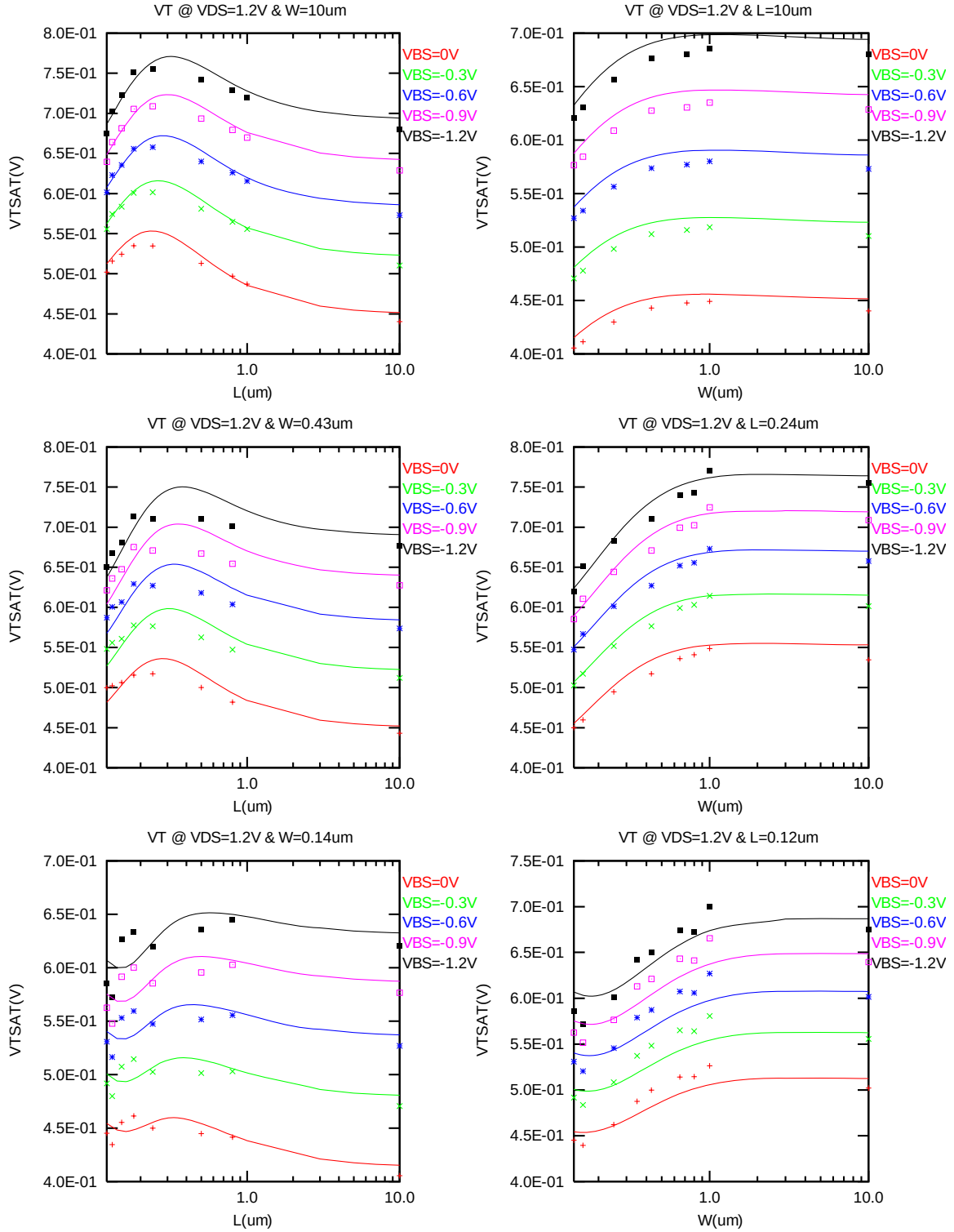


Figure 2-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = 1.2$ V) for NMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.2V$ for $V_{BS} = 0V$ and $V_{BS} = -1.2V$ at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

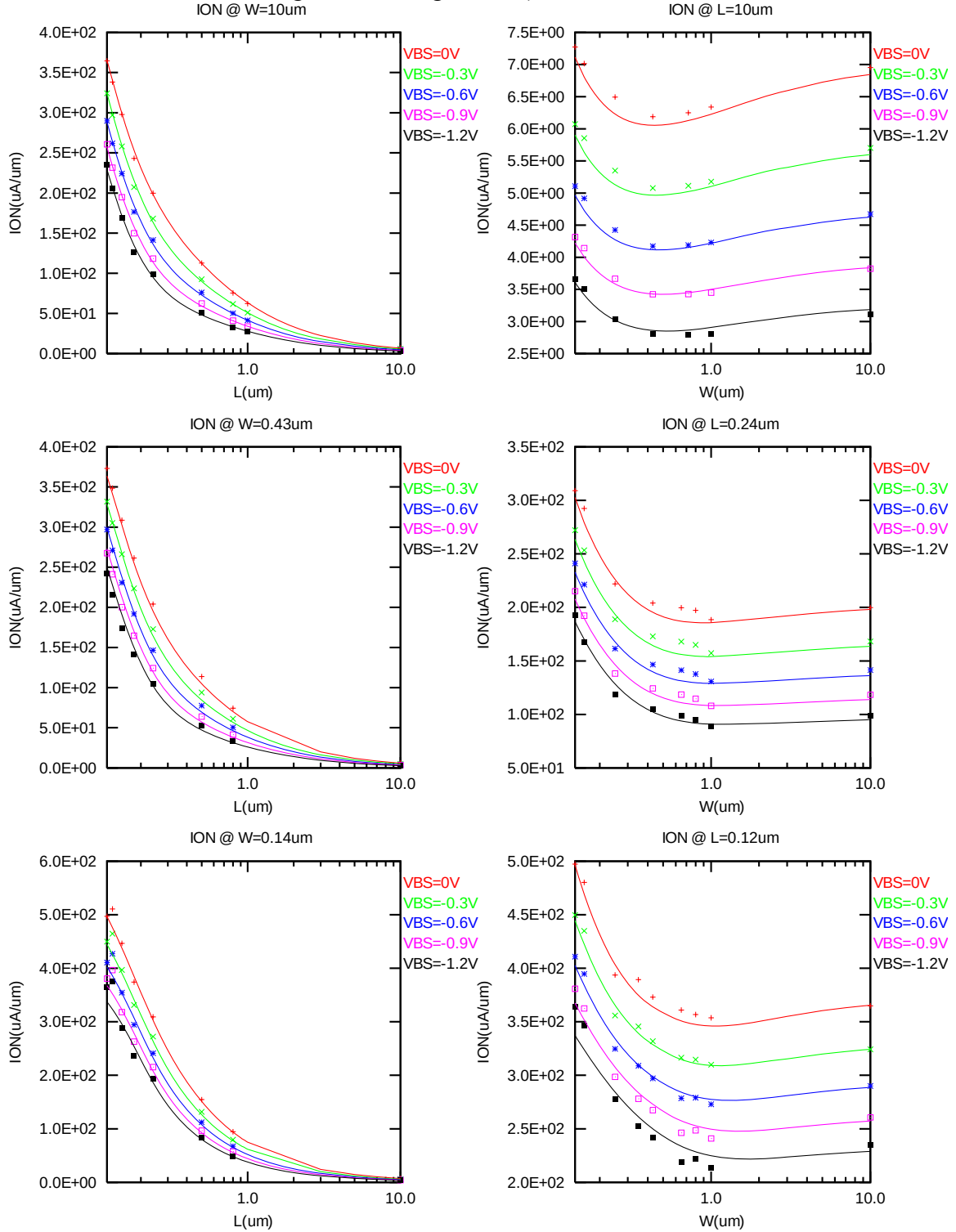


Figure 2-13 Saturation current I_{ON} vs channel length and channel width for NMOS

V_T vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot (W_{\text{DES}} \cdot 0.9) / (L_{\text{DES}} \cdot 0.9 + 0.012\mu\text{m})$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

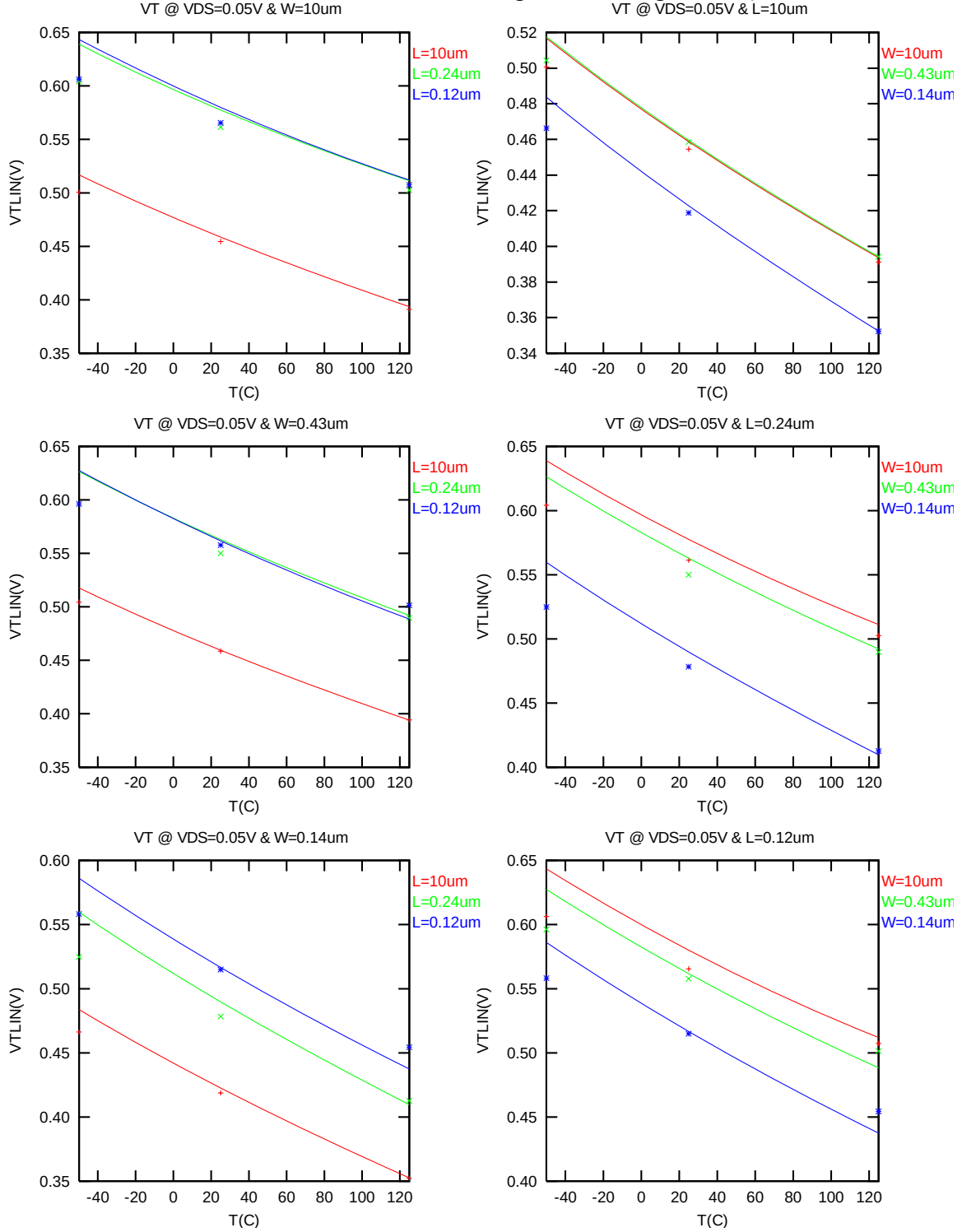


Figure 2-14 $V_{T\text{LIN}}$ vs temperature for NMOS

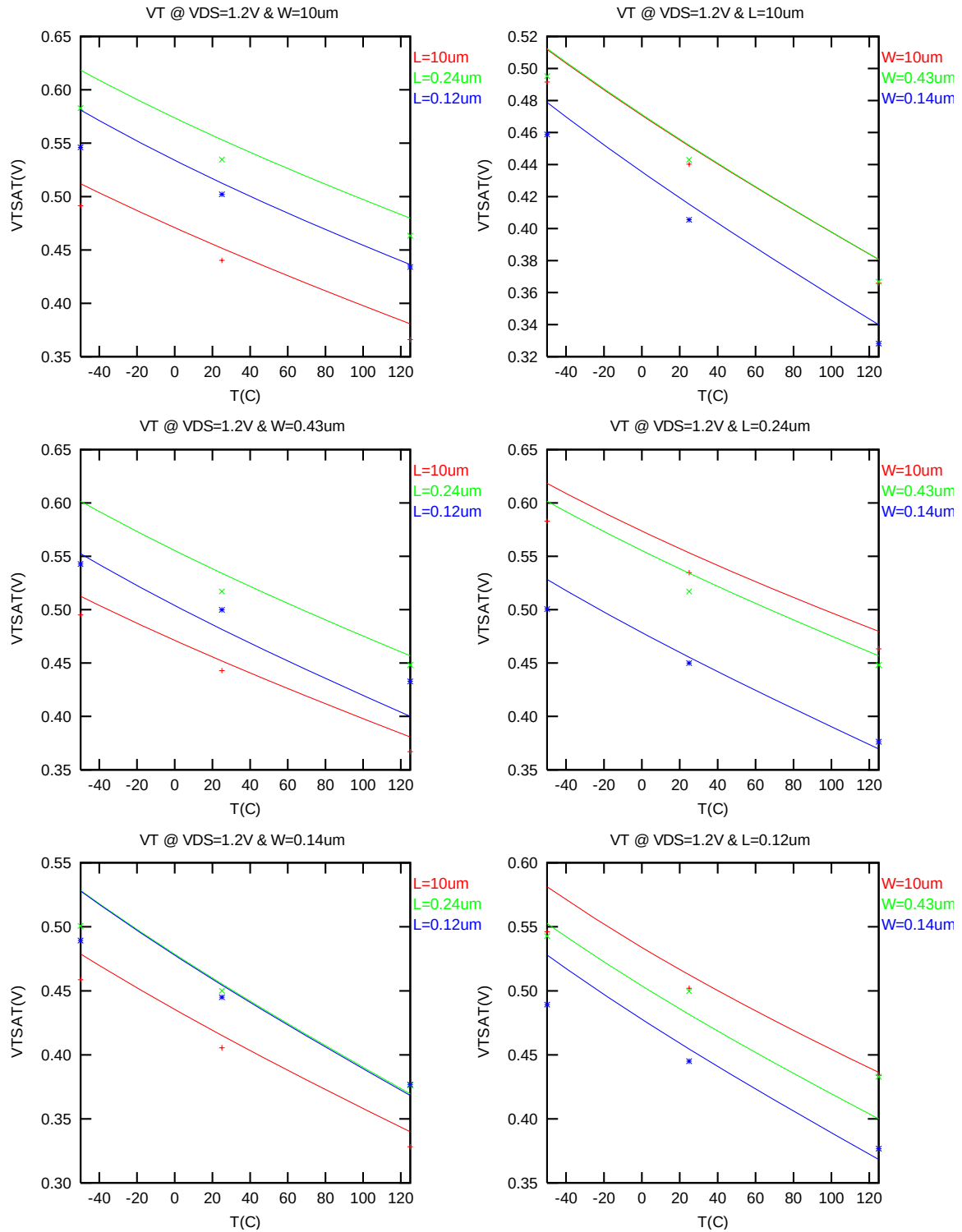


Figure 2-15 V_{TSAT} vs temperature for NMOS

I_{ON} vs Temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2$ V, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

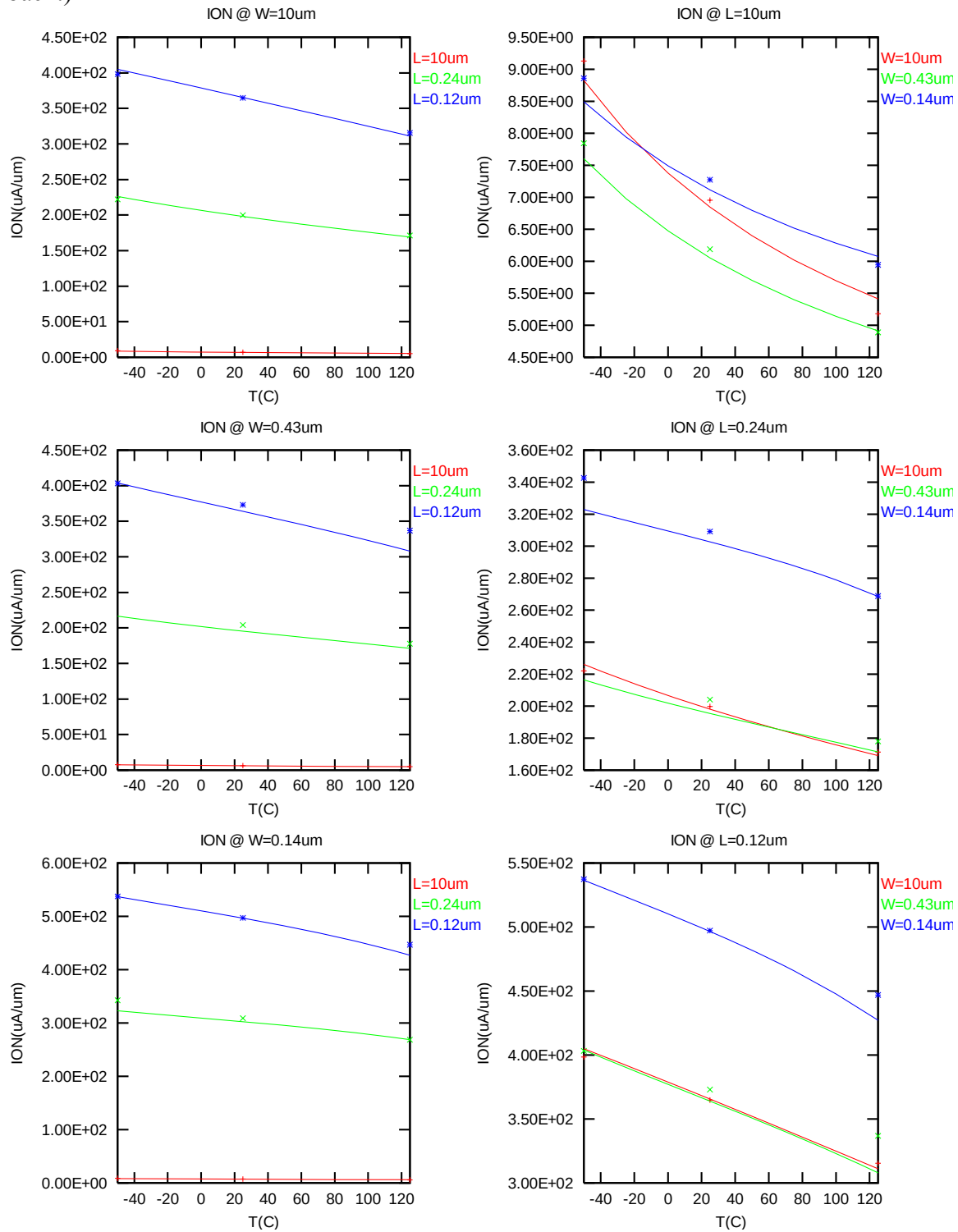


Figure 2-16 Saturation Current vs temperature for NMOS

2.3.4 I_{gate} Extraction plot

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 25C

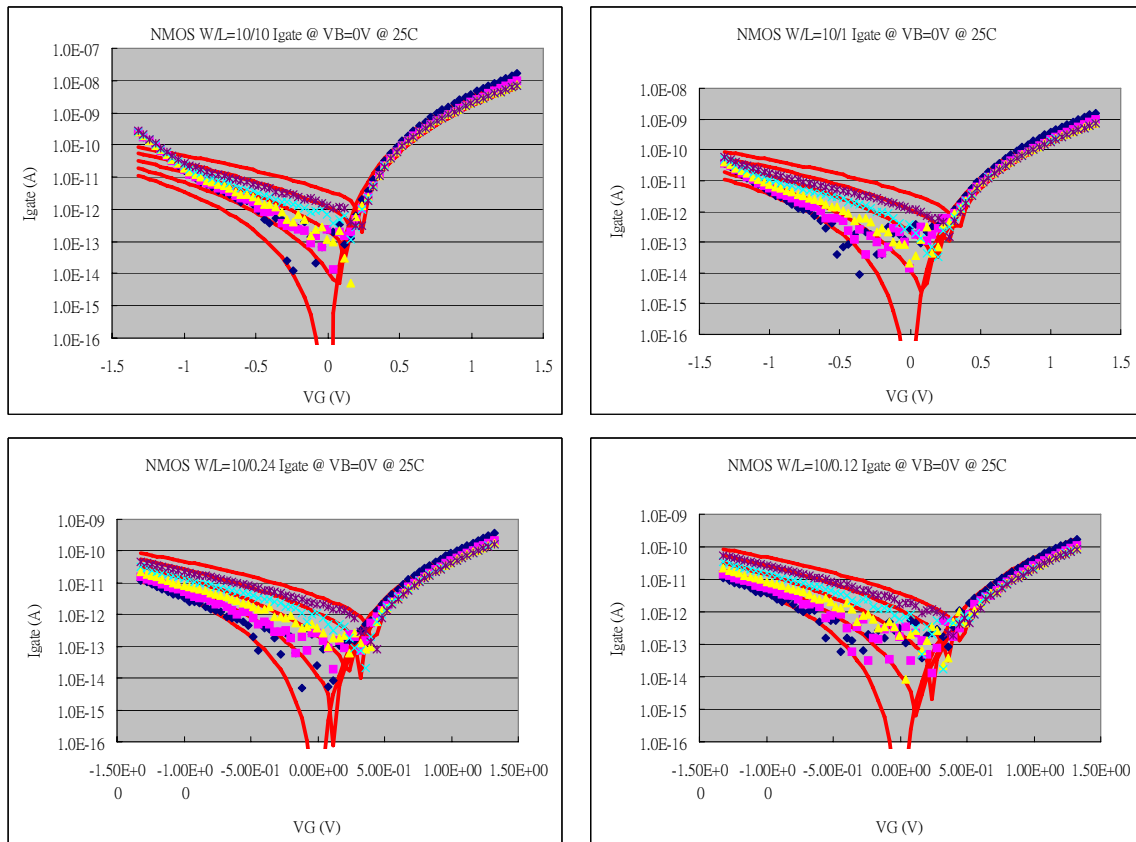


Figure 2-17 Measured and Simulated I_{GATE} for NMOS @ 25C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 125C

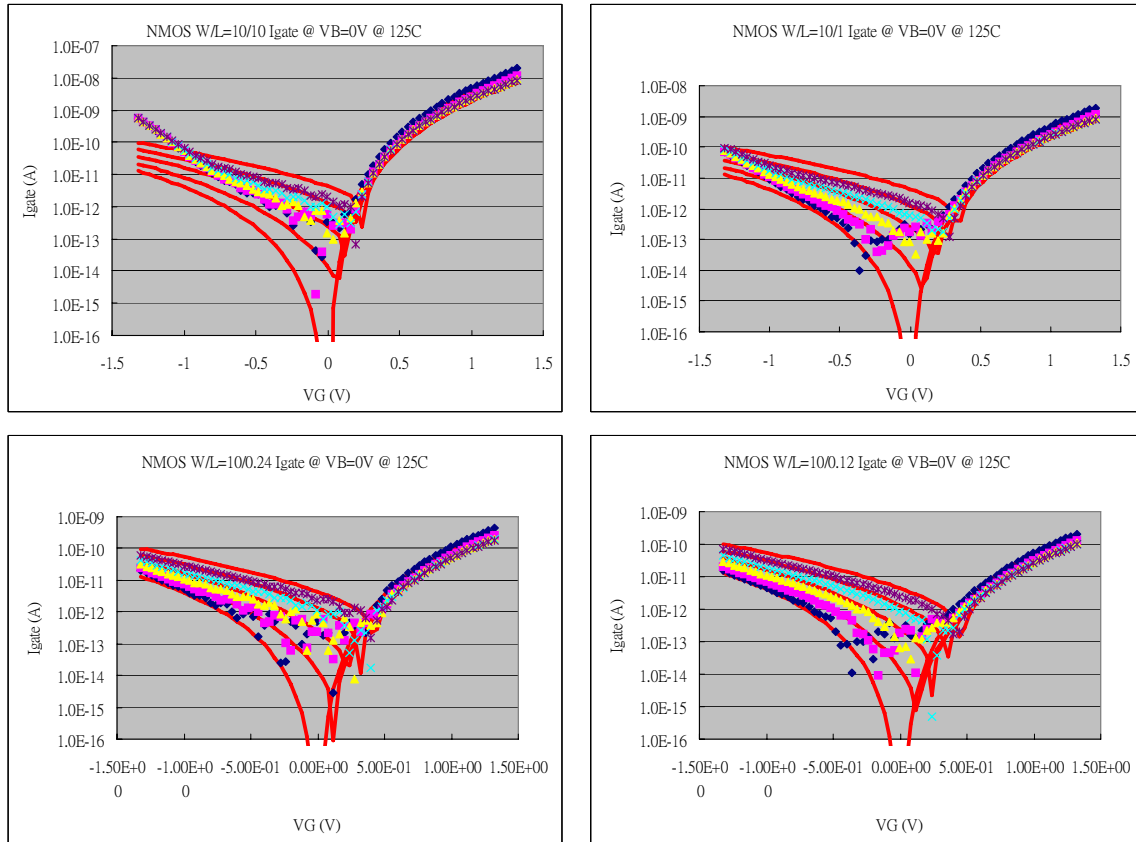


Figure 2-18 Measured and Simulated I_{GATE} for NMOS @ 125C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ -50C

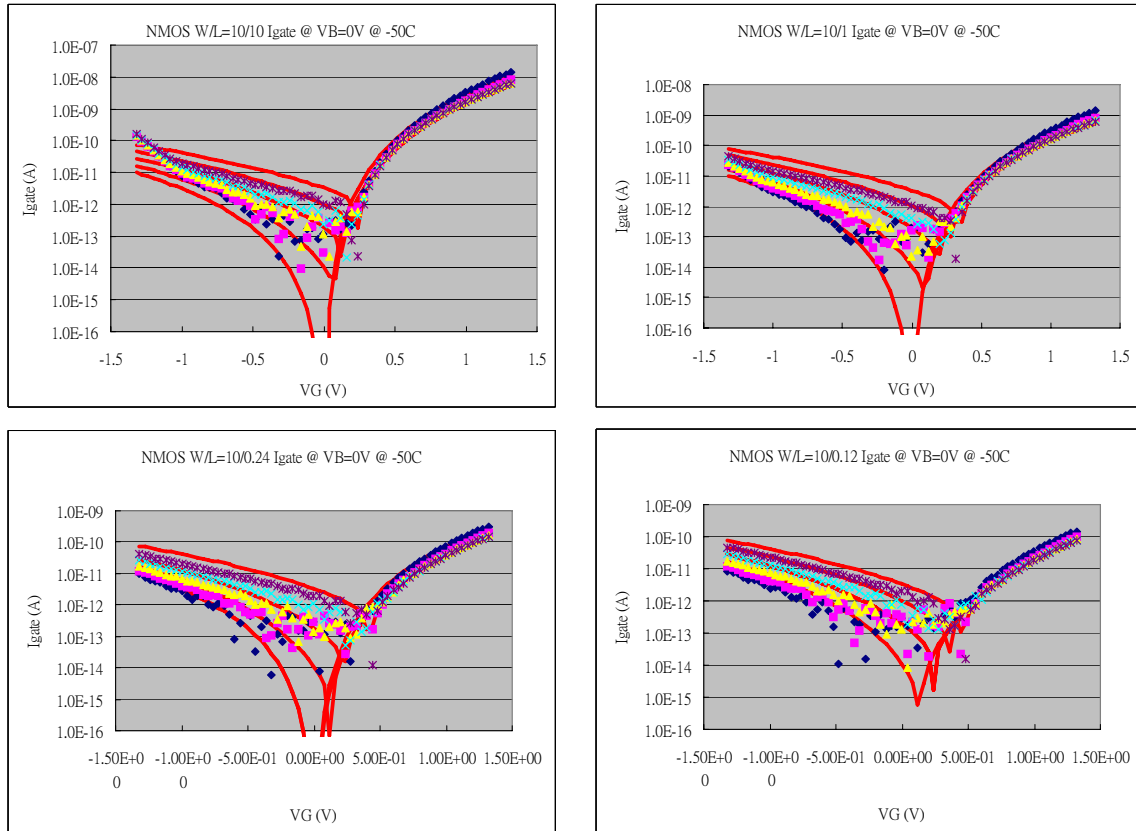


Figure 2-19 Measured and Simulated I_{GATE} for NMOS @ -50C

2.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

- a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- b. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- c. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- a. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- b. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-4 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D Accuracy

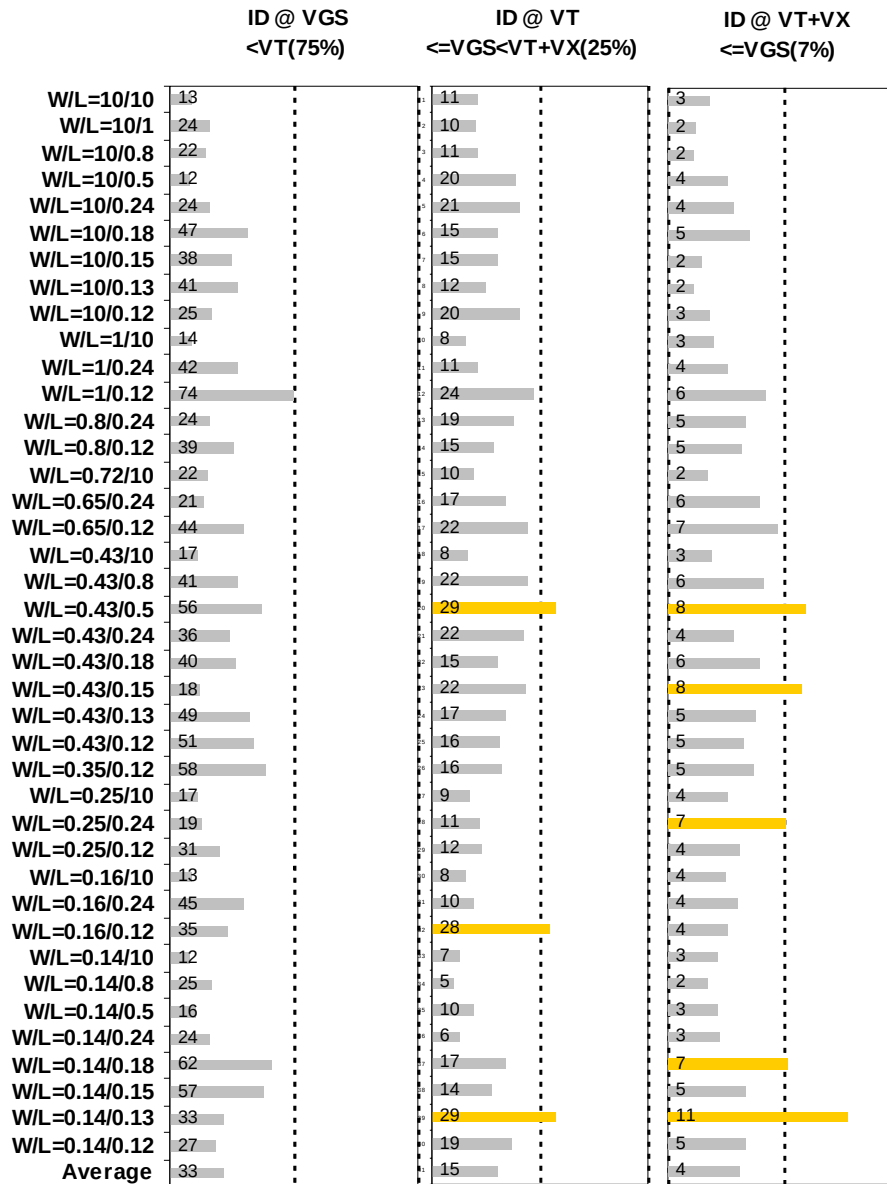


Figure 2-20 I_D accuracy at 25 °C for NMOS

G_M and G_{DS} Accuracy

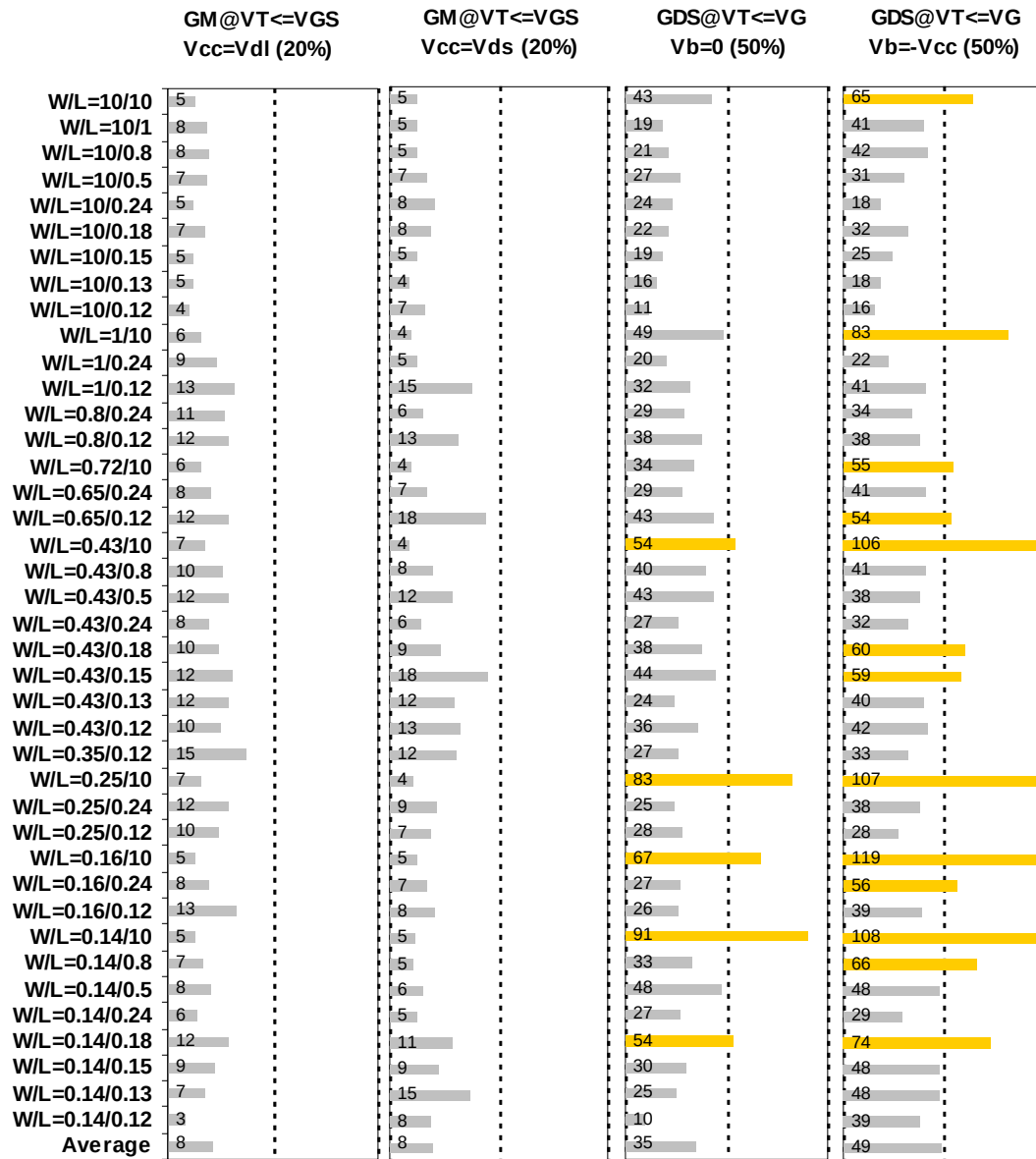


Figure 2-21 G_M and G_{DS} accuracy at 25 °C for NMOS

2.4 Parasitic source / drain junction capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-5 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.83E-02	Poly finger structure

2.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-6 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 1.3 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

2.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-7 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	7.20E-04	CJSW	F/m	9.94E-11	CJSWG	F/m	4.11E-10
PB	V	5.60E-01	PBSW	V	5.40E-01	PBSWG	V	1.11E+00
MJ	--	2.58E-01	MJSW	--	1.30E-01	MJSWG	--	7.17E-01
TCJ	1/K	9.56E-04	TCJSW	1/K	7.32E-04	TCJSWG	1/K	1.44E-03
TPB	V/K	1.63E-03	TPBSW	V/K	3.10E-04	TPBSWG	V/K	1.82E-03

2.4.4 T_{OX} Extraction

C_{GG} , C_{GC} vs. V_{BIAS} Plots

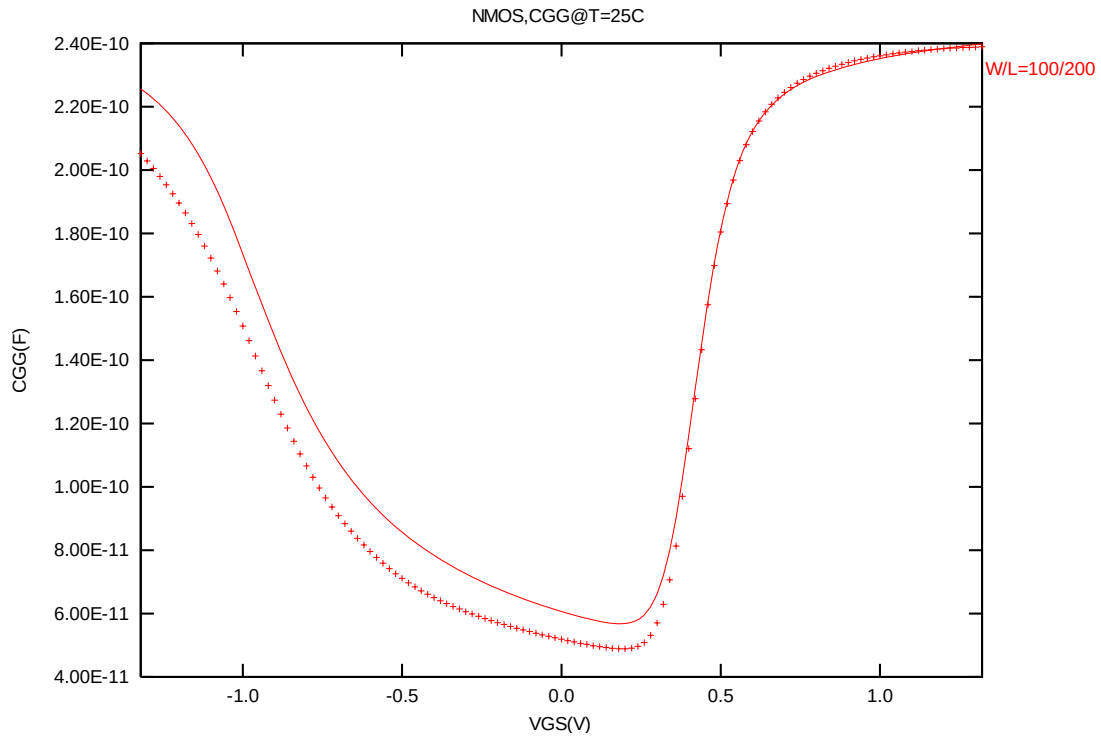


Figure 2-22 Measured and Simulated C_{GG} curves

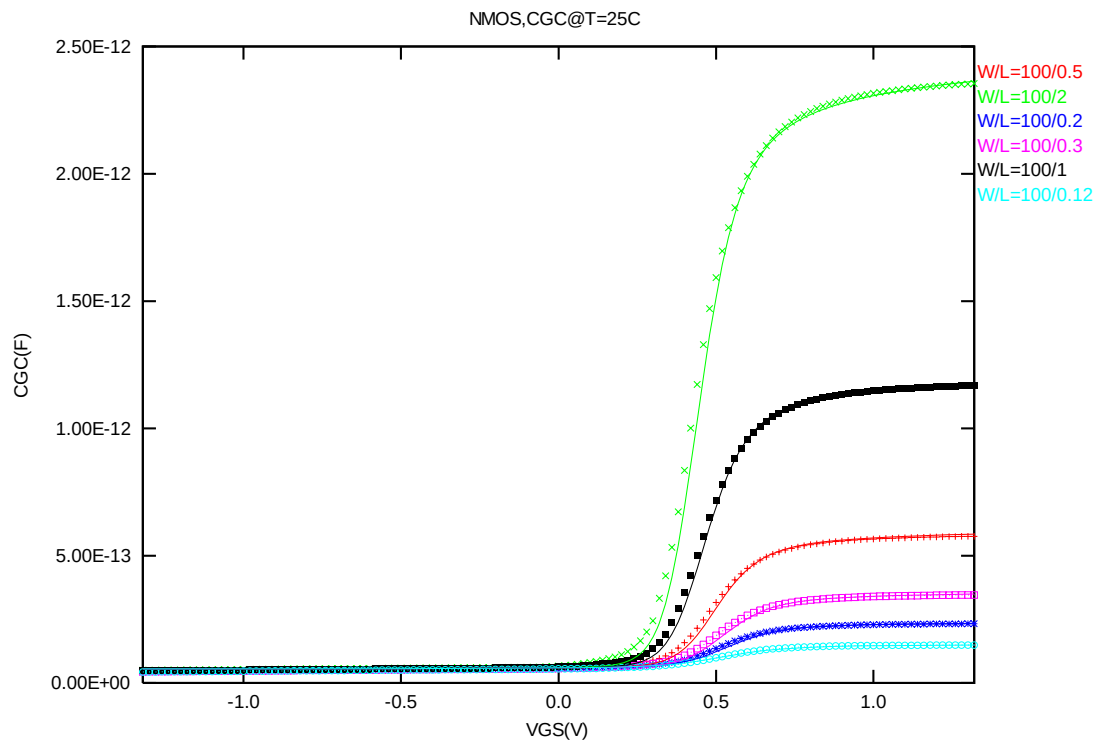


Figure 2-23 Measured and Simulated C_{GC} curves

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

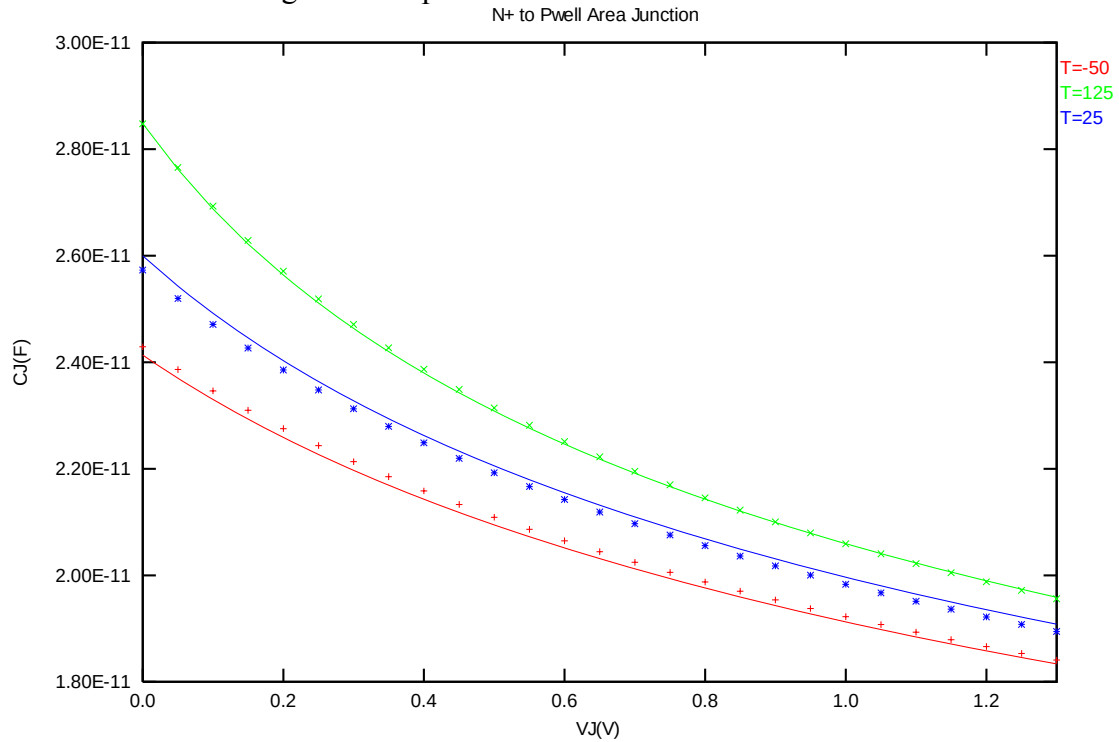


Figure 2-24 Bottom area junction capacitance at -50°C , 25°C and 125°C for NMOS

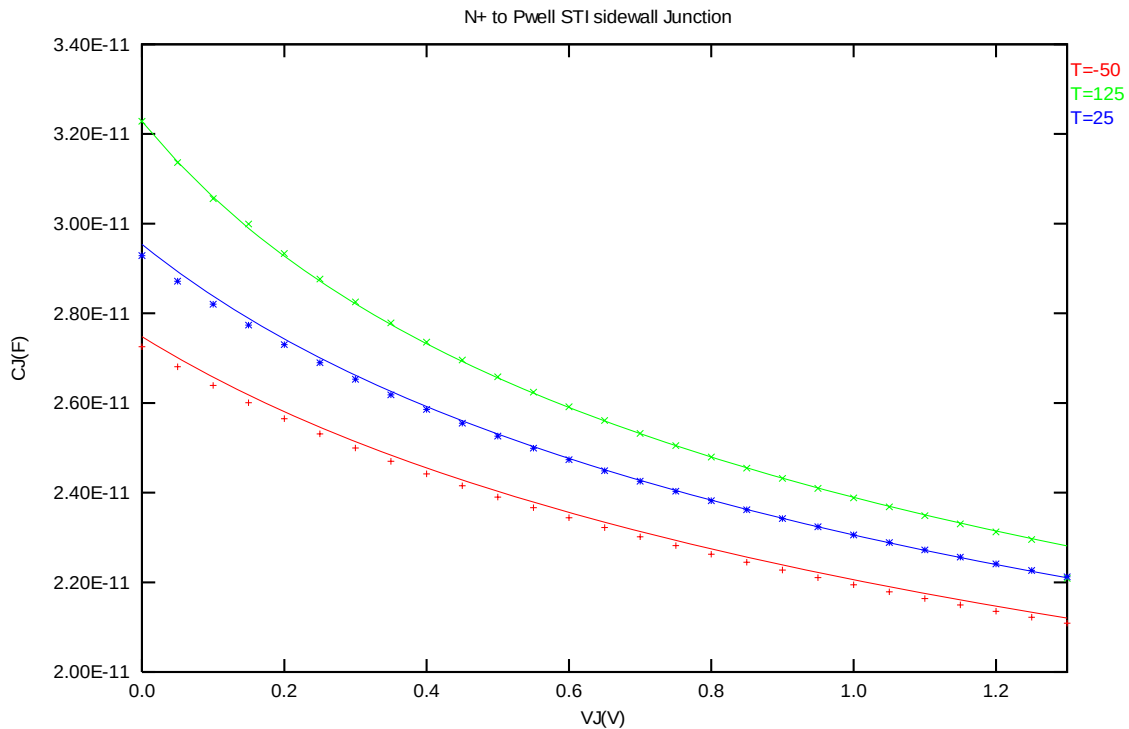


Figure 2-25 STI bounded sidewall junction capacitance at -50°C , 25°C and 125°C for NMOS

3 1.2 V p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGSML	QGSML	QGSML	QGSML
Wafer Number	#12	#12	#12	#12

3.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 3-2 Electrical Parameters for PMOS

All parameters are given for T=25 °C, V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.492	-0.494	-0.497	-0.497
I _{ON}	V _{DS} = -1.2 V	μA/ μm	-1.286	-1.29	-1.28	-1.29

Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.521	-0.518	-0.518	-0.521
V _{TSAT}	V _{DS} = -1.2 V	V	-0.460	-0.459	-0.456	-0.460
Body Effect V _T shift	V _{DS} = -1.2 V V _{BS} = 0 ~ -1.2 V	V	--	-0.236	-0.236	-0.236
DIBL V _T shift	V _{DS} = -0.05~-1.2 V	V	-0.061	-0.059	-0.062	-0.061
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-130	-127.49	-126.37	-130.00
I _{OFF}	V _{DS} = -1.2 V, V _{GS} =0 V	A/μm	-5e-12	-3.3E-12	-3.9E-12	-3.3E-12

Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.459	-0.457	-0.456	-0.456
V _{TSAT}	V _{DS} = -1.2 V	V	--	-0.444	-0.441	-0.441
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-1.843	-1.84	-1.86	-1.86

Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.497	-0.497	-0.493	-0.496
V _{TSAT}	V _{DS} = -1.2 V	V	-0.446	-0.449	-0.455	-0.458
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-157.1	-155.12	-155.50	-155.59
I _{OFF}	V _{DS} = -1.2 V, V _{GS} =0 V	A/μm	-7.664e-12	-3.6E-12	-1.8E-11	-2.2E-12

Capacitance

C _J	V _J = 0 V	F/m ²	9.67e-4	9.65E-04	9.68E-04	9.68E-04
C _{JSW}	V _J = 0 V	F/m	1.22e-10	1.16E-10	1.19E-10	1.19E-10
C _{JSWG}	V _J = 0 V	F/m	5.43e-10	5.09E-10	5.32E-10	5.32E-10

3.3 MOSFET DC model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 3-3 Geometry of PMOS devices measured for parameter extraction

WL	10	1	0.8	0.5	0.24	0.18	0.15	0.13	0.12
10	X	X	X	X	X	X	X	X	X
1	X				X				X
0.8					X				X
0.72	X								
0.65					X				X
0.43	X		X	X	X	X	X	X	X
0.35									X
0.25	X				X				X
0.16	X				X				X
0.14	X		X	X	X	X	X	X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.05	-1.2	-1.15
V_{GS} (V)	0.5	-1.35	-0.025
V_{BS} (V)	0	1.2	0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-1.35	-0.025
V_{GS} (V)	-0.6	-1.2	-0.15
V_{BS} (V)	0	1.2	1.2

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to the physical dimensions ($L_{DES} * 0.9 + 12nm$ and $W_{DES} * 0.9$) respectively.

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

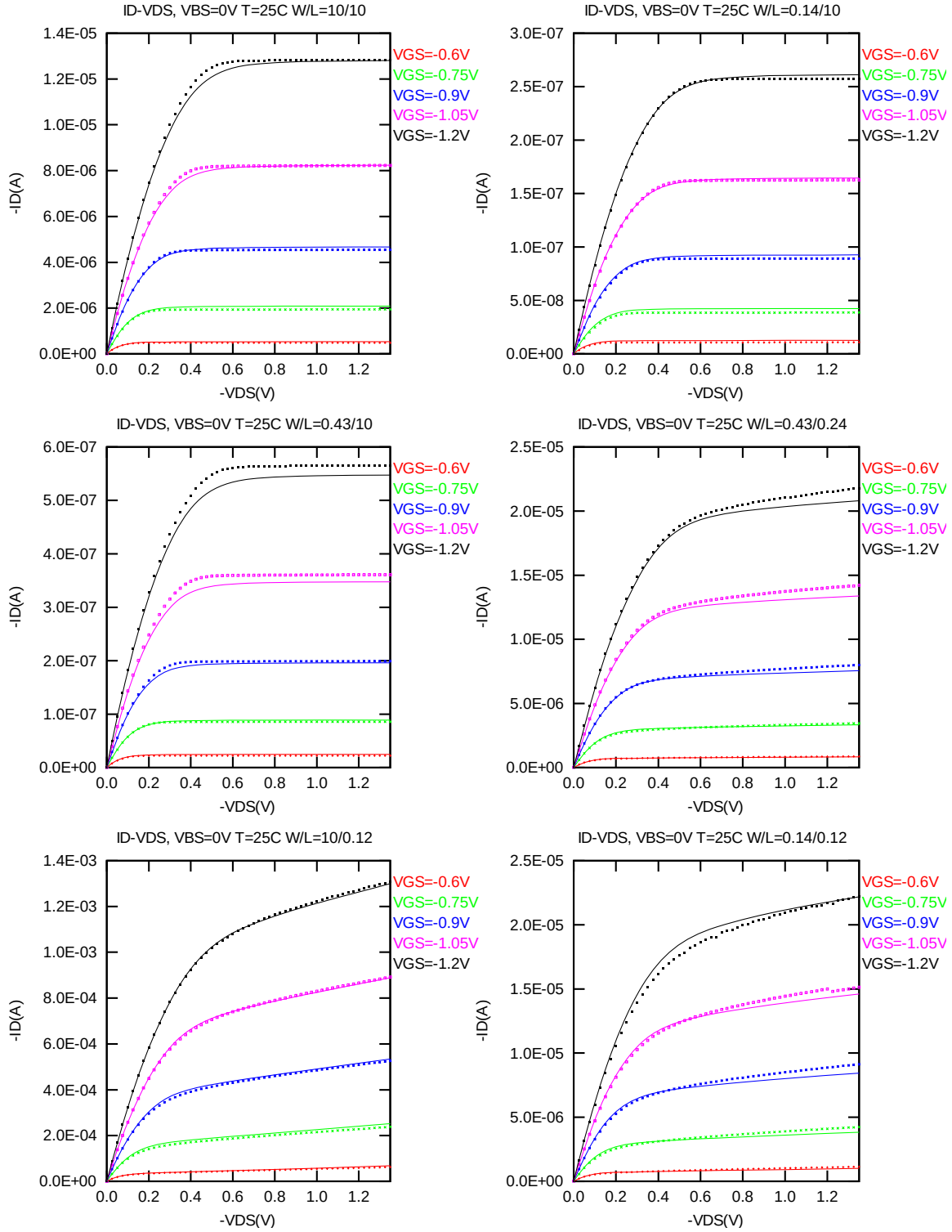


Figure 3-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for PMOS

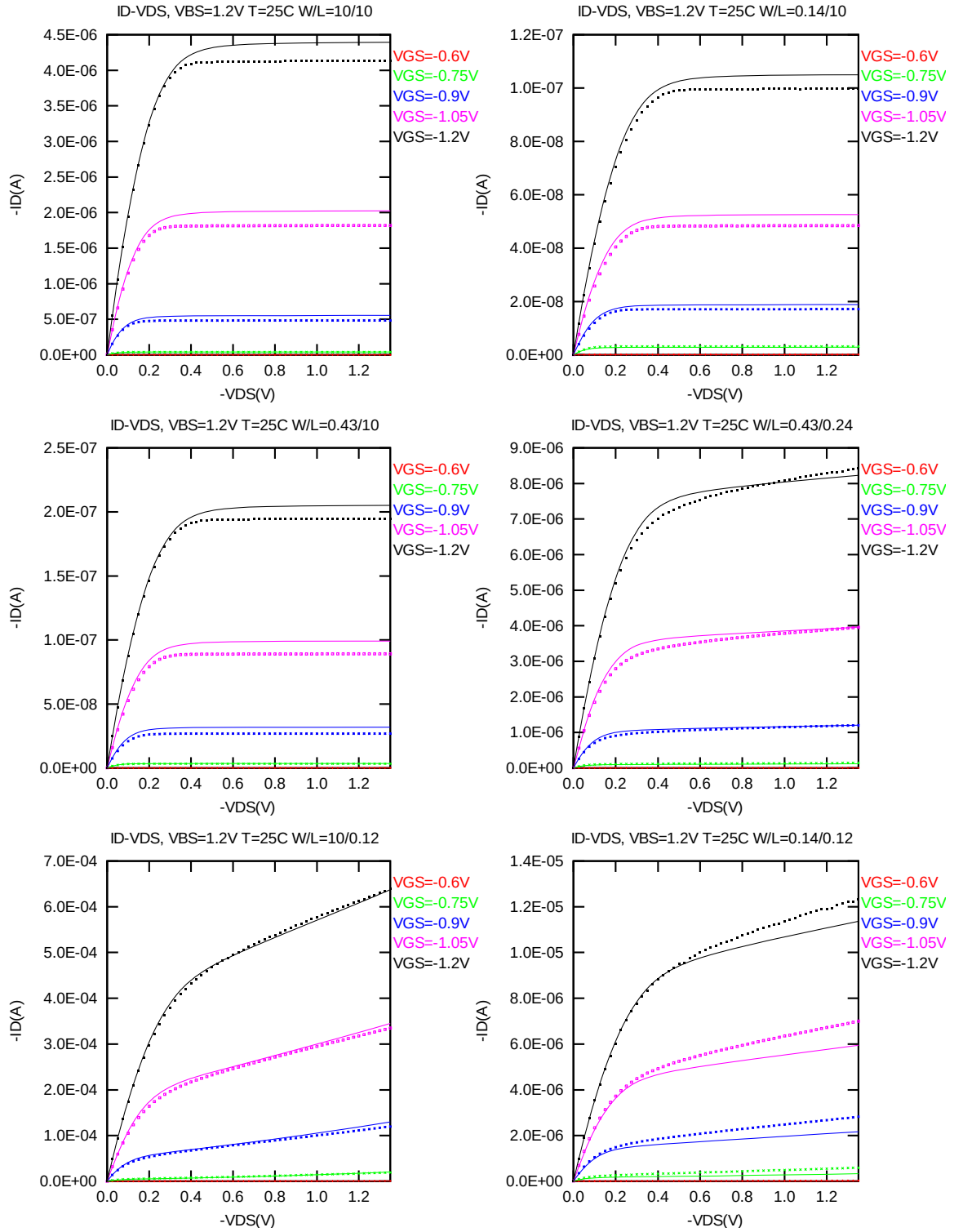


Figure 3-2 I_D vs. V_{DS} at $V_{BS}=1.2$ V for PMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

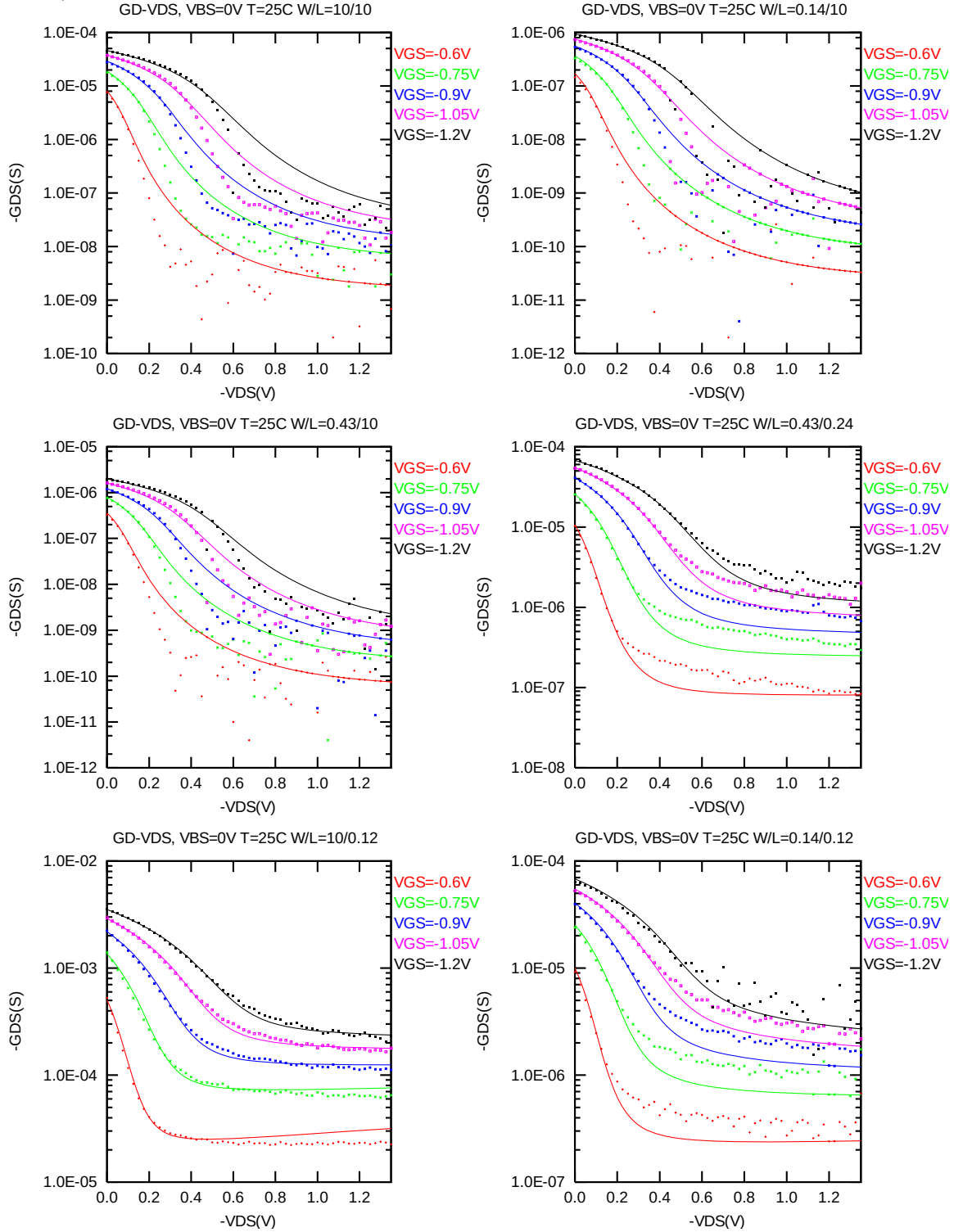


Figure 3-3 G_{DS} vs. V_{DS} at $V_{BS} = 0V$ for PMOS

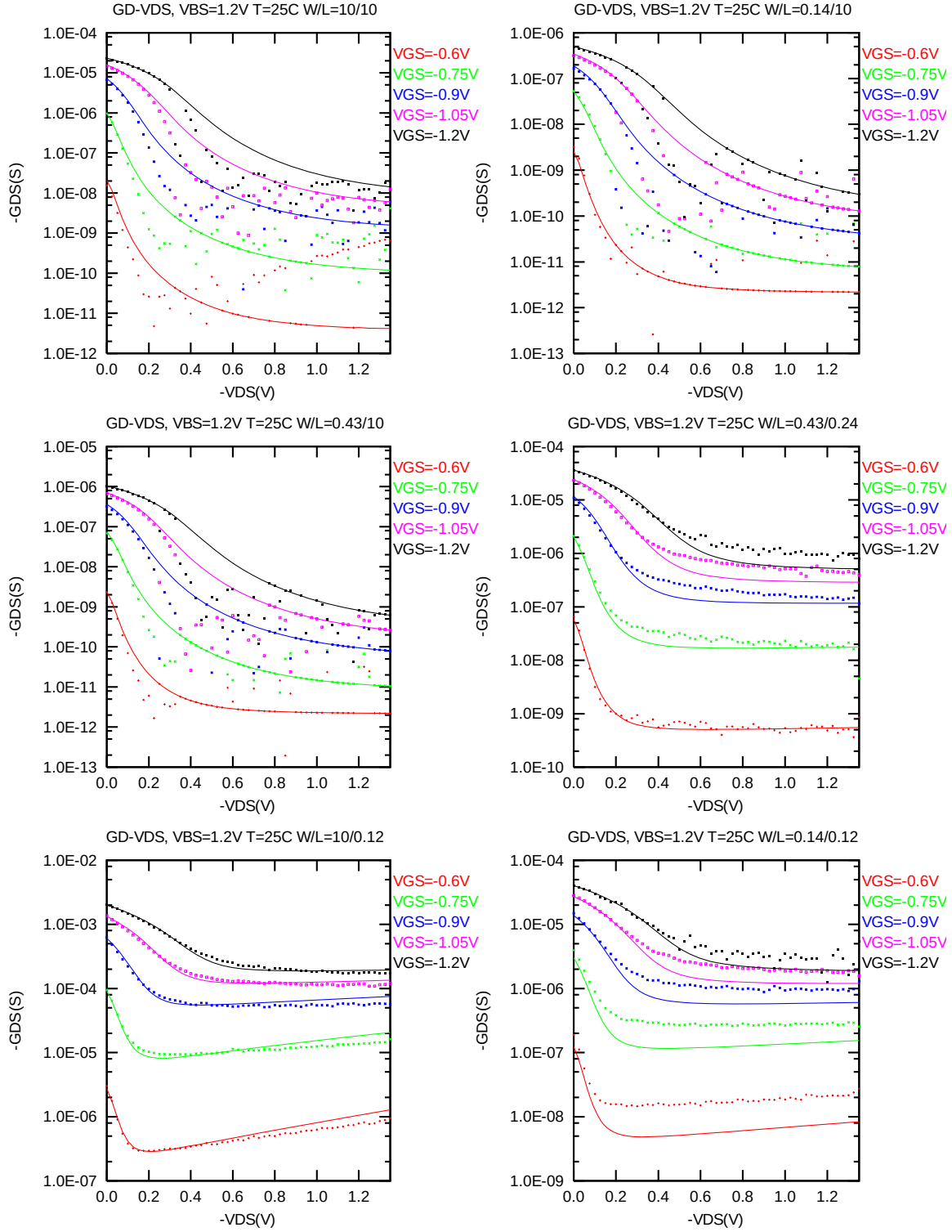


Figure 3-4 G_{DS} vs. V_{DS} at $V_{BS} = 1.2V$ for PMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

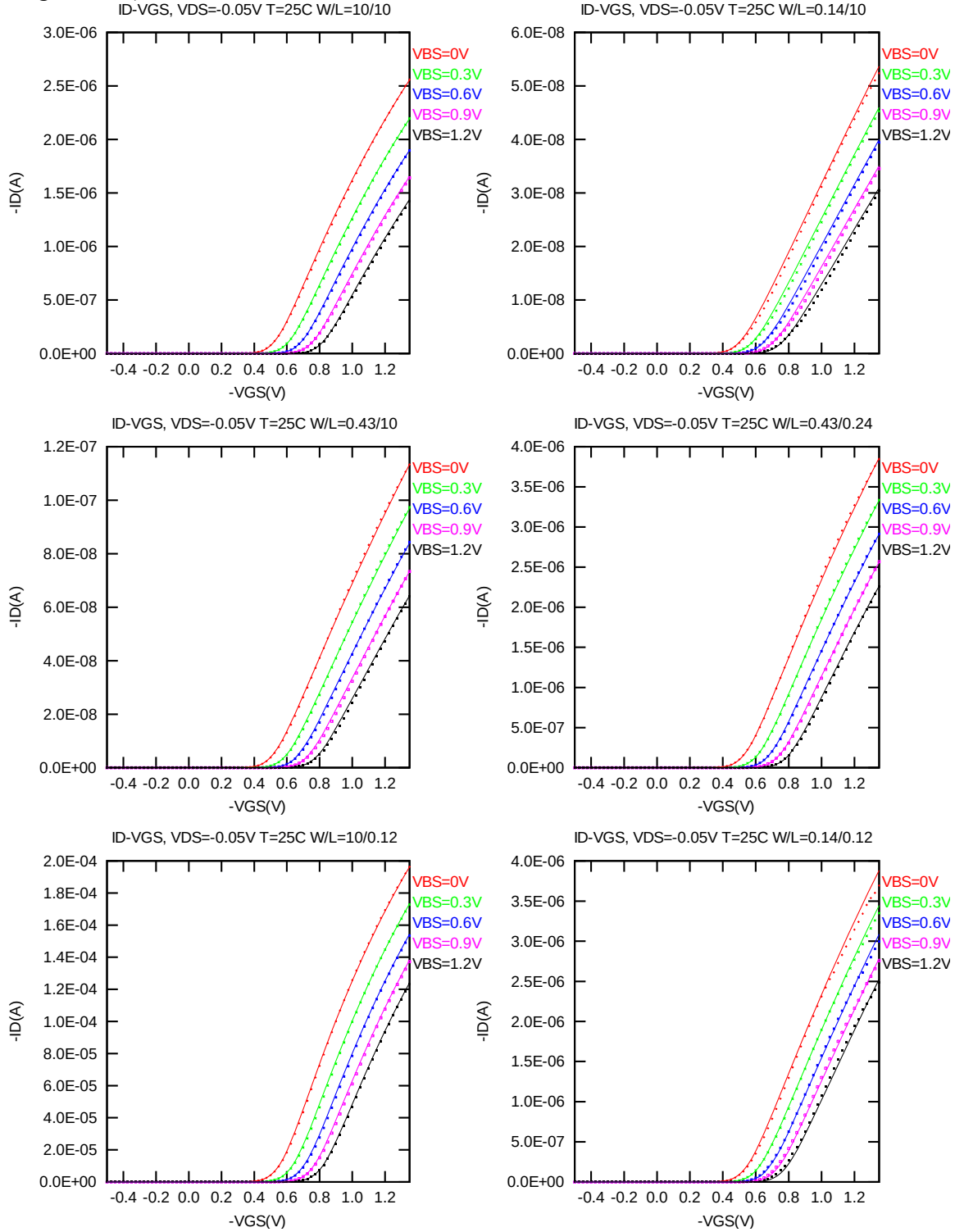


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = -0.05V$ for PMOS

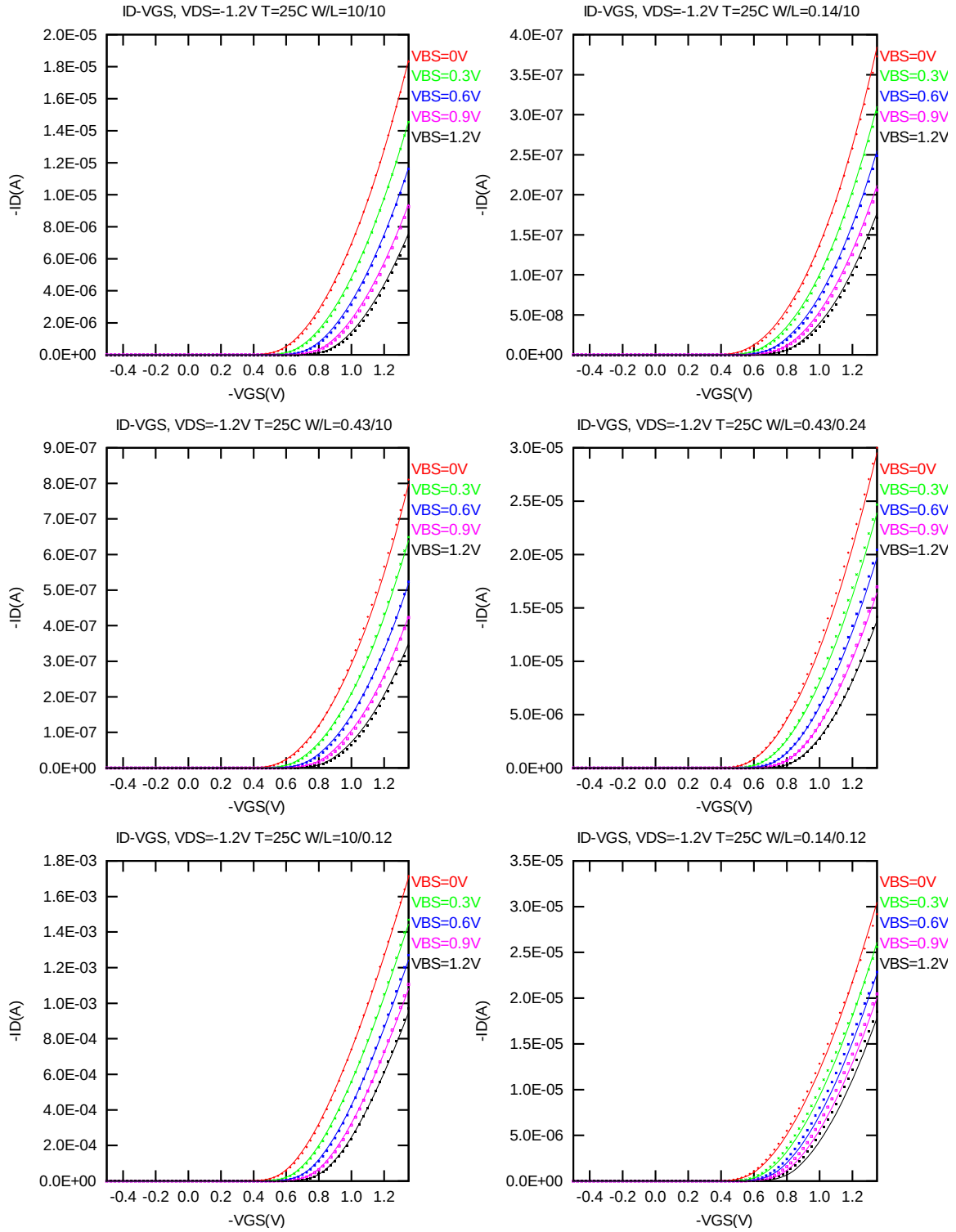


Figure 3-6 I_D vs. V_{GS} at $V_{DS} = -1.2V$ for PMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 °C. The simulation result is with $gmindc=1e-13$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

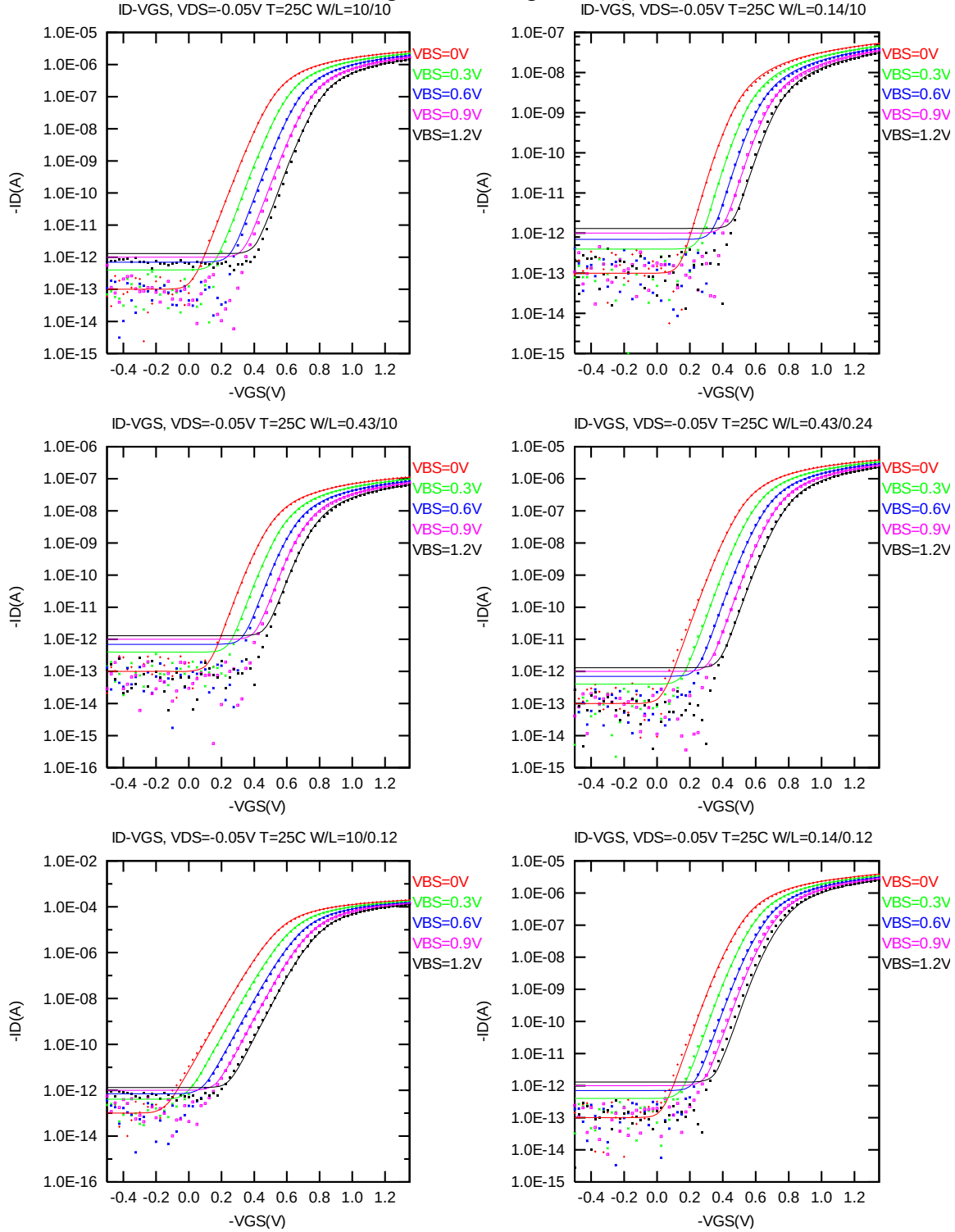


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.05V$ for PMOS

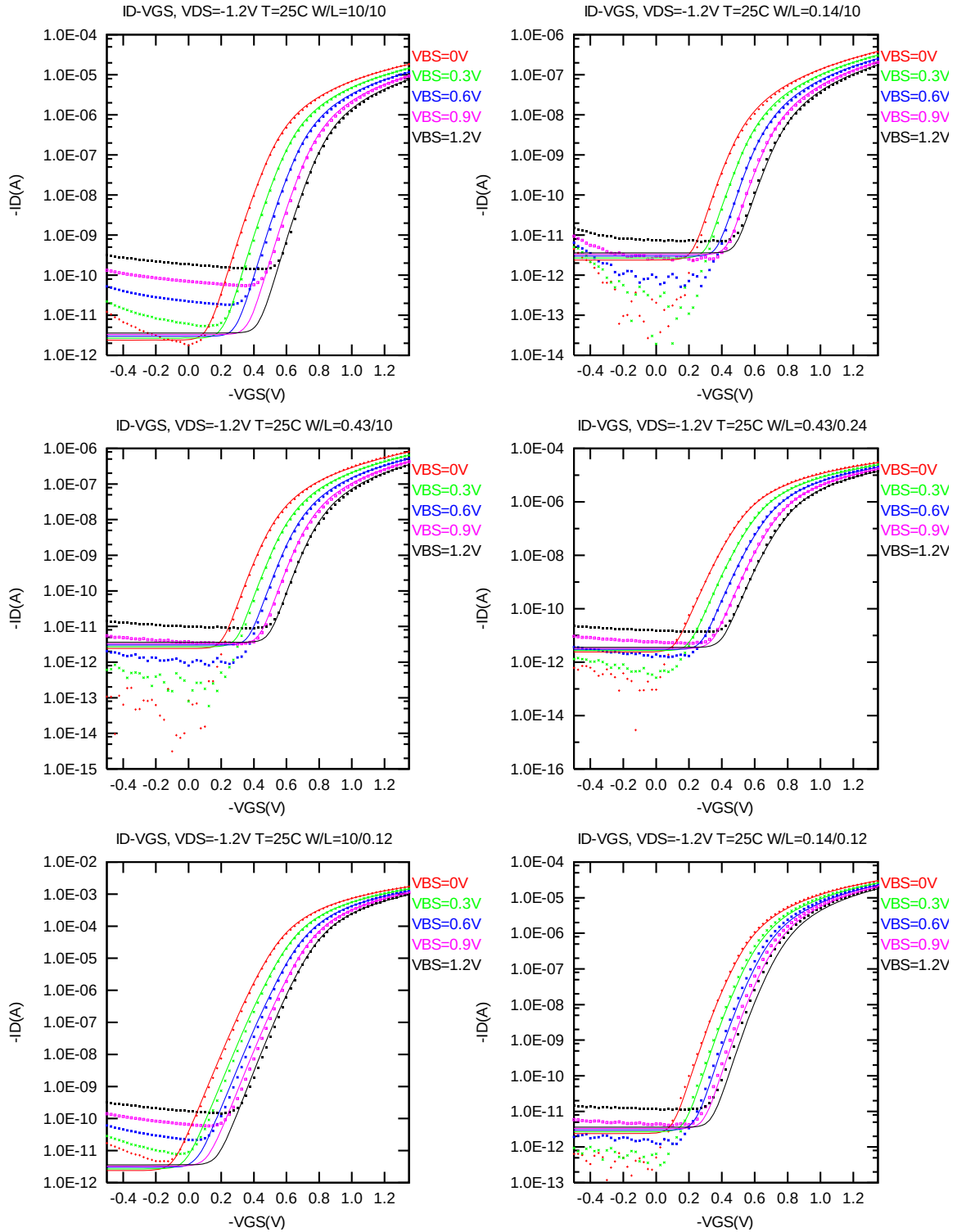


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -1.2V$ for PMOS

G_M vs. V_{GS} Plots

The following plots show the measured and simulated transfer conductance curves in linear scale for selected devices at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

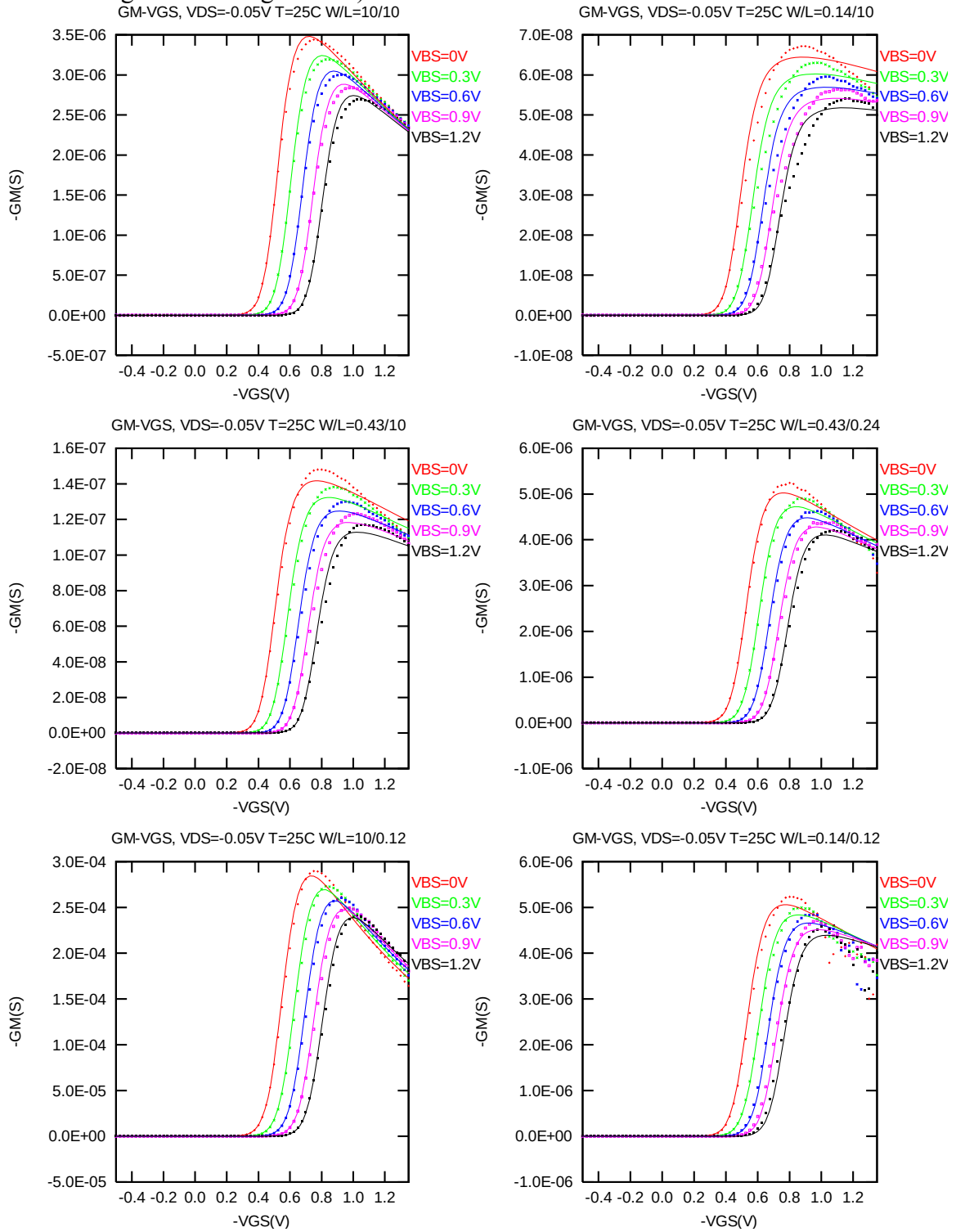


Figure 3-9 G_M vs. V_{GS} at $V_{DS} = -0.05 V$ for PMOS

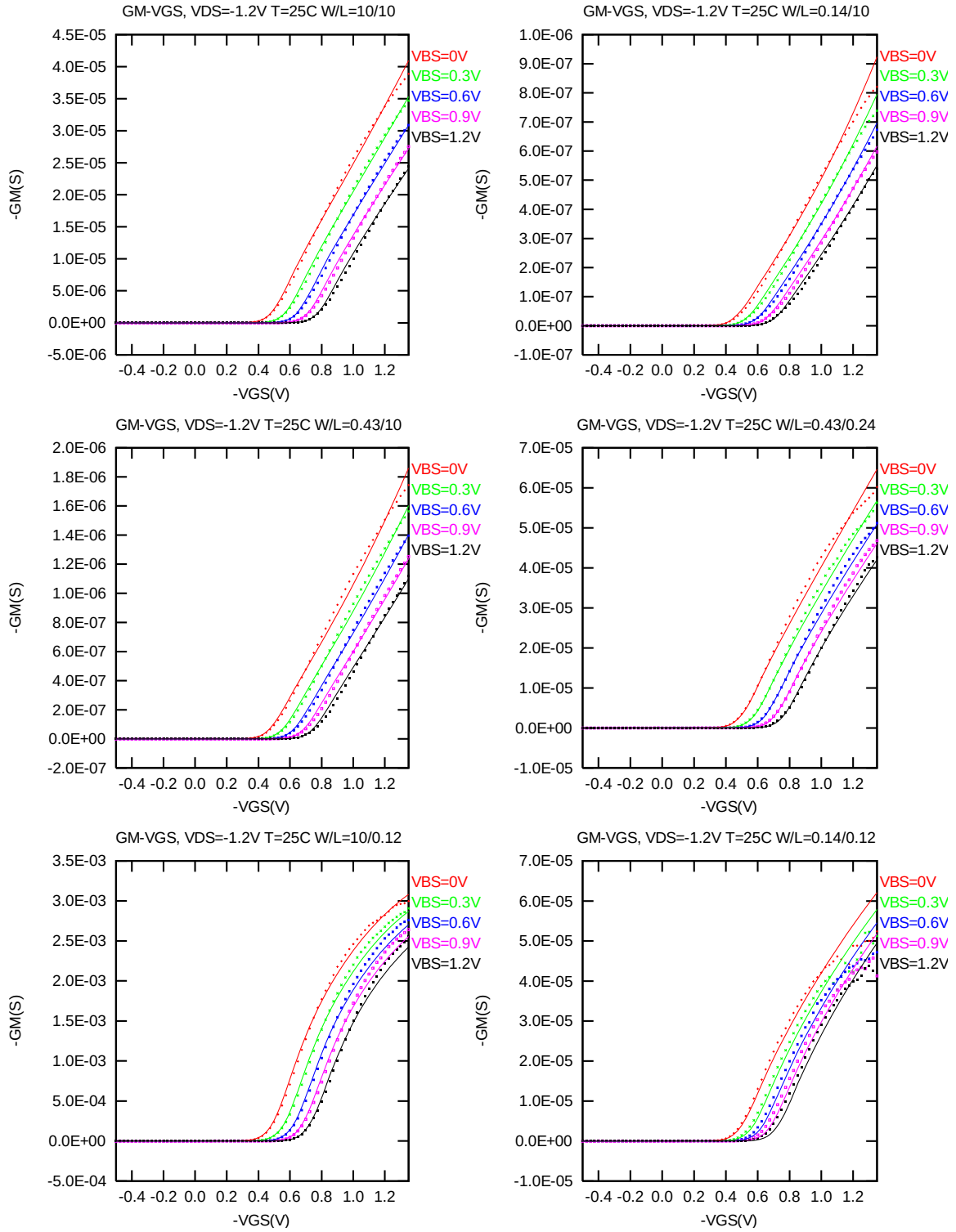


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = 1.2V$ for PMOS

V_T vs. L_{DES} & V_T vs. W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$ at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

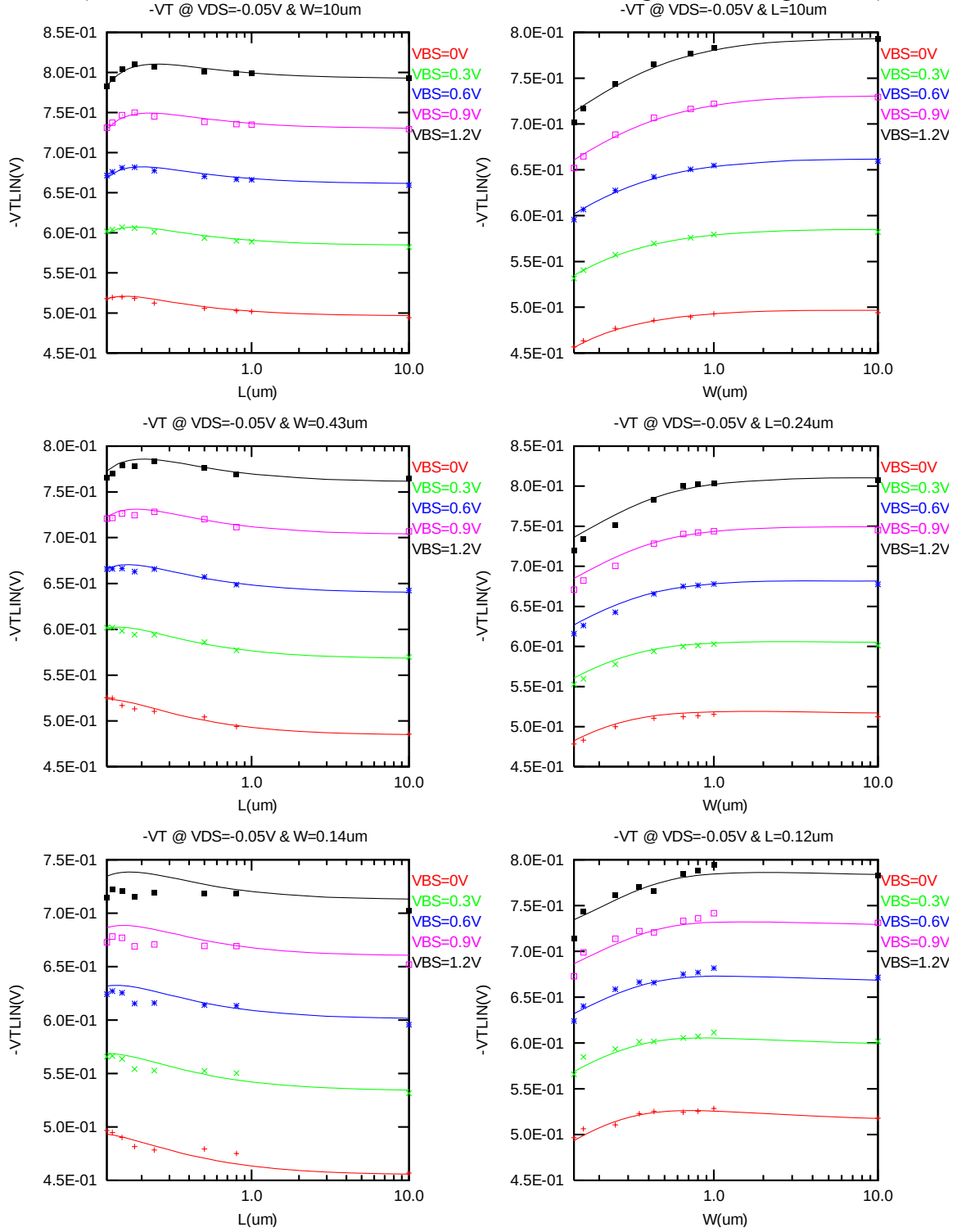


Figure 3-11 V_T vs. channel length & channel width for low drain bias ($V_{DS} = -0.05V$) for PMOS

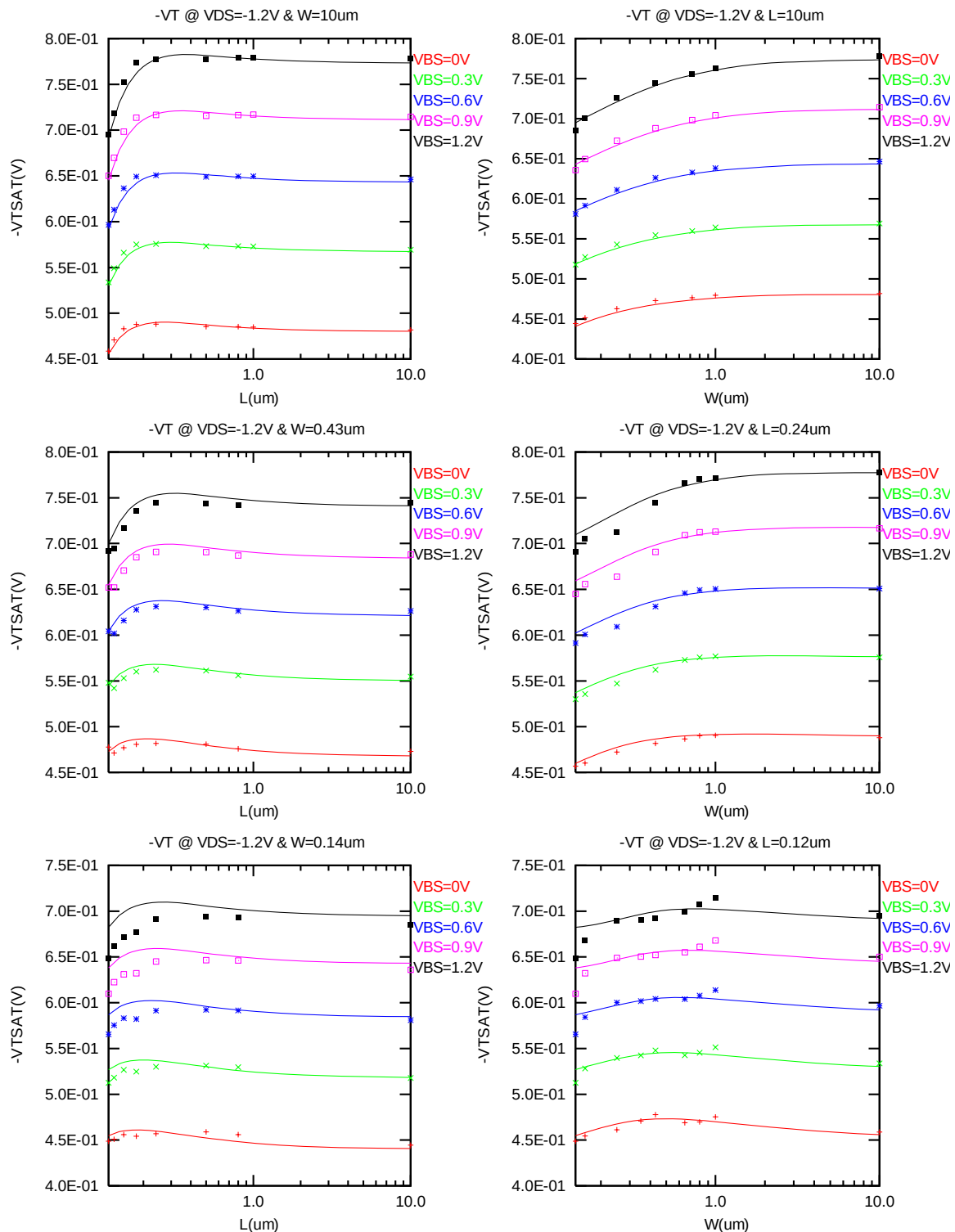


Figure 3-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = -1.2$ V) for PMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -1.2$ for $V_{BS} = 0$ V to $V_{BS} = 1.2$ V at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

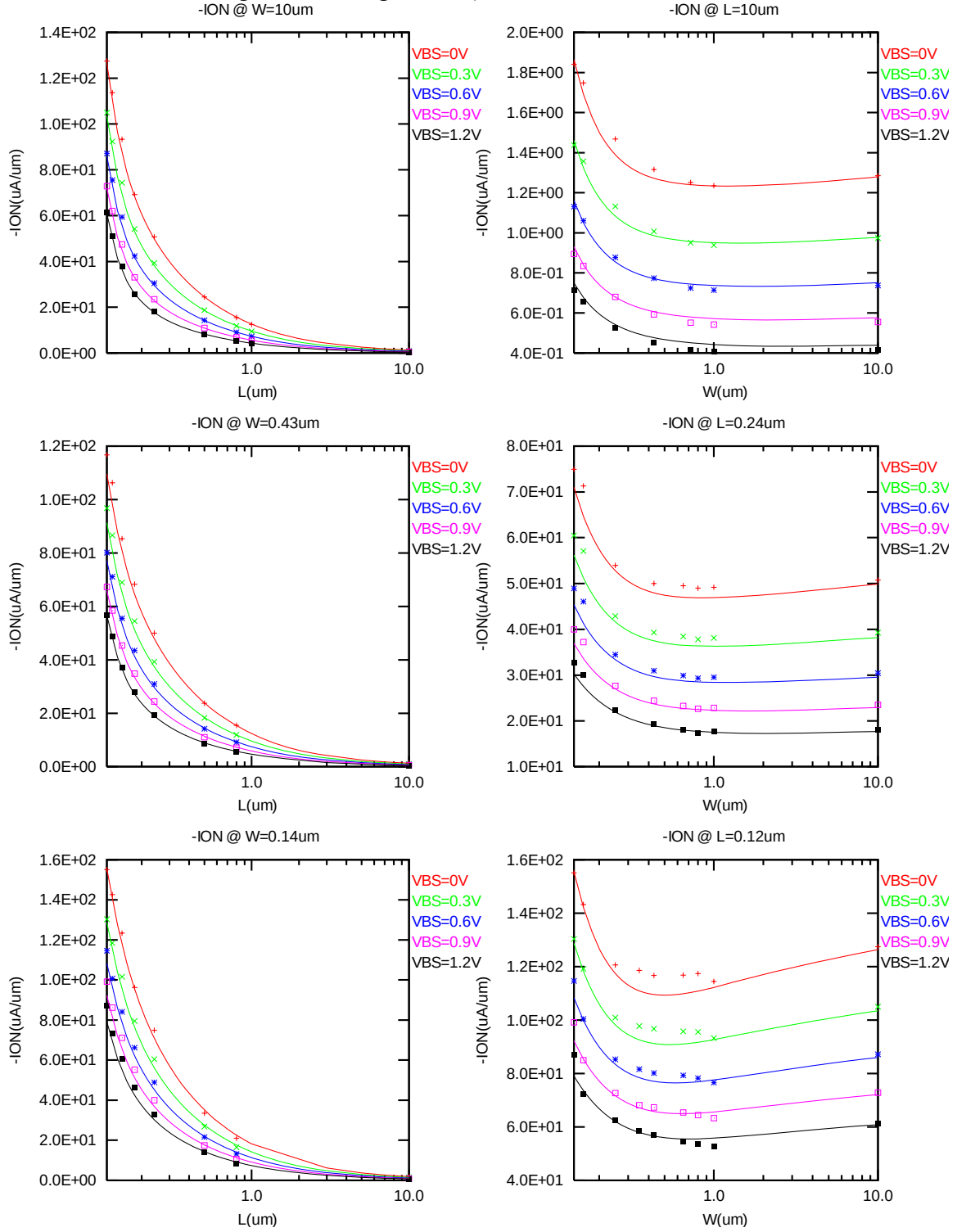


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

V_T vs. Temperature

The following plots give the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot (W_{\text{DES}} \cdot 0.9) / (L_{\text{DES}} \cdot 0.9 + 0.012\mu\text{m})$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

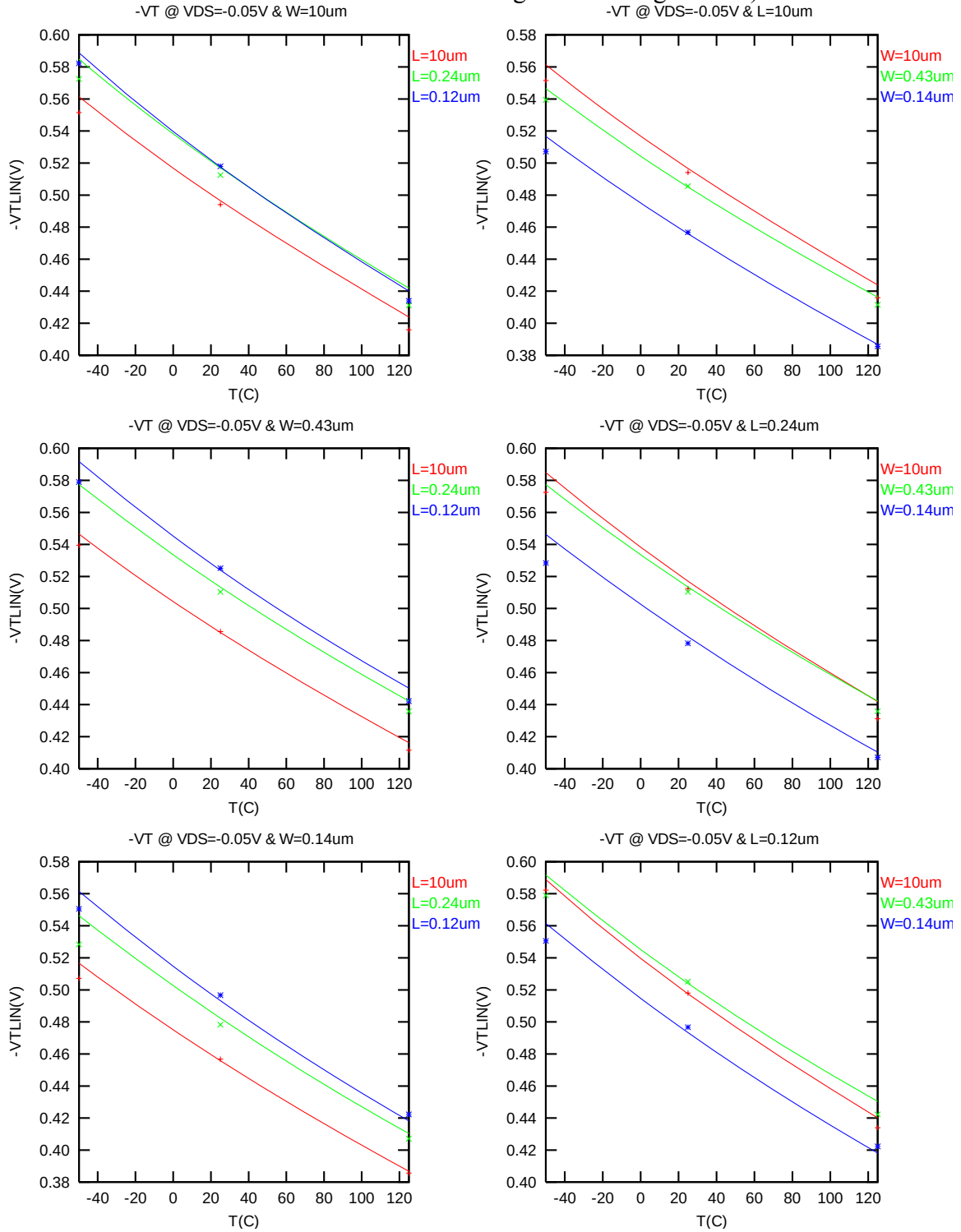


Figure 3-14 $V_{T\text{LIN}}$ vs. temperature for PMOS

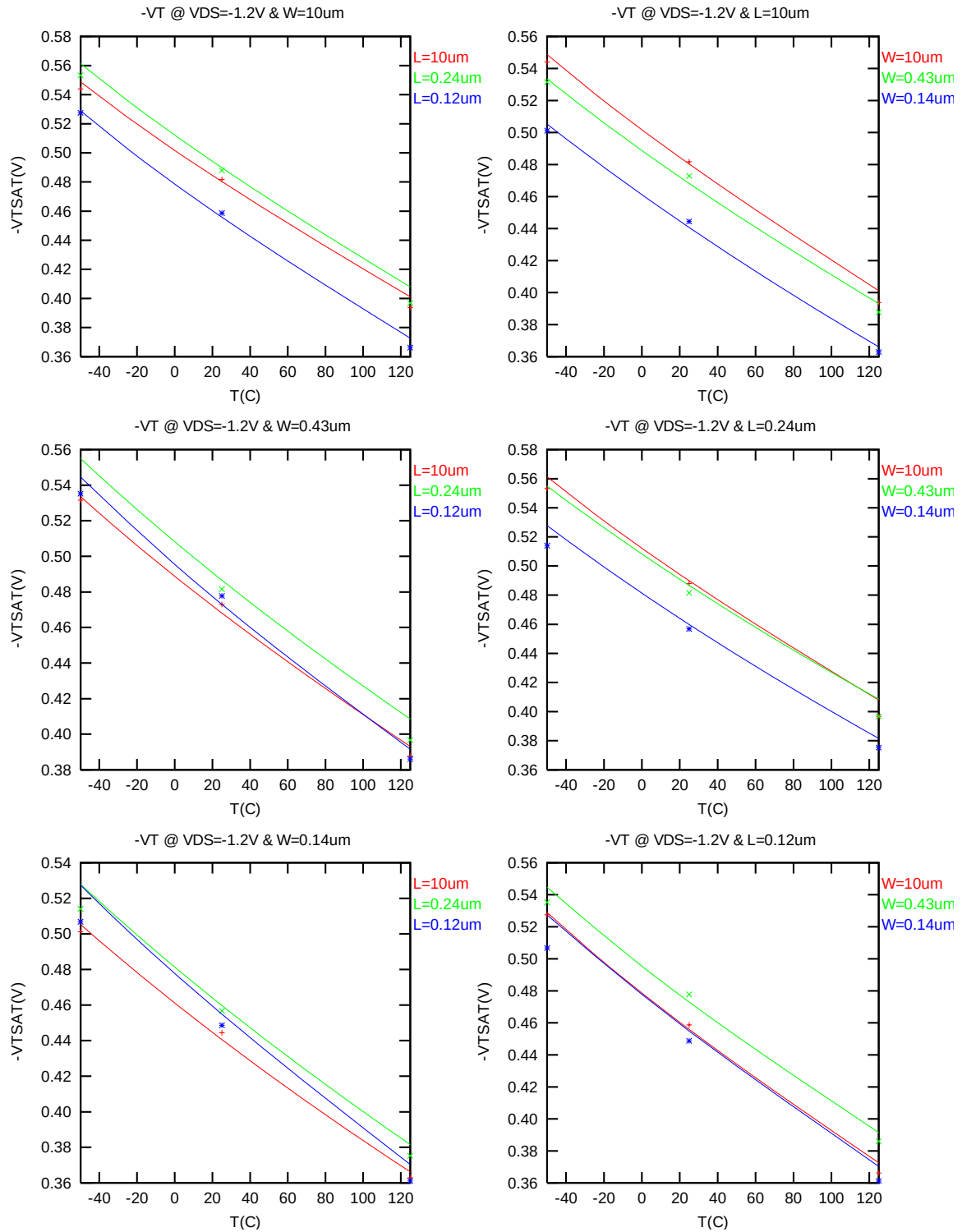


Figure 3-15 $V_{T_{SAT}}$ vs temperature for PMOS

I_{ON} vs Temperature

The following plot shows the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS}=-1.2$ V, $V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

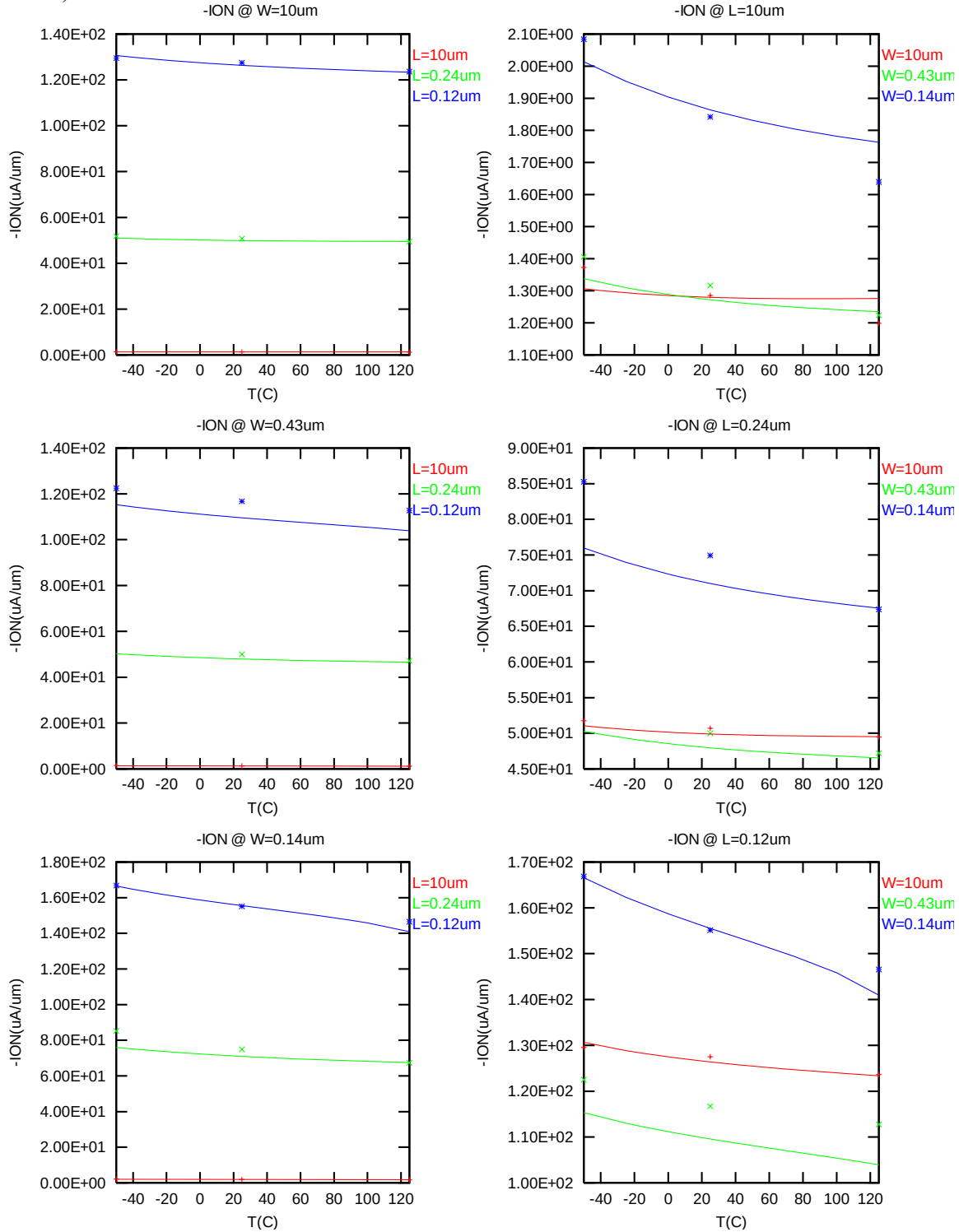


Figure 3-16 Saturation Current vs. temperature for PMOS

3.3.4 I_{gate} Extraction plot

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 25C

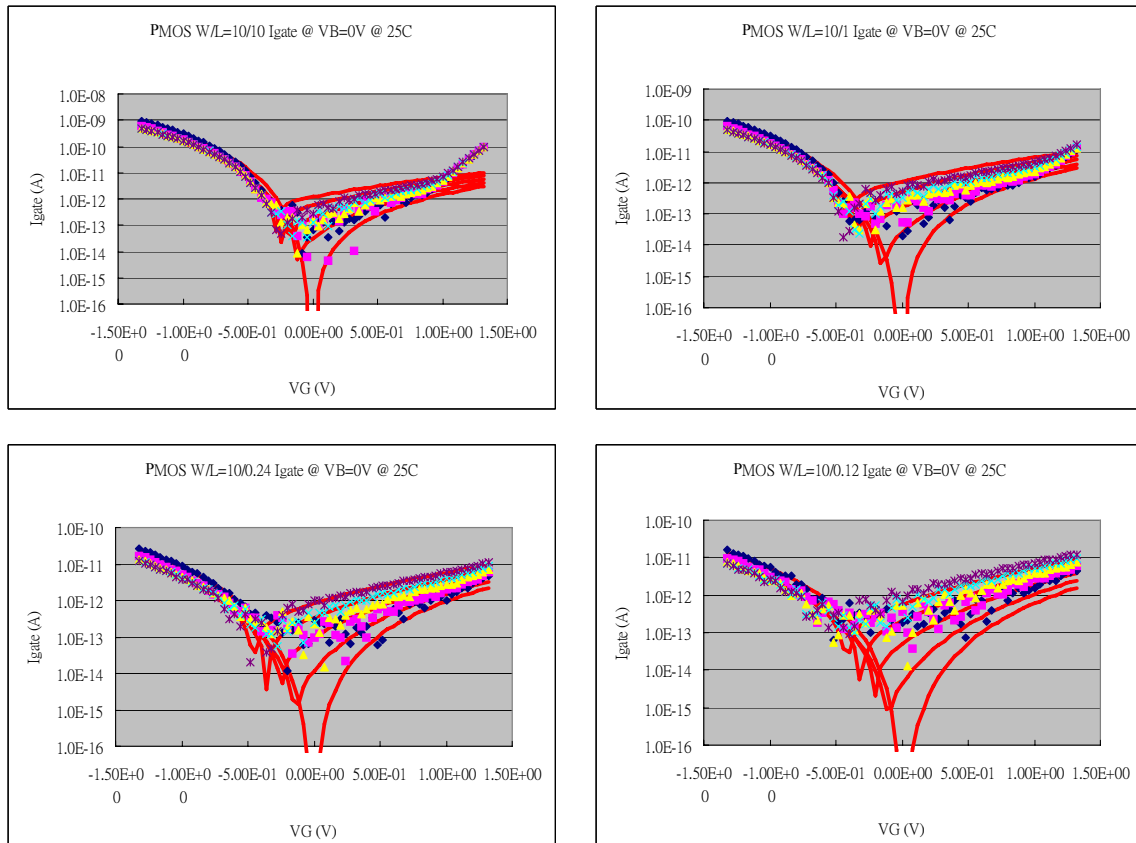


Figure 3-17 Measured and Simulated I_{GATE} for PMOS @25C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 125C

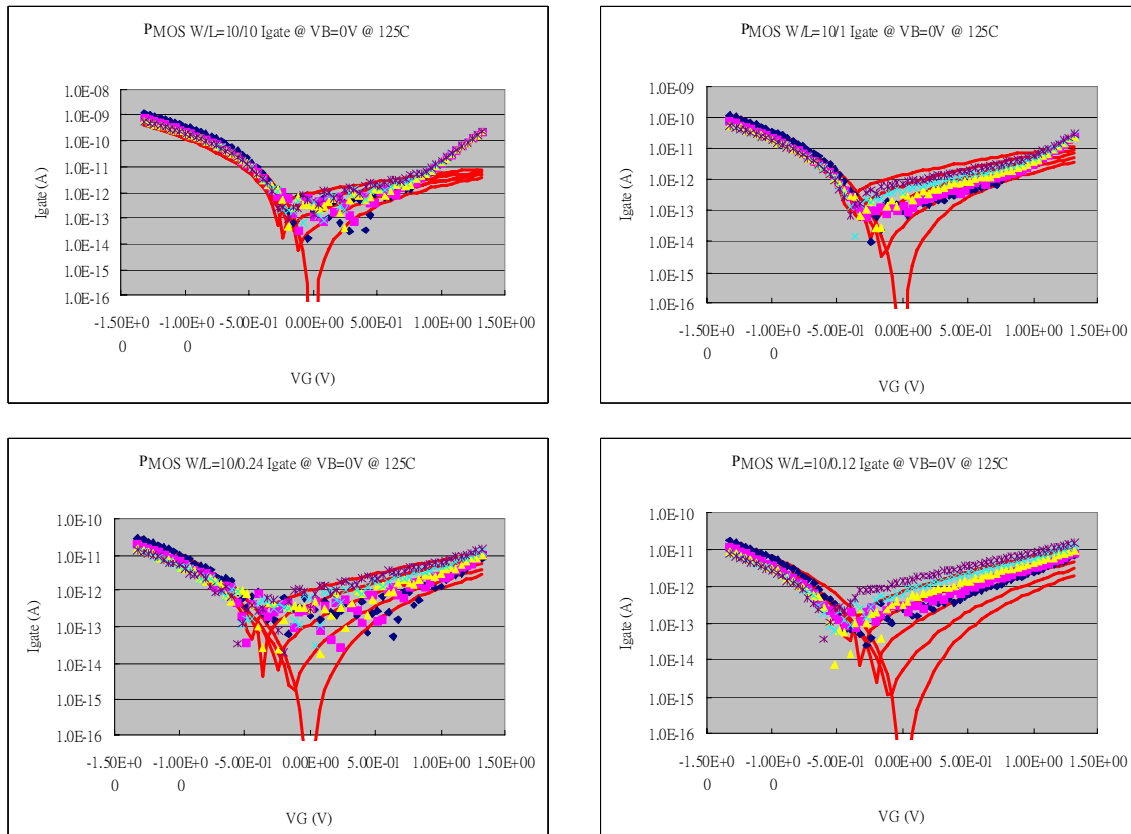


Figure 3-18 Measured and Simulated I_{GATE} for PMOS @ 125C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ $-50C$

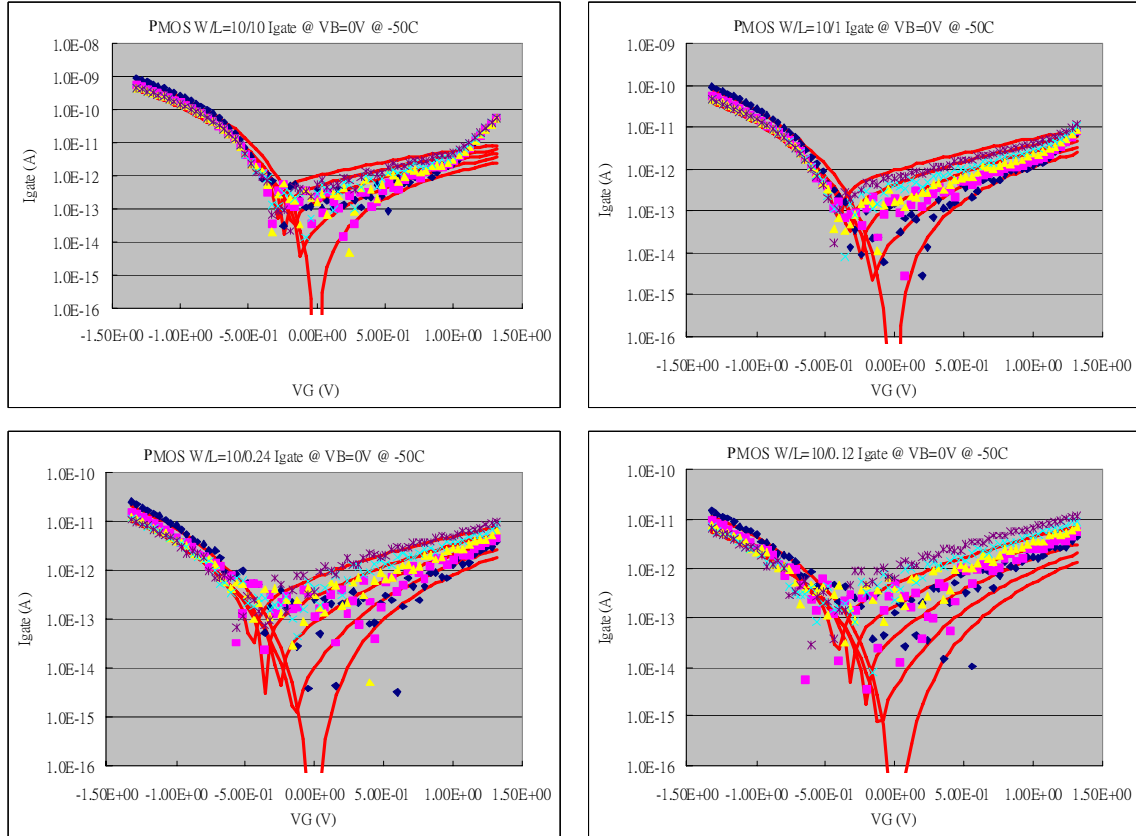


Figure 3-19 Measured and Simulated I_{GATE} for PMOS @ $-50C$

3.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

- c. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- d. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- e. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = -1.2 \text{ V}$

2) G_M and G_{DS} accuracy criteria

- f. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- g. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D Accuracy

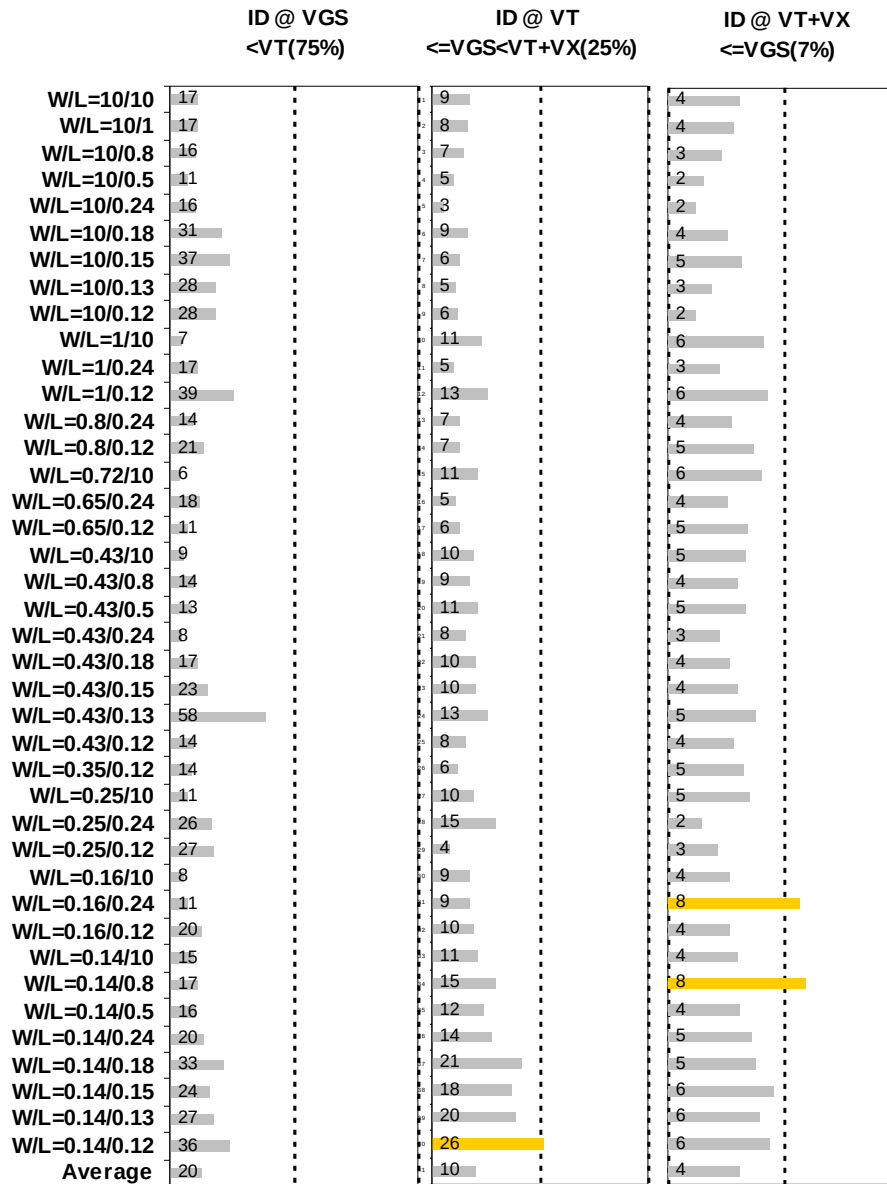


Figure 3-20 I_D accuracy at 25 °C for PMOS

G_M and G_{DS} Accuracy

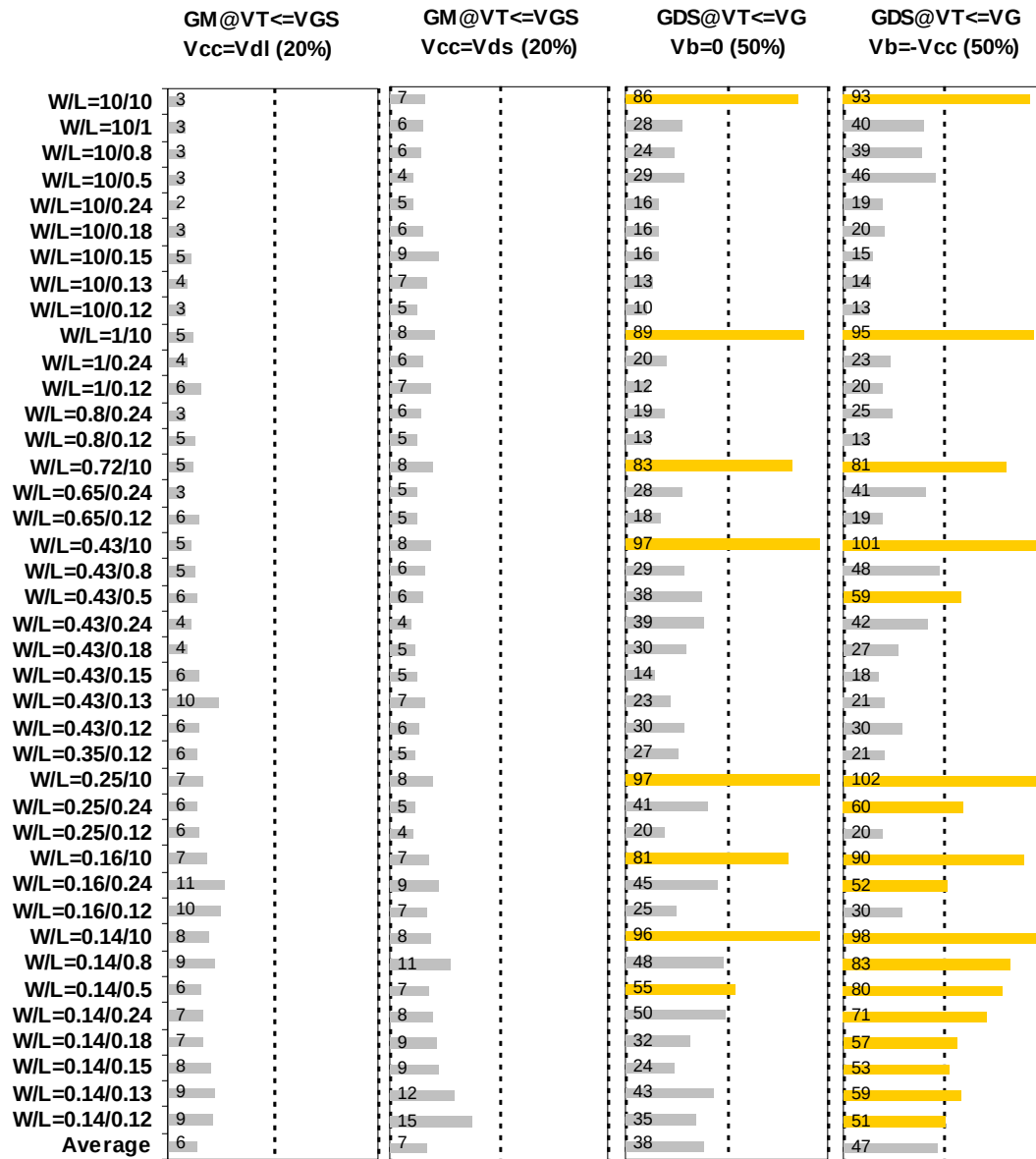


Figure 3-21 G_M and G_{DS} accuracy at 25 °C for PMOS

3.4 Parasitic source / drain junction capacitance model

3.4.1 Test devices

The following test structures were used for parameter extraction.

Table 3-6 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.83E-02	Poly finger structure

3.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-7 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 1.2 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

3.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 3-8 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction			STI Bounded Sidewall			Poly Gate Sidewall		
Capacitance Model			Junction Capacitance Model			Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	9.69E-04	CJSW	F/m	1.18E-10	CJSWG	F/m	5.31E-10
PB	V	6.87E-01	PBSW	V	4.50E-01	PBSWG	V	5.00E-01
MJ	--	3.08E-01	MJSW	--	1.00E-01	MJSWG	--	3.91E-01
TCJ	1/K	1.00E-03	TCJSW	1/K	5.24E-04	TCJSWG	1/K	5.29E-04
TPB	V/K	1.71E-03	TPBSW	V/K	2.59E-03	TPBSWG	V/K	9.24E-04

3.4.4 T_{OX} Extraction

C_{GG} , C_{GC} vs. V_{BIAS} Plots

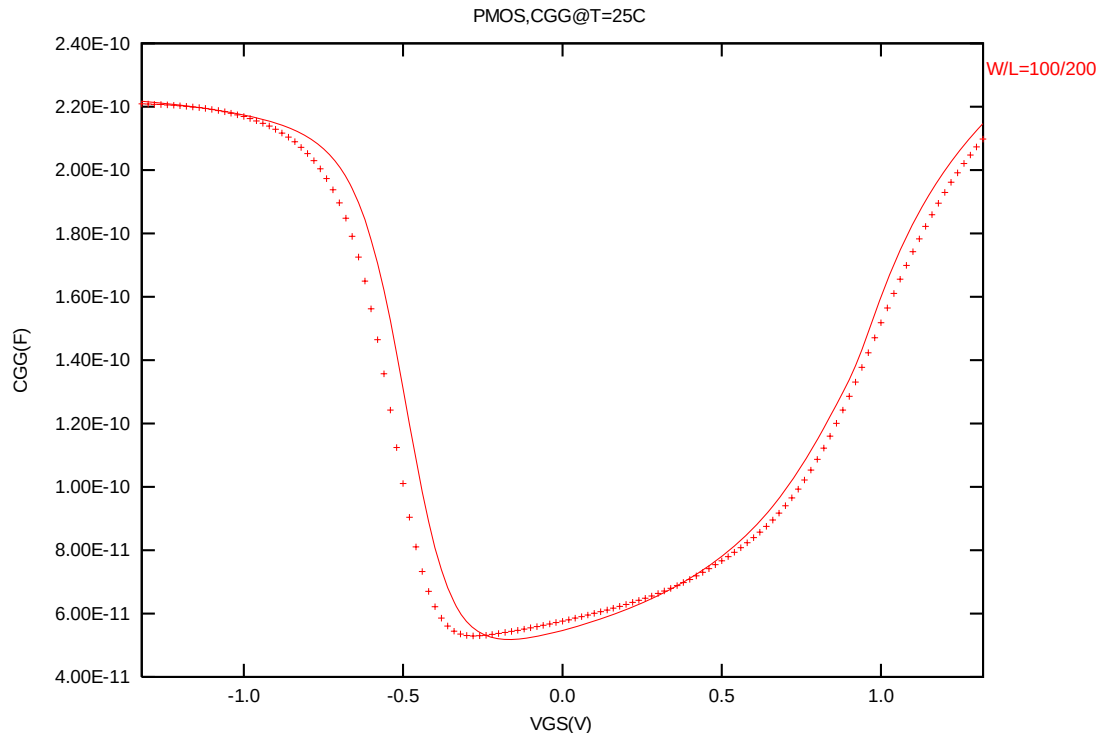


Figure 3-22 Measured and Simulated C_{GG} curves

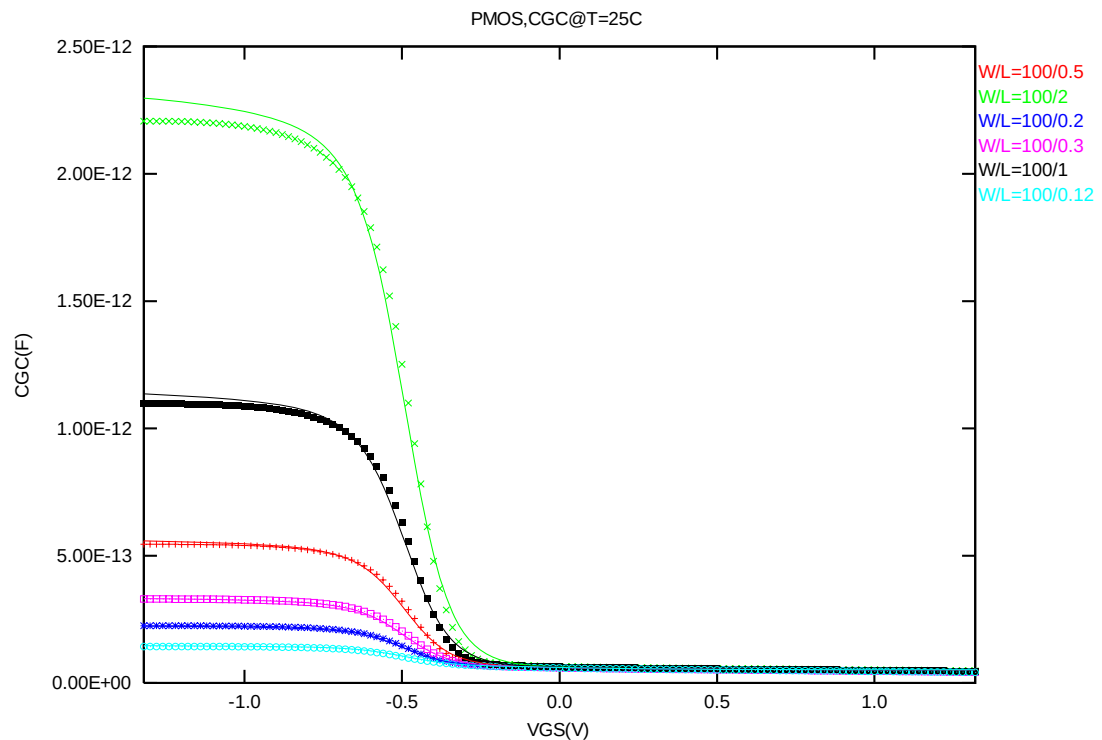


Figure 3-23 Measured and Simulated C_{GC} curves

C_J , C_{JSW} vs. V_{BIAS} Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

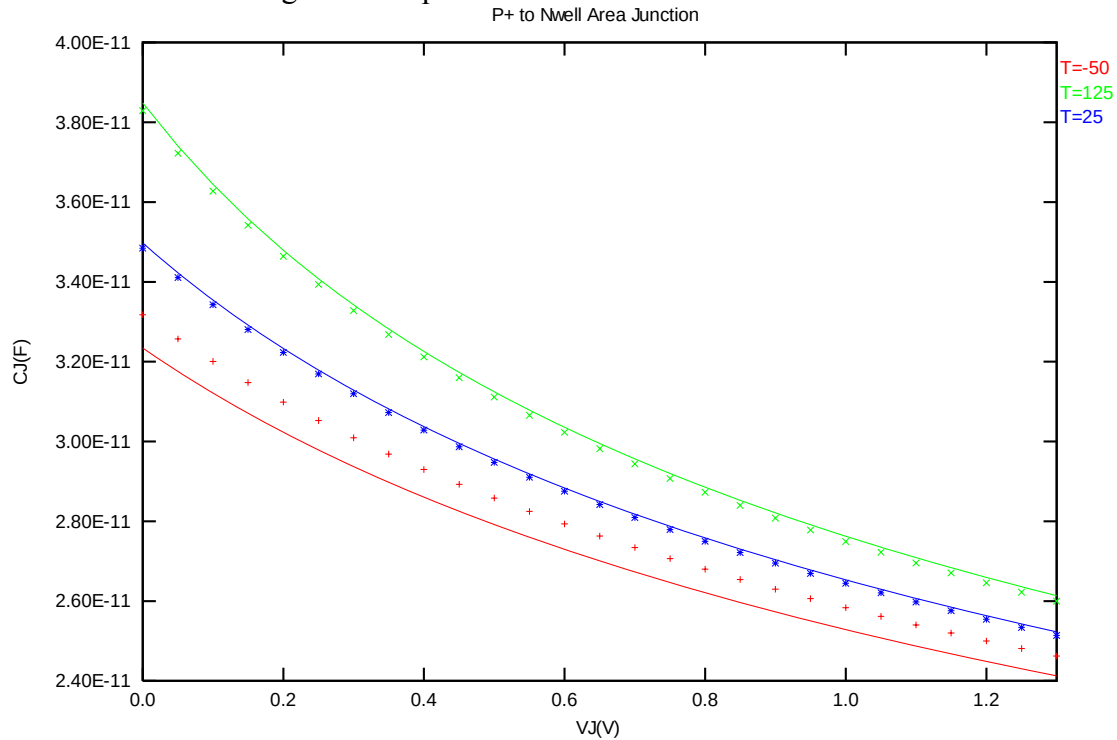


Figure 3-24 Bottom area junction capacitance at -50°C , 25°C and 125°C for PMOS

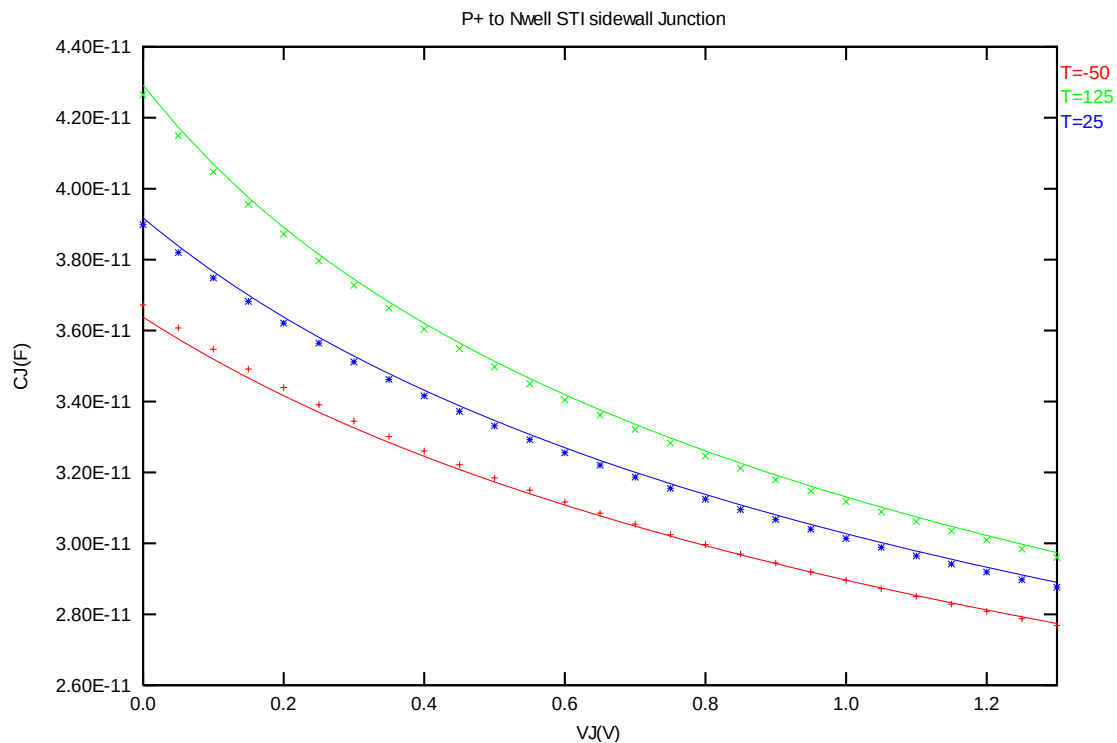


Figure 3-25 STI bounded sidewall junction capacitance at -50°C , 25°C and 125°C for PMOS

4 Dynamic Performance

4.1 Test circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 4-1 Ring Oscillator (Inverter) structure description for SA=0.34μm

R.O. (Inverter) Structure description	
Contact size	0.16 um × 0.16 um
Polysilicon gate to contact spacing	0.12 um
Contact to diffusion edge spacing	0.06 um
R _{load}	1E-5 Ohm
Lumped parasitic wiring capacitance	3.33E-15 F/stage
Inverter	
Logic gate type	Inverter
Channel length	0.12 um (with 12nm sizing back)
Channel width for n-ch MOSFET	2.5 um × 2
Channel width for p-ch MOSFET	3.75 um × 2

The simulation results at different temperatures and using different worst-case models are listed below.

Table 4-2 Ring Oscillator simulation results (ps/gate)

Inverter, R=0, C=0f, Fanout=1, Vcc=1.2V				Inverter, R=0, C=0f, Fanout=1, T=25C			
Temp(C)	TT	FF	SS	Vcc(V)	TT	FF	SS
-50	27.467	22.548	35.056	0.9	65.181	47.36	97.754
0	28.783	23.852	36.285	1	46.517	35.953	64.094
25	29.395	24.486	36.855	1.1	35.994	29.047	46.941
85	30.869	26.01	38.125	1.2	29.395	24.486	36.855
125	31.888	27.092	39.006	1.32	24.21	20.757	29.249

5 Worst case model

5.1 Parameter Variations

Table 5-1 Corner parameters

	n_12_1110e	TT	SS	FF	SNFP	FNSP
1	xl	-9.00E-09	+1.05E-09	-6.06E-10	+2.79E-10	-1.01E-10
2	xw	1.40E-08	-1.64E-08	+1.16E-08	-7.51E-09	+6.40E-09
3	tox	2.64E-09	+8.00E-11	-8.00E-11	+0.00E+00	+0.00E+00
4	wwl	-1.80E-17	-2.50E-19	+8.81E-19	-2.81E-19	+3.98E-19
5	cgso	1.87E-10	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	1.87E-10	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	4.51E-01	+2.01E-02	-2.00E-02	+1.90E-02	-1.90E-02
8	pvth0	-3.07E-16	+8.96E-16	-9.20E-16	+4.76E-16	-4.71E-16
9	k3	2.21E+00	+6.56E+00	-6.56E+00	+3.10E+00	-3.03E+00
10	nlx	1.80E-07	+8.56E-09	-8.81E-09	+6.57E-09	-6.40E-09
11	u0	3.23E-02	-8.03E-04	+5.76E-04	-5.42E-04	+4.98E-04
12	rdsw	3.80E+01	+1.80E+00	-7.00E-01	+7.00E-01	-8.00E-01
13	ags	7.79E-01	+1.81E-01	-1.71E-01	+8.55E-02	-8.05E-02
14	wags	0.00E+00	+6.15E-09	-6.05E-09	+3.07E-09	-3.03E-09
15	vsat	1.62E+05	+9.90E+03	-7.08E+03	+2.50E+03	-1.65E+03
16	pvsat	-8.92E-10	+6.10E-10	-3.80E-11	+2.00E-10	-7.50E-11
17	cf	9.06E-11	-9.06E-12	+9.06E-12	+0.00E+00	+0.00E+00
18	cgdl	3.16E-11	-10.0%	+10.0%	+0.0%	+0.0%
19	cgs1	3.16E-11	-10.0%	+10.0%	+0.0%	+0.0%
20	cj	7.20E-04	+10.0%	-10.0%	+10.0%	-10.0%

21	cjsw	9.94E-11	+10.0%	-10.0%	+10.0%	-10.0%
22	cjswg	4.11E-10	+10.0%	-10.0%	+10.0%	-10.0%

	p_12_11110e	TT	SS	FF	SNFP	FNSP
1	xl	-8.00E-09	+3.00E-09	-6.00E-10	-3.00E-10	+1.50E-09
2	xw	4.00E-09	-1.66E-09	+2.20E-09	+1.10E-09	-8.31E-10
3	tox	2.83E-09	+8.49E-11	-8.49E-11	+0.00E+00	+0.00E+00
4	wwl	-5.78E-22	+6.85E-23	+7.05E-23	+1.27E-23	+2.51E-23
5	cgso	1.63E-10	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	1.63E-10	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	-4.73E-01	-2.01E-02	+2.01E-02	+1.97E-02	-1.93E-02
8	pvth0	-1.72E-15	-1.77E-15	+1.83E-15	+9.65E-16	-8.50E-16
9	k3	-3.71E+00	+2.32E+00	-2.43E+00	-1.09E+00	+1.18E+00
10	nlx	2.69E-08	+6.00E-09	-5.51E-09	-2.95E-09	+3.54E-09
11	u0	1.11E-02	-5.14E-04	+3.99E-04	+2.56E-04	-2.90E-04
12	wu0	-1.21E-09	+5.25E-11	-5.80E-11	-3.30E-11	+3.40E-11
13	pu0	0.00E+00	-2.26E-17	+1.70E-17	+7.70E-18	-8.50E-18
14	rdsw	1.52E+02	+1.03E+02	-8.83E+01	-5.21E+01	+3.38E+01
15	ags	6.75E-01	+1.68E-01	-1.69E-01	-7.82E-02	+7.17E-02
16	lags	6.81E-07	+3.48E-07	-2.19E-07	-1.01E-07	+2.09E-07
17	wags	-2.04E-09	+1.50E-08	-1.05E-08	-4.42E-09	+8.50E-09
18	pags	-7.50E-14	+1.05E-13	-1.00E-13	-5.46E-14	+3.70E-14
19	vsat	2.33E+05	+0.00E+00	+0.00E+00	+0.00E+00	+0.00E+00
20	pvsat	-9.59E-09	+0.00E+00	+0.00E+00	+0.00E+00	+0.00E+00
21	cf	8.90E-11	-8.90E-12	+8.90E-12	+0.00E+00	+0.00E+00
22	cgdl	8.36E-11	-10.0%	+10.0%	+0.0%	+0.0%

23	cgs1	8.36E-11	-10.0%	+10.0%	+0.0%	+0.0%
24	cj	9.69E-04	+10.0%	-10.0%	-10.0%	+10.0%
25	cjsw	1.18E-10	+10.0%	-10.0%	-10.0%	+10.0%
26	cjswg	5.31E-10	+10.0%	-10.0%	-10.0%	+10.0%

Threshold voltage in linear region V_{TLIN}

The threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$ for NMOS and $I_D = -70nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$ for PMOS at 25 °C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

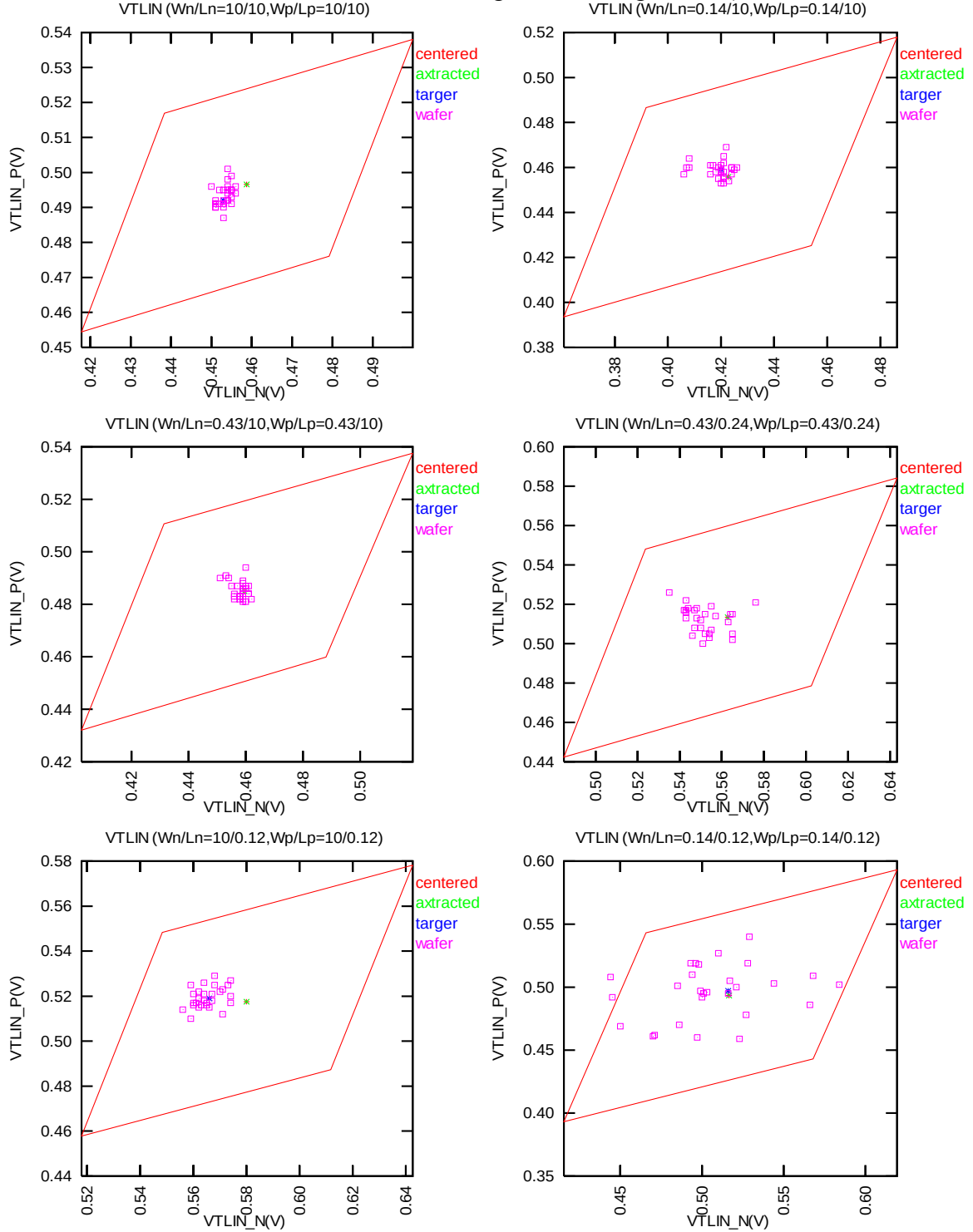


Figure 5-1 Corner graph of threshold voltage at low drain bias

➤ Saturation Current I_{ON}

(The dimensions are after 90% shrink and 12nm sizing channel length back.)

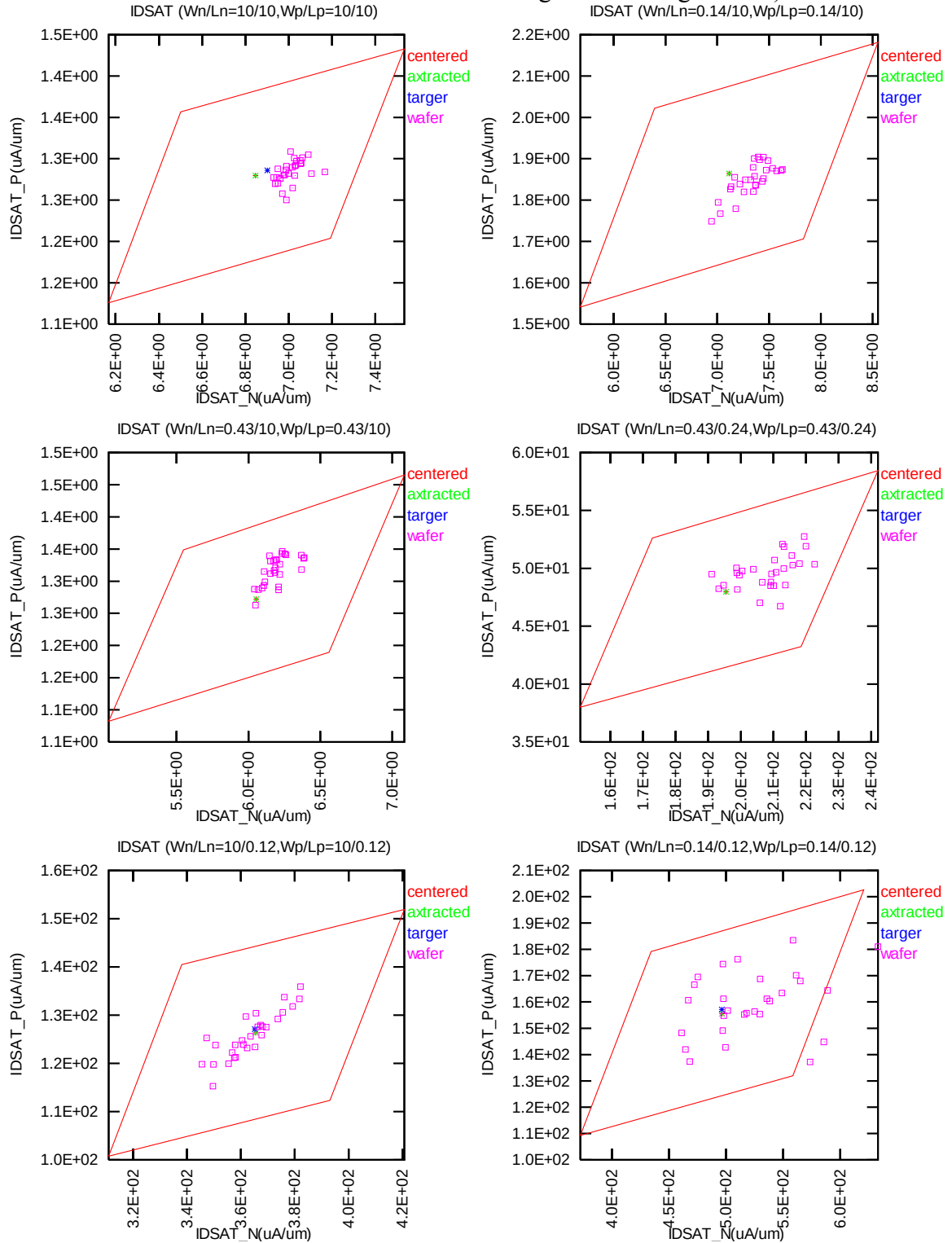


Figure 5-2 Corner graph of on-current

DEVICE PERFORMANCE LOOK UP TABLE (SIMULATION RESULTS FROM HSPICE VER.05.3):

**** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The EDR value of size W/L : 10um/0.12um would be the same as the model card simulation result in size of W/L : 10um/0.108um(After 90% shrink without 12nm sizing channel length back).**

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
VTH (V)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.517	0.512	0.459	0.452	0.394	0.381
SS	9/9	0.559	0.554	0.501	0.494	0.437	0.424
FF	9/9	0.477	0.472	0.418	0.411	0.352	0.339
SNFP	9/9	0.539	0.533	0.481	0.473	0.416	0.403
FNSP	9/9	0.496	0.491	0.438	0.431	0.373	0.360
TT	9/0.108	0.641	0.578	0.577	0.510	0.510	0.434
SS	9/0.108	0.704	0.642	0.642	0.574	0.575	0.499
FF	9/0.108	0.578	0.516	0.514	0.447	0.445	0.370
SNFP	9/0.108	0.673	0.611	0.610	0.543	0.543	0.467
FNSP	9/0.108	0.609	0.546	0.545	0.477	0.477	0.401
TT	0.144/9	0.486	0.481	0.425	0.417	0.355	0.342
SS	0.144/9	0.549	0.543	0.489	0.481	0.421	0.406
FF	0.144/9	0.424	0.420	0.363	0.355	0.291	0.279
SNFP	0.144/9	0.517	0.512	0.457	0.449	0.388	0.374
FNSP	0.144/9	0.454	0.449	0.393	0.386	0.322	0.310
TT	0.144/0.108	0.585	0.526	0.516	0.452	0.437	0.366
SS	0.144/0.108	0.686	0.632	0.618	0.559	0.541	0.475
FF	0.144/0.108	0.483	0.421	0.413	0.346	0.333	0.258
SNFP	0.144/0.108	0.637	0.581	0.568	0.508	0.490	0.423
FNSP	0.144/0.108	0.533	0.471	0.463	0.397	0.384	0.310

Table 5-2 Corner window of NMOS VT

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.561	0.549	0.497	0.480	0.424	0.401
SS	9/9	0.604	0.591	0.540	0.523	0.469	0.445
FF	9/9	0.519	0.507	0.454	0.438	0.380	0.358
SNFP	9/9	0.539	0.527	0.474	0.458	0.401	0.379
FNSP	9/9	0.583	0.571	0.519	0.502	0.446	0.423
TT	9/0.108	0.593	0.533	0.522	0.460	0.444	0.377
SS	9/0.108	0.654	0.595	0.584	0.523	0.509	0.441
FF	9/0.108	0.534	0.476	0.462	0.403	0.383	0.318
SNFP	9/0.108	0.562	0.503	0.491	0.430	0.413	0.346
FNSP	9/0.108	0.625	0.566	0.554	0.493	0.477	0.411
TT	0.144/9	0.518	0.506	0.457	0.442	0.388	0.368
SS	0.144/9	0.583	0.571	0.523	0.507	0.455	0.433
FF	0.144/9	0.455	0.443	0.393	0.378	0.323	0.303
SNFP	0.144/9	0.486	0.474	0.425	0.410	0.355	0.335
FNSP	0.144/9	0.551	0.539	0.490	0.475	0.422	0.401
TT	0.144/0.108	0.565	0.531	0.497	0.458	0.422	0.373
SS	0.144/0.108	0.669	0.635	0.603	0.564	0.530	0.482
FF	0.144/0.108	0.463	0.429	0.394	0.356	0.318	0.269
SNFP	0.144/0.108	0.512	0.478	0.444	0.405	0.368	0.320
FNSP	0.144/0.108	0.617	0.583	0.550	0.512	0.476	0.428

Table 5-3 Corner window of PMOS VT

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT (μA/um)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.21	9.76	0.35	7.57	0.49	5.99
SS	9/9	0.09	8.62	0.20	6.81	0.33	5.51
FF	9/9	0.40	10.90	0.55	8.34	0.68	6.47
SNFP	9/9	0.14	9.16	0.27	7.19	0.41	5.77
FNSP	9/9	0.30	10.36	0.44	7.96	0.58	6.22
TT	9/0.108	2.32	403.99	7.68	364.36	16.77	309.59
SS	9/0.108	0.56	341.10	2.55	309.39	7.56	268.16
FF	9/0.108	8.58	466.97	19.38	419.32	31.86	350.36
SNFP	9/0.108	1.12	372.20	4.44	336.82	11.41	289.03
FNSP	9/0.108	4.64	435.56	12.61	391.84	23.59	330.22
TT	0.144/9	0.32	9.34	0.48	7.82	0.65	6.67
SS	0.144/9	0.11	7.19	0.22	6.24	0.37	5.55
FF	0.144/9	0.71	11.55	0.87	9.41	1.04	7.77
SNFP	0.144/9	0.19	8.24	0.34	7.03	0.50	6.13
FNSP	0.144/9	0.49	10.47	0.65	8.61	0.83	7.21
TT	0.144/0.108	7.34	528.82	19.52	488.24	37.66	417.32
SS	0.144/0.108	0.68	379.47	3.54	367.62	11.60	345.65
FF	0.144/0.108	40.32	680.58	61.80	608.61	83.01	487.58
SNFP	0.144/0.108	2.26	452.42	8.68	427.92	21.81	382.54
FNSP	0.144/0.108	19.51	605.33	37.37	548.48	58.48	452.94

Table 5-4 Corner window of NMOS Ion

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IDSAT (μA/um)	-VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.02	1.46	0.06	1.43	0.11	1.43
SS	9/9	0.01	1.26	0.03	1.26	0.07	1.29
FF	9/9	0.05	1.66	0.10	1.60	0.16	1.57
SNFP	9/9	0.04	1.57	0.08	1.52	0.13	1.49
FNSP	9/9	0.02	1.35	0.04	1.35	0.09	1.36
TT	9/0.108	1.52	133.00	4.44	129.42	9.01	127.38
SS	9/0.108	0.39	103.51	1.67	103.25	4.44	104.62
FF	9/0.108	4.44	163.01	9.28	155.87	15.49	150.32
SNFP	9/0.108	2.73	148.61	6.60	142.69	11.98	138.40
FNSP	9/0.108	0.76	117.58	2.74	116.38	6.40	116.78
TT	0.144/9	0.05	2.18	0.10	2.02	0.15	1.91
SS	0.144/9	0.02	1.76	0.04	1.68	0.08	1.64
FF	0.144/9	0.13	2.60	0.18	2.36	0.25	2.17
SNFP	0.144/9	0.09	2.40	0.14	2.19	0.20	2.03
FNSP	0.144/9	0.03	1.96	0.06	1.85	0.11	1.78
TT	0.144/0.108	2.21	163.15	5.58	152.52	10.52	137.84
SS	0.144/0.108	0.18	108.49	0.98	106.61	3.13	103.39
FF	0.144/0.108	11.73	220.35	17.33	198.43	23.39	168.96
SNFP	0.144/0.108	5.78	192.26	10.55	175.51	16.37	153.58
FNSP	0.144/0.108	0.67	134.87	2.52	129.54	6.09	121.60

Table 5-5 Corner window of PMOS Ion

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IOFF (A)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	1.06E-14	2.40E-13	4.65E-13	7.37E-13	6.32E-11	7.07E-11
SS	9/9	1.00E-14	2.38E-13	1.41E-13	3.84E-13	2.46E-11	2.76E-11
FF	9/9	1.32E-14	2.42E-13	1.59E-12	1.98E-12	1.62E-10	1.82E-10
SNFP	9/9	1.03E-14	2.40E-13	2.40E-13	4.92E-13	3.79E-11	4.25E-11
FNSP	9/9	1.14E-14	2.40E-13	9.06E-13	1.22E-12	1.05E-10	1.18E-10
TT	9/0.108	1.47E-14	2.68E-13	4.96E-12	2.25E-11	8.86E-10	3.17E-09
SS	9/0.108	1.04E-14	2.42E-13	8.42E-13	3.96E-12	2.29E-10	8.07E-10
FF	9/0.108	5.83E-14	5.36E-13	2.93E-11	1.31E-10	3.40E-09	1.22E-08
SNFP	9/0.108	1.12E-14	2.47E-13	1.89E-12	8.60E-12	4.27E-10	1.51E-09
FNSP	9/0.108	2.64E-14	3.43E-13	1.29E-11	5.87E-11	1.82E-09	6.55E-09
TT	0.144/9	1.00E-14	2.40E-13	3.39E-14	2.66E-13	2.62E-12	3.16E-12
SS	0.144/9	1.00E-14	2.40E-13	1.36E-14	2.44E-13	6.33E-13	9.36E-13
FF	0.144/9	1.05E-14	2.41E-13	1.67E-13	4.13E-13	1.08E-11	1.24E-11
SNFP	0.144/9	1.00E-14	2.40E-13	1.88E-14	2.50E-13	1.24E-12	1.62E-12
FNSP	0.144/9	1.02E-14	2.40E-13	7.44E-14	3.11E-13	5.50E-12	6.39E-12
TT	0.144/0.108	1.02E-14	2.41E-13	2.22E-13	1.25E-12	3.99E-11	1.52E-10
SS	0.144/0.108	1.00E-14	2.40E-13	2.03E-14	2.82E-13	4.04E-12	1.36E-11
FF	0.144/0.108	2.18E-14	3.31E-13	4.63E-12	2.49E-11	4.08E-10	1.70E-09
SNFP	0.144/0.108	1.00E-14	2.40E-13	5.20E-14	4.23E-13	1.18E-11	4.16E-11
FNSP	0.144/0.108	1.17E-14	2.52E-13	1.08E-12	5.73E-12	1.35E-10	5.44E-10

Table 5-6 Corner window of NMOS Ioff (gminc=1E-13)

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	1.00E-14	2.36E-13	6.10E-14	3.00E-13	1.14E-11	1.53E-11
SS	9/9	9.99E-15	2.36E-13	2.52E-14	2.56E-13	5.93E-12	8.01E-12
FF	9/9	1.03E-14	2.40E-13	1.80E-13	4.57E-13	2.54E-11	3.38E-11
SNFP	9/9	1.02E-14	2.36E-13	1.10E-13	3.64E-13	1.75E-11	2.32E-11
FNSP	9/9	1.00E-14	2.36E-13	3.64E-14	2.72E-13	7.85E-12	1.06E-11
TT	9/0.108	1.69E-14	2.97E-13	5.65E-12	2.90E-11	8.52E-10	3.11E-09
SS	9/0.108	1.07E-14	2.45E-13	1.05E-12	5.51E-12	2.34E-10	8.45E-10
FF	9/0.108	6.38E-14	6.52E-13	2.69E-11	1.29E-10	2.83E-09	9.86E-09
SNFP	9/0.108	3.20E-14	4.14E-13	1.34E-11	6.72E-11	1.64E-09	5.87E-09
FNSP	9/0.108	1.19E-14	2.54E-13	2.20E-12	1.12E-11	4.18E-10	1.50E-09
TT	0.144/9	1.00E-14	2.40E-13	1.06E-14	2.41E-13	2.19E-13	5.13E-13
SS	0.144/9	1.00E-14	2.40E-13	1.01E-14	2.40E-13	1.27E-13	3.92E-13
FF	0.144/9	1.00E-14	2.40E-13	1.54E-14	2.47E-13	6.96E-13	1.14E-12
SNFP	0.144/9	1.00E-14	2.40E-13	1.19E-14	2.42E-13	3.72E-13	7.15E-13
FNSP	0.144/9	1.00E-14	2.40E-13	1.02E-14	2.40E-13	1.53E-13	4.26E-13
TT	0.144/0.108	1.00E-14	2.40E-13	3.93E-14	3.16E-13	6.37E-12	1.61E-11
SS	0.144/0.108	1.00E-14	2.40E-13	1.10E-14	2.43E-13	5.83E-13	1.52E-12
FF	0.144/0.108	1.18E-14	2.45E-13	8.06E-13	2.28E-12	7.66E-11	1.95E-10
SNFP	0.144/0.108	1.02E-14	2.41E-13	1.77E-13	6.73E-13	2.34E-11	5.94E-11
FNSP	0.144/0.108	1.00E-14	2.40E-13	1.52E-14	2.53E-13	1.78E-12	4.45E-12

Table 5-7 Corner window of PMOS Ioff (gminde=1E-13)

6 Shallow Trench Isolation (STI) or LOD Stress Effect

6.1 Simulation results for NMOS at SA=0.38 μ m (SAREF=2.2 μ m)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 6-1 Simulation results for NMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.459
I _{ON}	V _{DS} = 1.2 V	μA/ μm	6.81

Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.591
V _{TSAT}	V _{DS} = 1.2 V	V	0.523
Body Effect V _T shift	V _{DS} = 1.2 V V _{BS} = 0 ~ -1.2 V	V	0.176
DIBL V _T shift	V _{DS} = 0.05~1.2 V	V	0.068
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	341.81
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	1.7E-12

Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.423
V _{TSAT}	V _{DS} = 1.2 V	V	0.416
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	7.07

Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.528
V _{TSAT}	V _{DS} = 1.2 V	V	0.466
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	462.19
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	6.4E-12

6.1.1 Model verification plots for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$. The dimension is after 90% shrink and 12nm sizing channel length back.

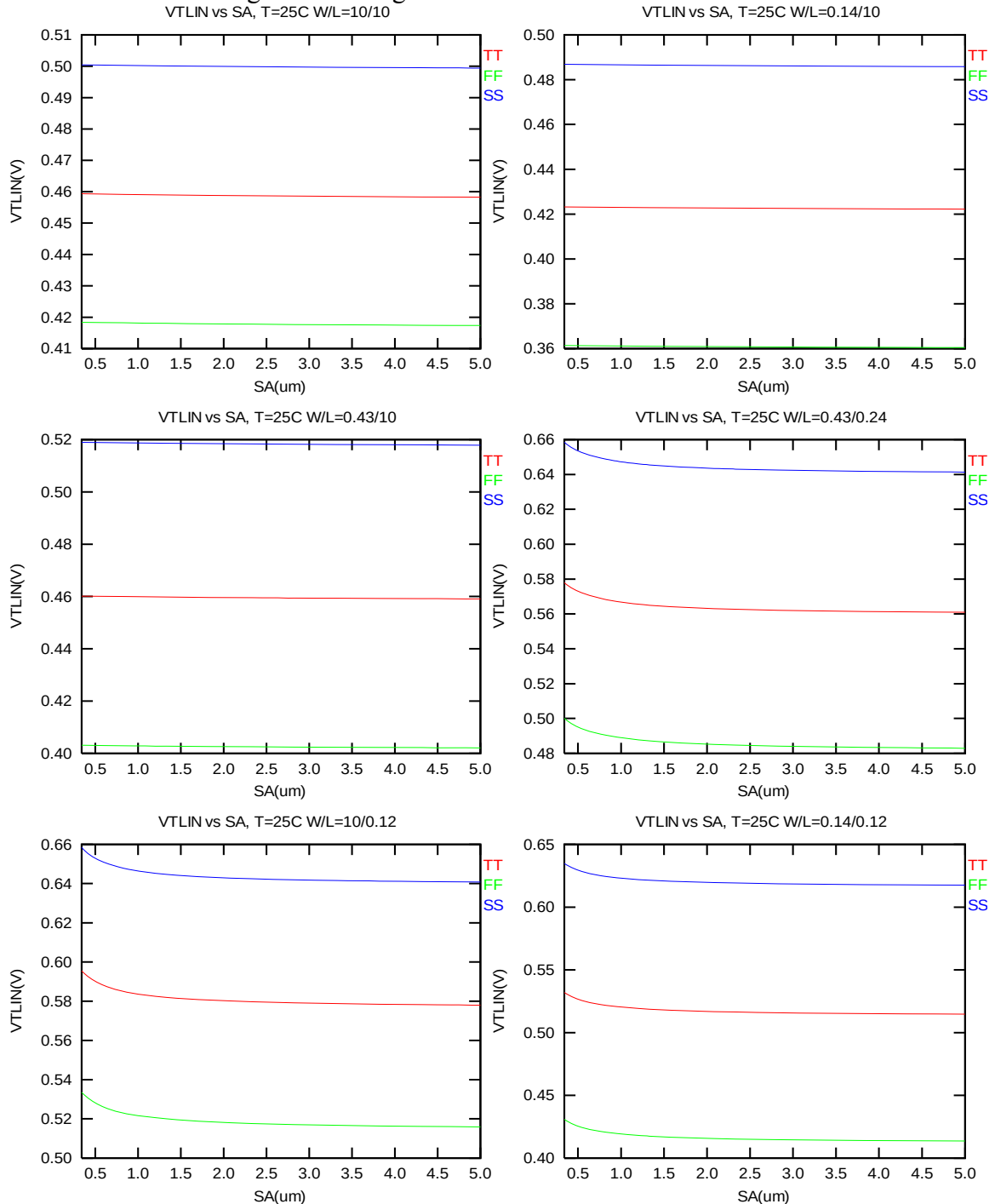


Figure 6-1 V_T vs. SA for low drain bias ($V_{DS} = 0.05 V$) for NMOS

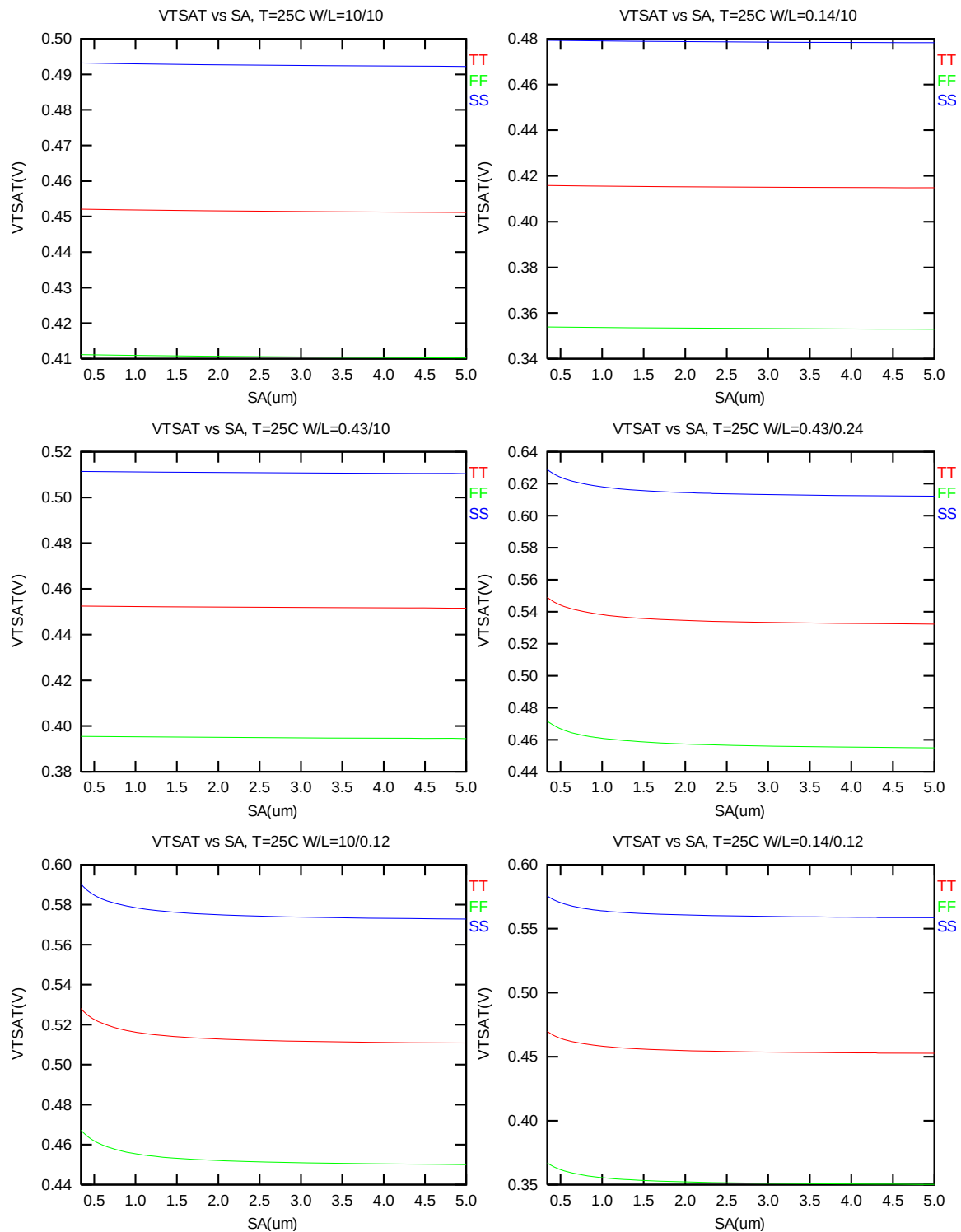


Figure 6-2 V_T vs. SA for high drain bias ($V_{DS} = 1.2$ V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=1.2$ V for $V_{BS}=0$ at 25°C .

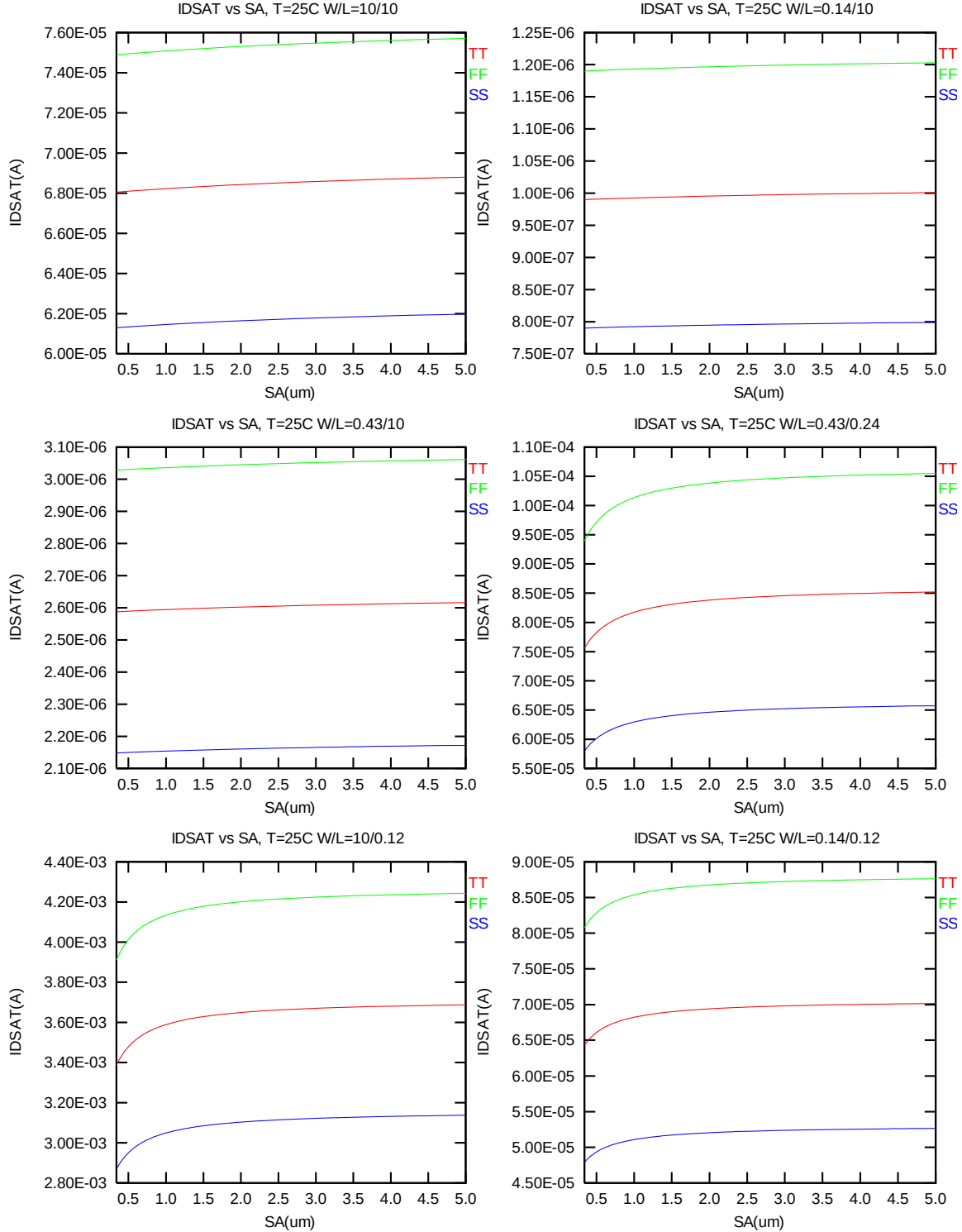


Figure 6-3 Saturation current I_{ON} vs. SA for NMOS

6.2 Simulation results for PMOS at SA=0.38μm (SAREF=2.2μm)

The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
-----------	-----------	------	------------------

Long channel device ($W_{DES}/L_{DES} = 10\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.496
I_{ON}	$V_{DS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-1.30

Wide and short channel device ($W_{DES}/L_{DES} = 10\text{ }\mu\text{m} / 0.12\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.517
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.455
Body Effect V_T shift	$V_{DS} = -1.2\text{ V}$ $V_{BS} = 0 \sim -1.2\text{ V}$	V	-0.235
DIBL V_T shift	$V_{DS} = -0.05 \sim -1.2\text{ V}$	V	-0.061
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-141.25
I_{OFF}	$V_{DS} = -1.2\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-3.6E-12

Narrow and long channel device ($W_{DES}/L_{DES} = 0.14\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.455
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.440
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-1.87

Narrow and short channel device ($W_{DES}/L_{DES} = 0.14\text{ }\mu\text{m} / 0.12\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.492
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.453
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-169.02
I_{OFF}	$V_{DS} = -1.2\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-2.3E-12

Table 6-2 Simulation results for PMOS

6.2.1 Model verification plots for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = -70nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu m)$. The dimension is after 90% shrink and 12nm sizing channel length back.

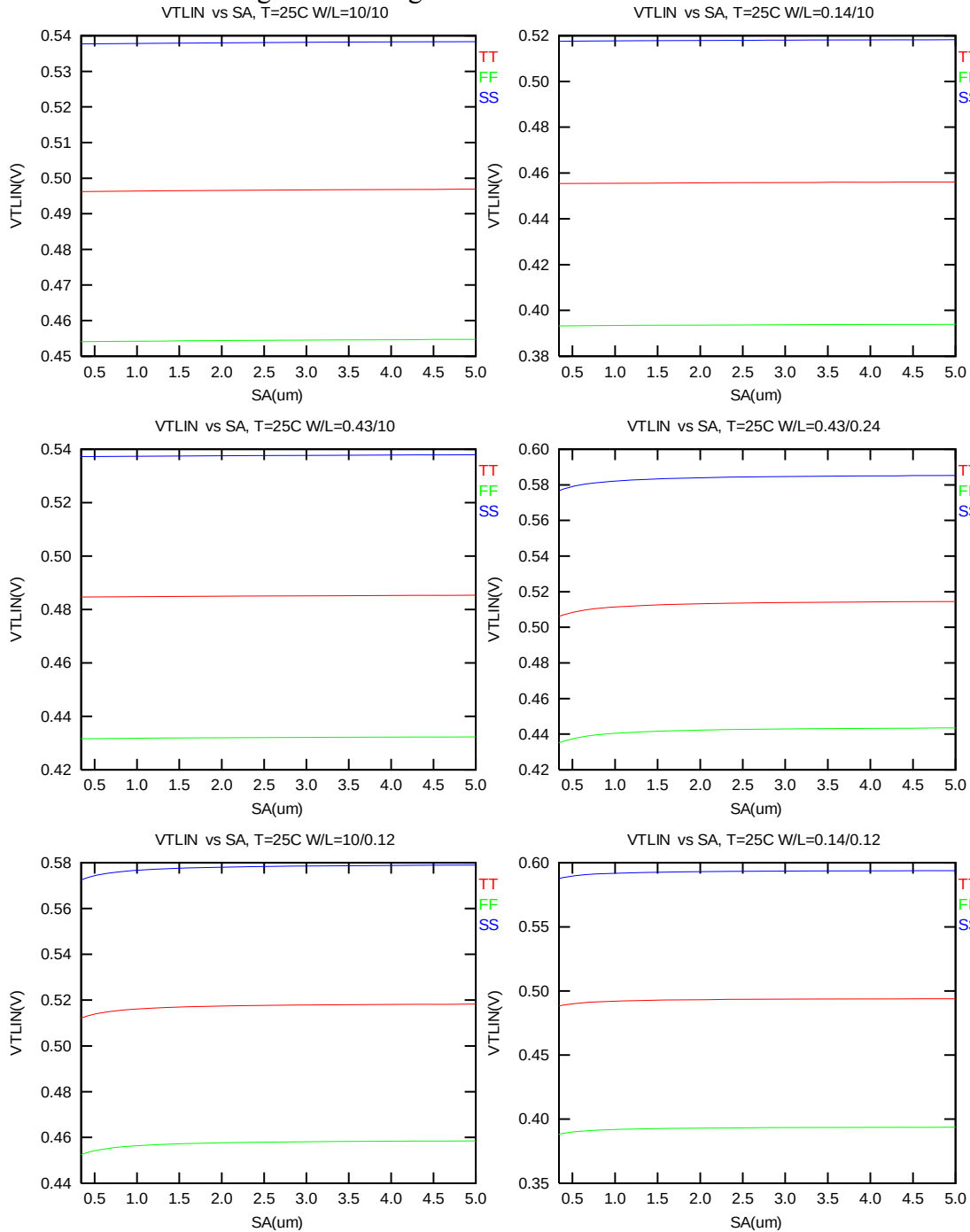


Figure 6-4 V_T vs. SA for low drain bias ($V_{DS} = -0.05 V$) for PMOS

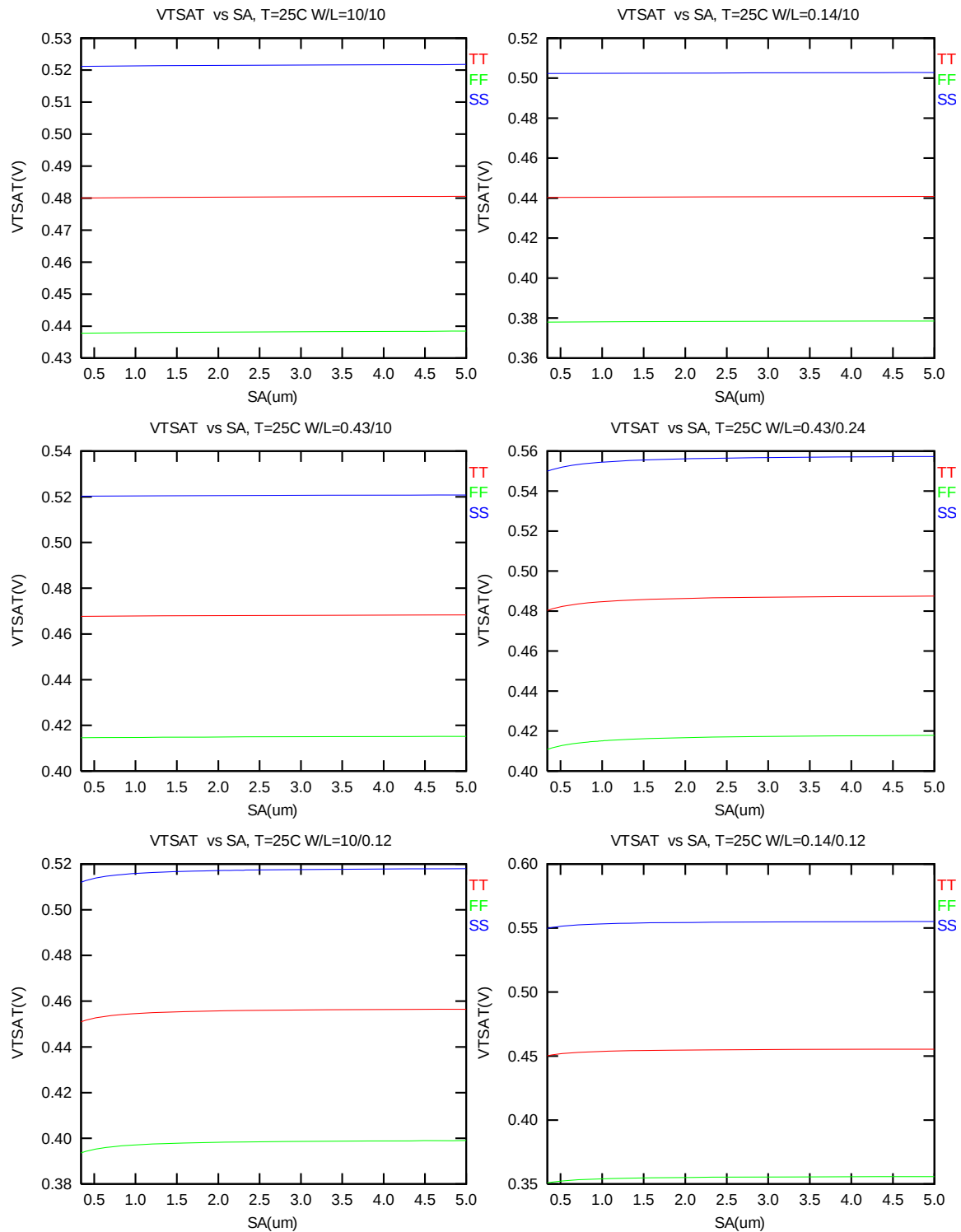


Figure 6-5 V_T vs. SA for high drain bias ($V_{DS} = -1.2$ V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}= -1.2$ V for $V_{BS}=0$ at 25°C.

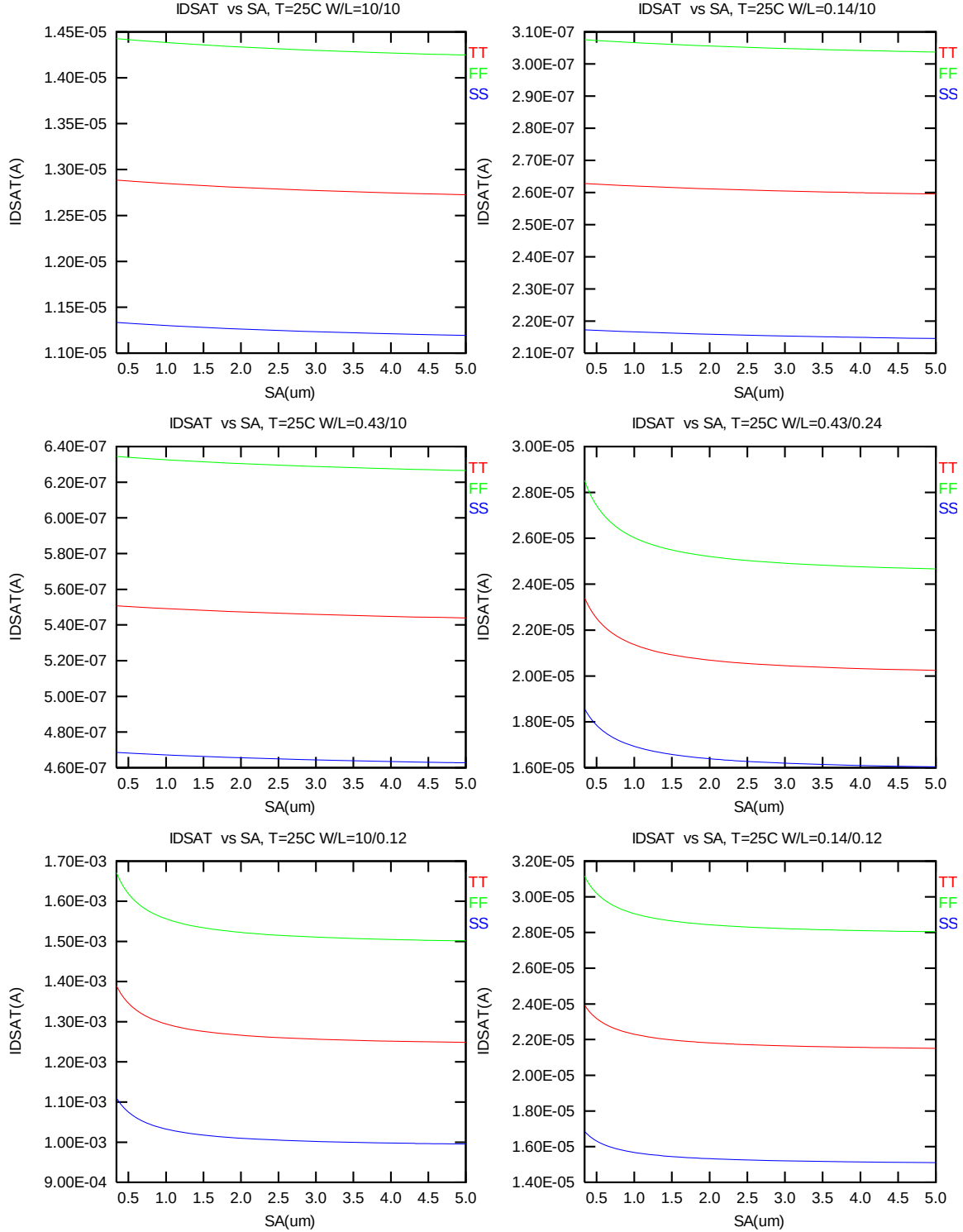


Figure 6-6 Saturation current I_{ON} vs. SA for PMOS

6.3 V_T and I_{ON} variation v. SA for both N/PMOS

Delta V_T is defined as $V_T(SA) - V_T(2.2\mu m)$ for NMOS and $-[V_T(SA) - V_T(2.2\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SA) - I_{ON}(2.2\mu m)] / I_{ON}(2.2\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SA

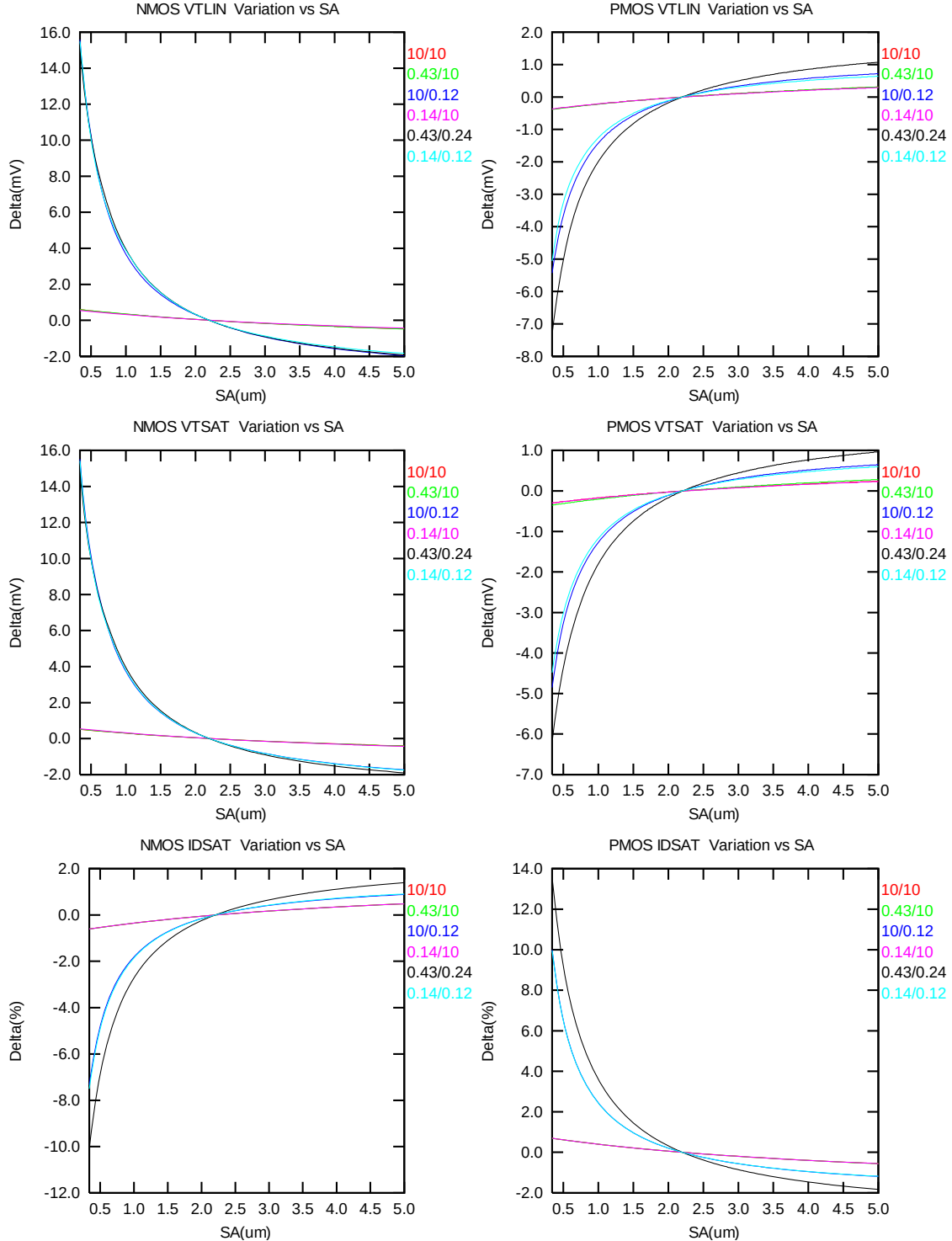


Figure 6-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

7 Noise Characteristic

7.1 Verification plots

NMOS, $W=10\mu\text{m}$ $L=0.12\mu\text{m}$

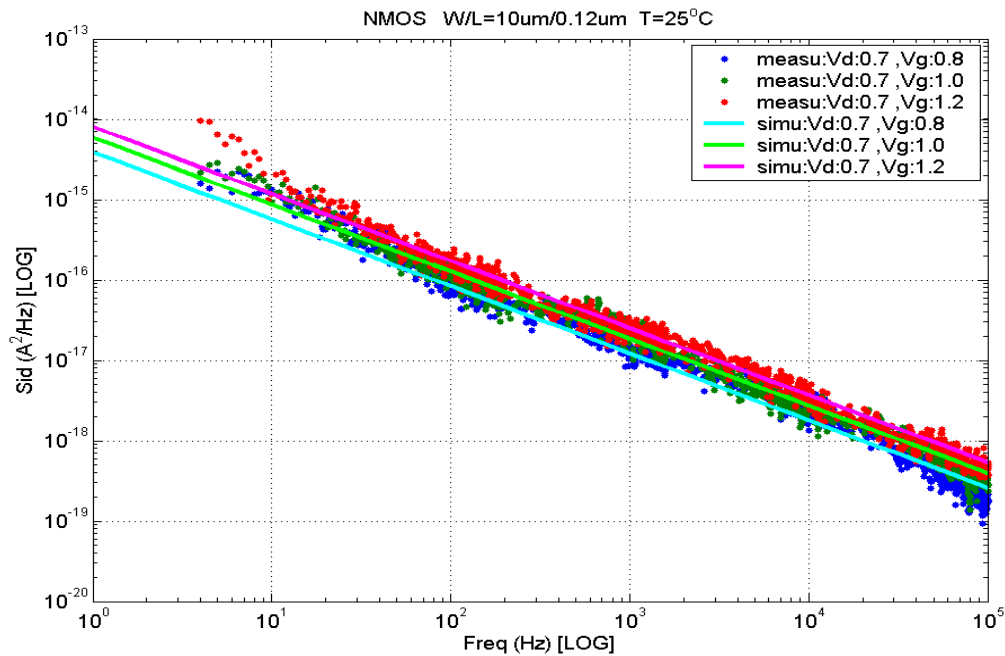


Figure 7-1 NMOS $W/L=10\mu\text{m}/0.12\mu\text{m}$ @ $V_d=0.7\text{V}$

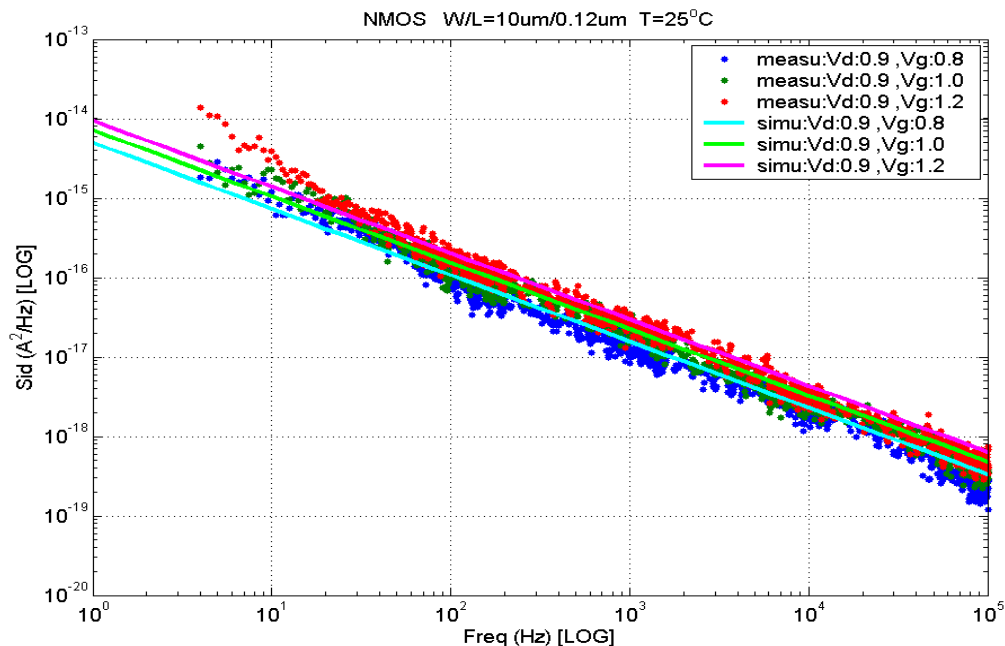


Figure 7-2 NMOS $W/L=10\mu\text{m}/0.12\mu\text{m}$ @ $V_d=0.9\text{V}$

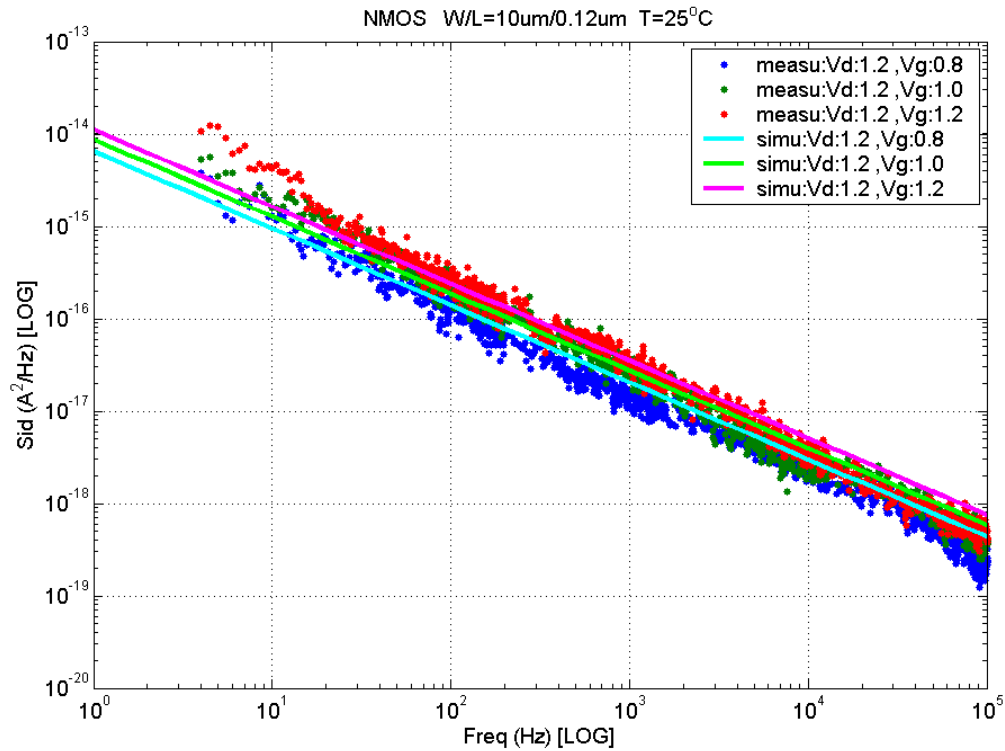


Figure 7-3 NMOS W/L=10μm/0.12μm @ Vd=1.2V

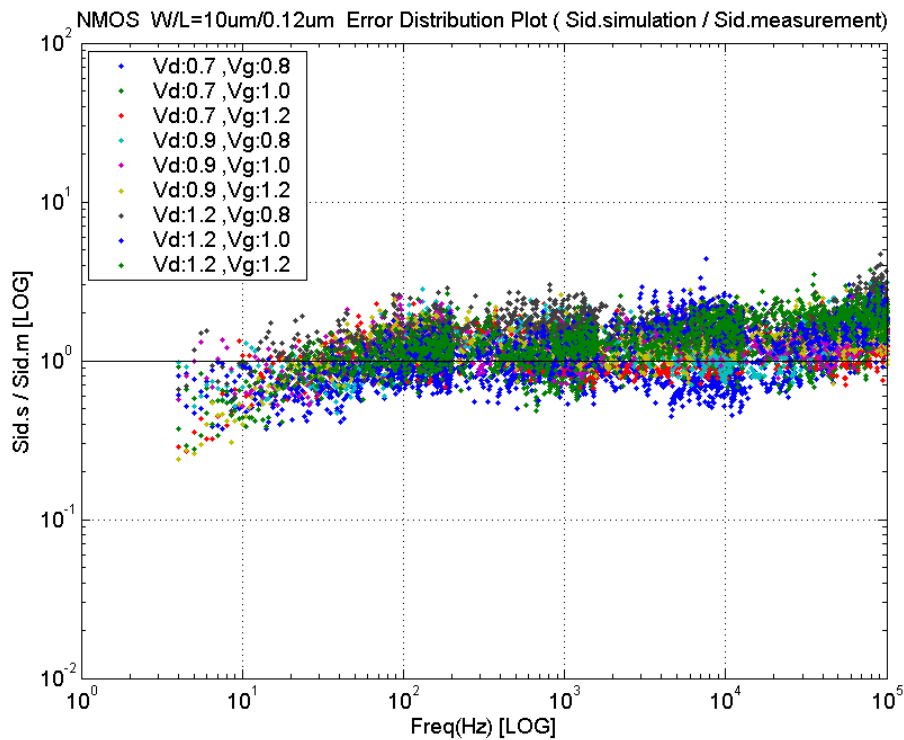


Figure 7-4 NMOS W/L=10μm/0.12μm Error Distribution Plot

NMOS, $W=10\mu\text{m}$ $L=10\mu\text{m}$

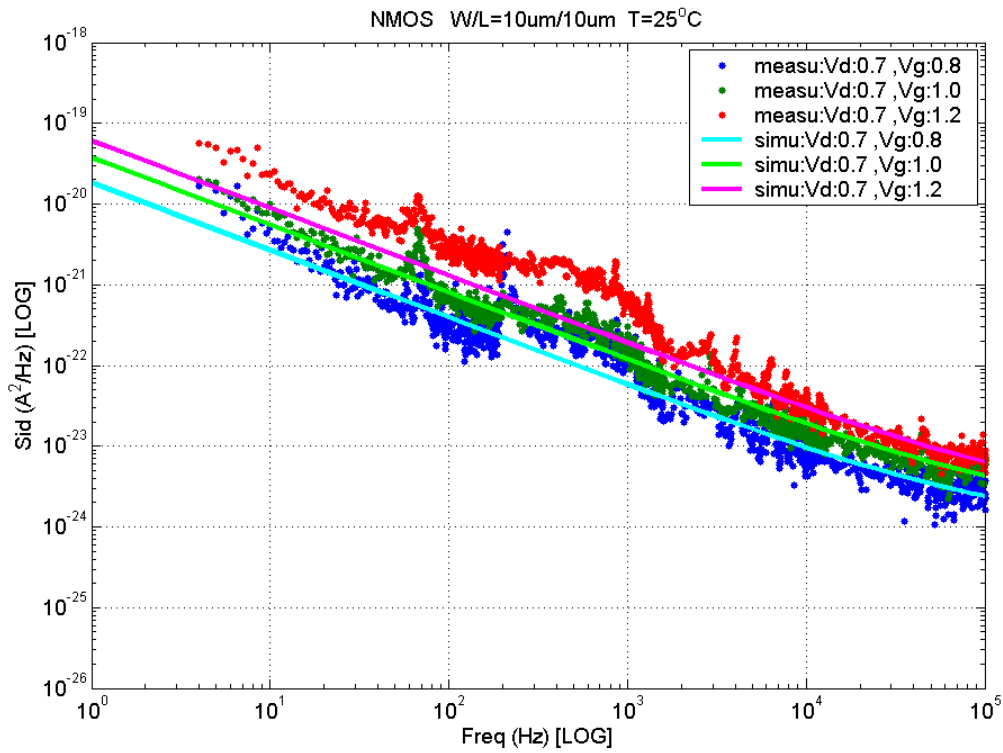


Figure 7-5 NMOS $W/L=10\mu\text{m}/10\mu\text{m}$ @ $V_d=0.7\text{V}$

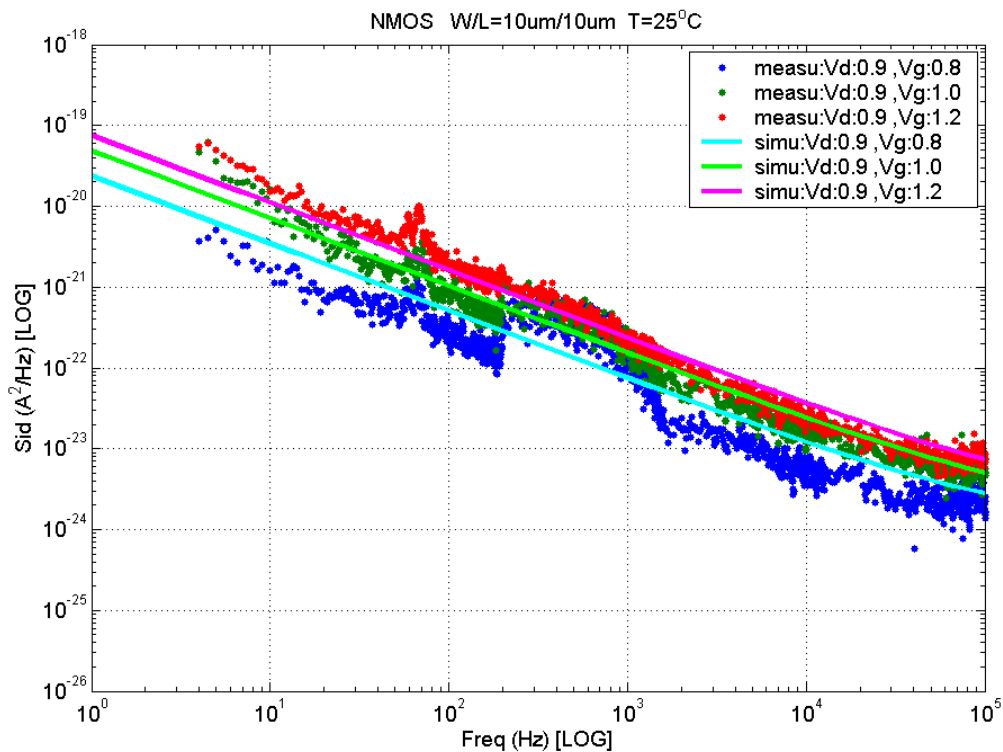


Figure 7-6 NMOS $W/L=10\mu\text{m}/10\mu\text{m}$ @ $V_d=0.9\text{V}$

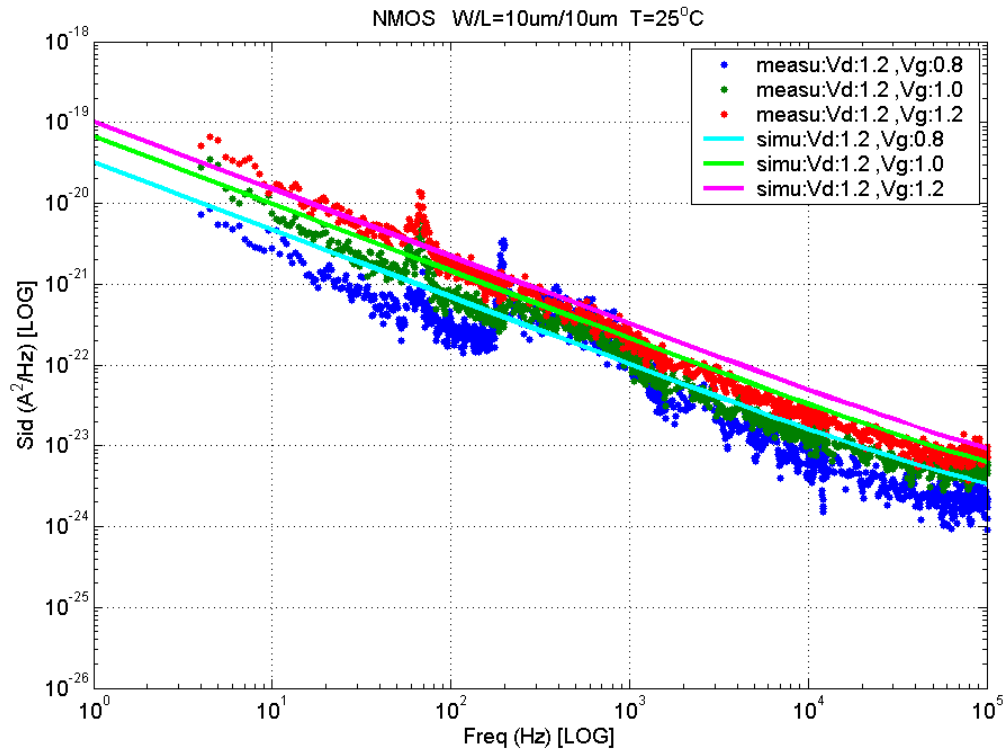


Figure 7-7 NMOS W/L=10 μ m/10 μ m @ Vd=1.2V

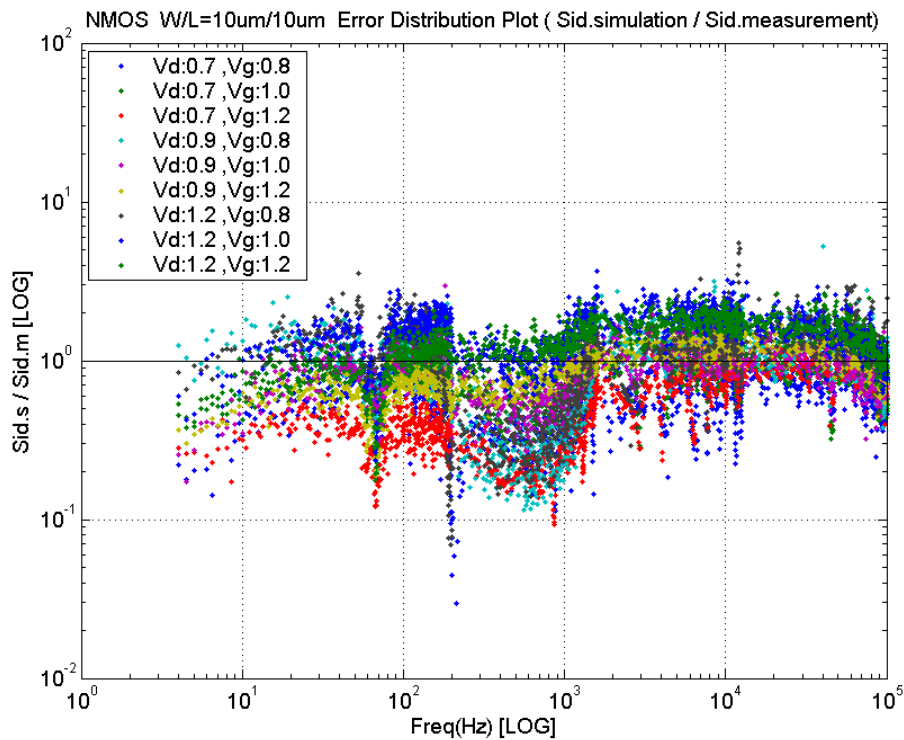


Figure 7-8 NMOS W/L=10 μ m/10 μ m Error Distribution Plot

PMOS, $W=10\mu\text{m}$ $L=0.12\mu\text{m}$

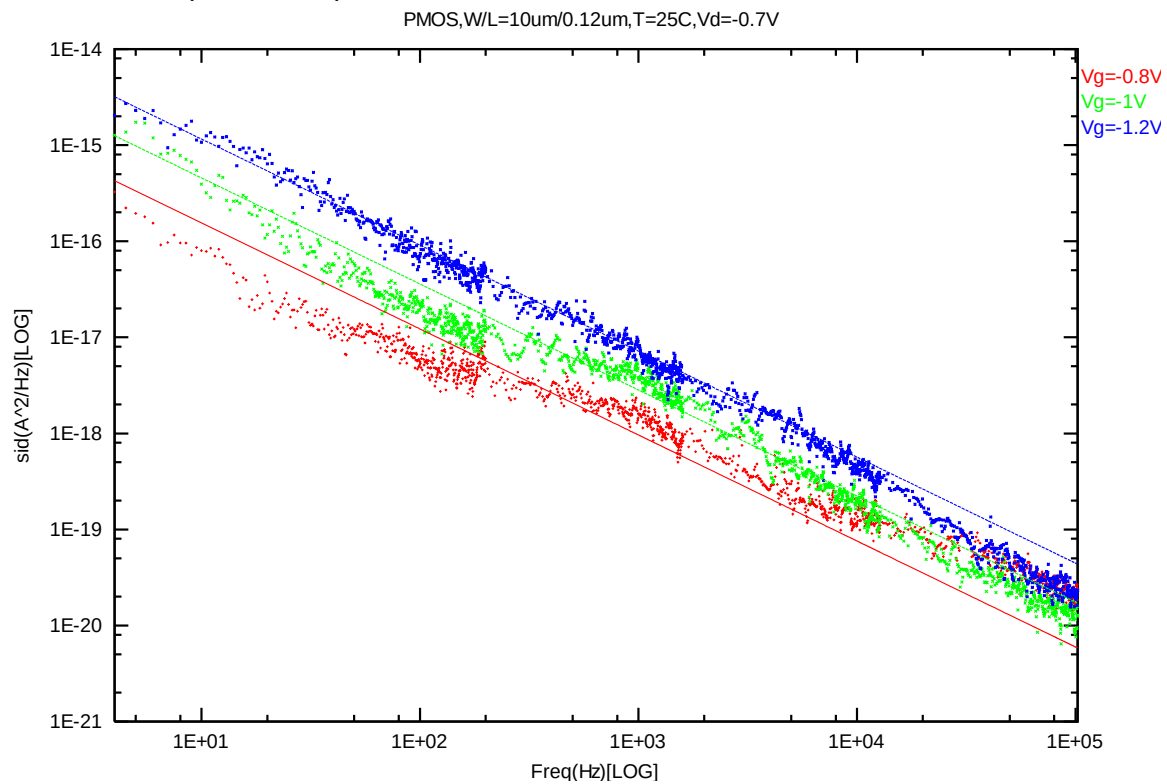


Figure 7-9 PMOS W/L=10 μm /0.12 μm @ Vd= -0.7V

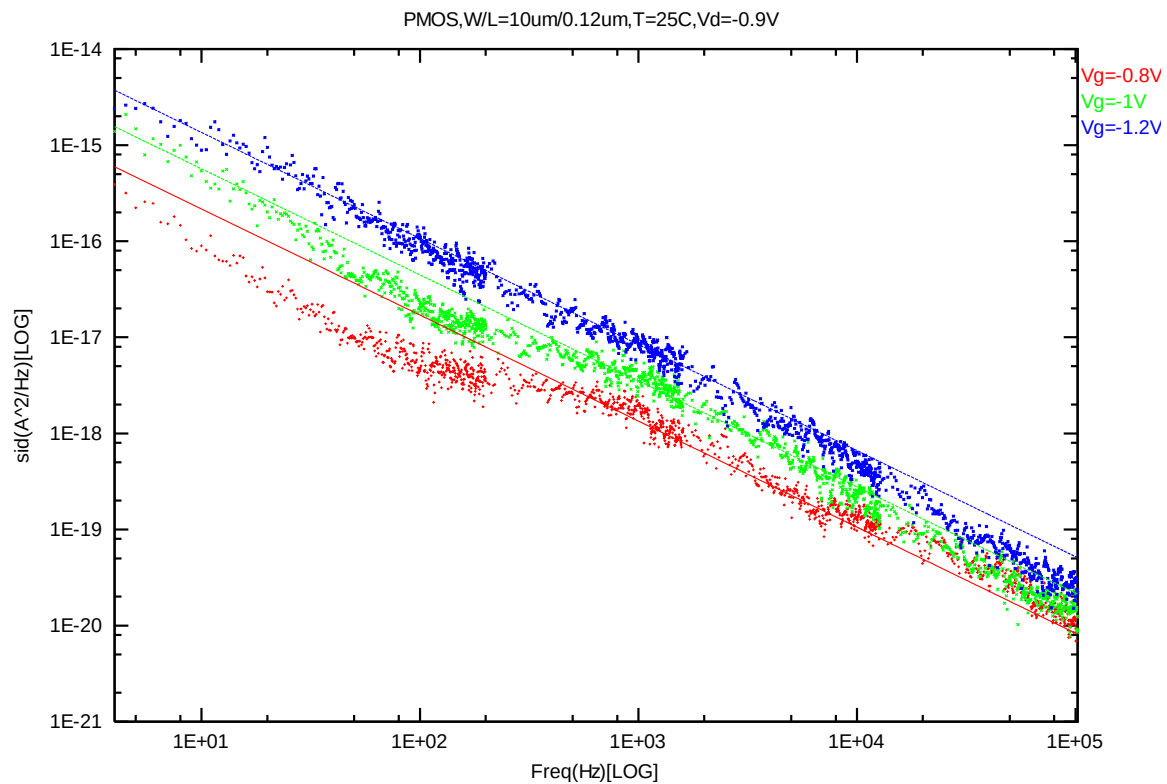


Figure 7-10 PMOS W/L=10 μm /0.12 μm @ Vd= -0.9V

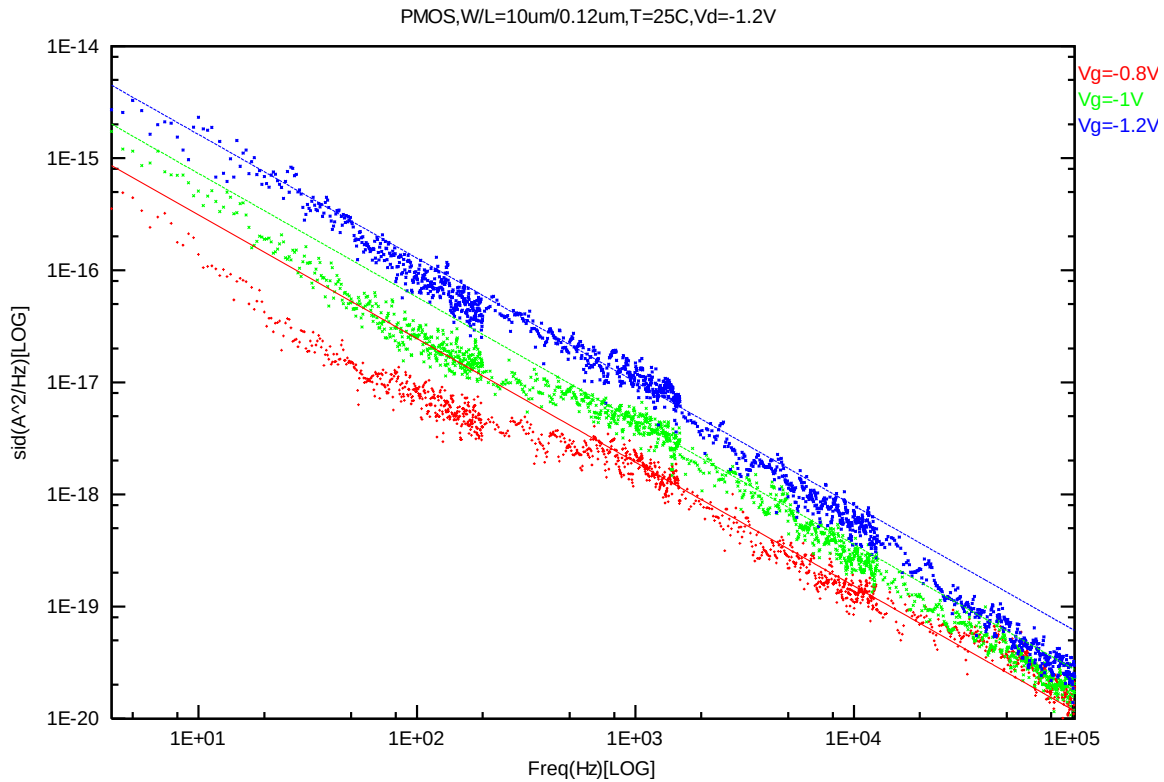


Figure 7-11 PMOS W/L=10μm/0.12μm @ Vd= -1.2V

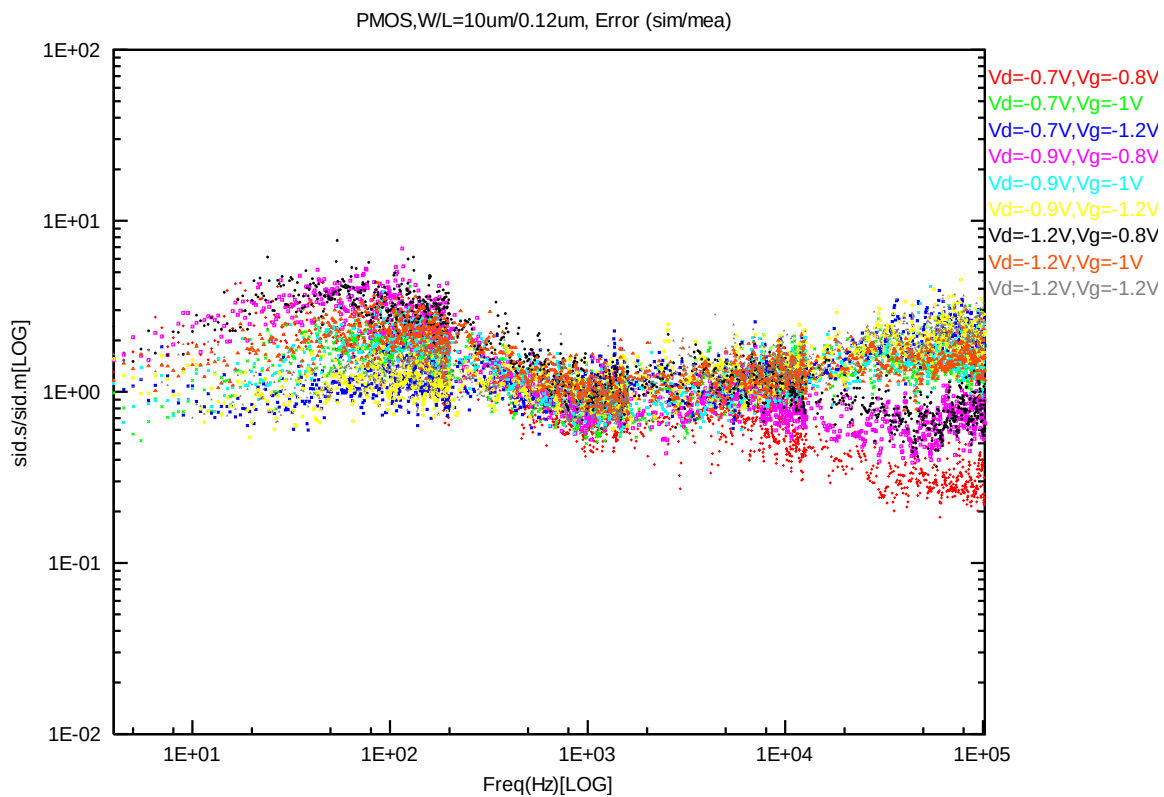


Figure 7-12 PMOS W/L=10μm/0.12μm Error Distribution Plot

PMOS, W=10μm L=10μm

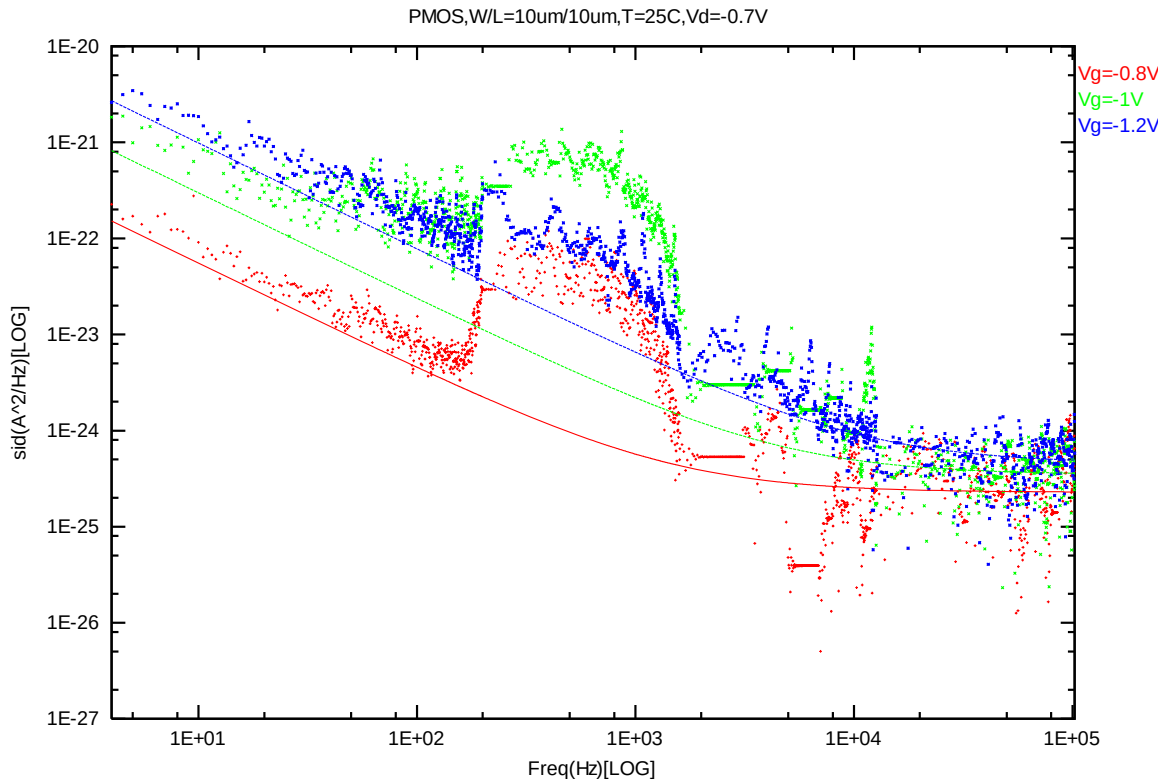


Figure 7-13 PMOS W/L=10μm/10μm @ Vd= -0.7V

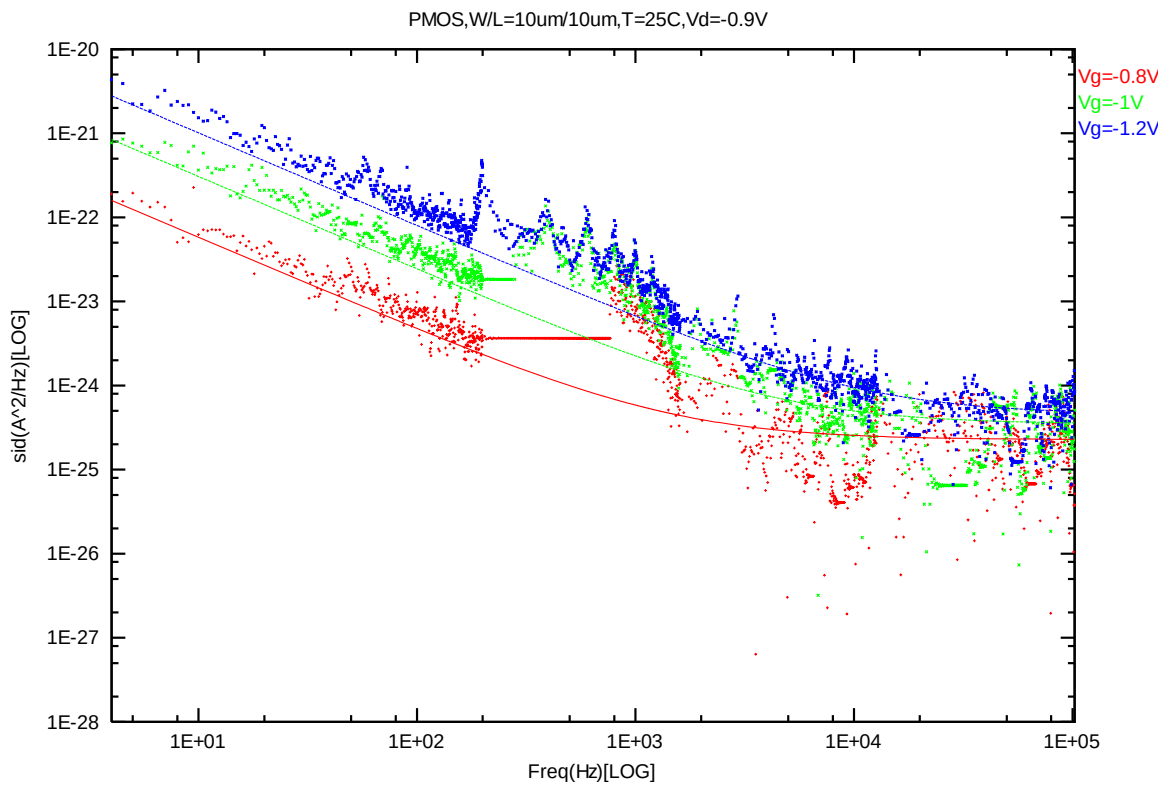


Figure 7-14 PMOS W/L=10μm/10μm @ Vd= -0.9V

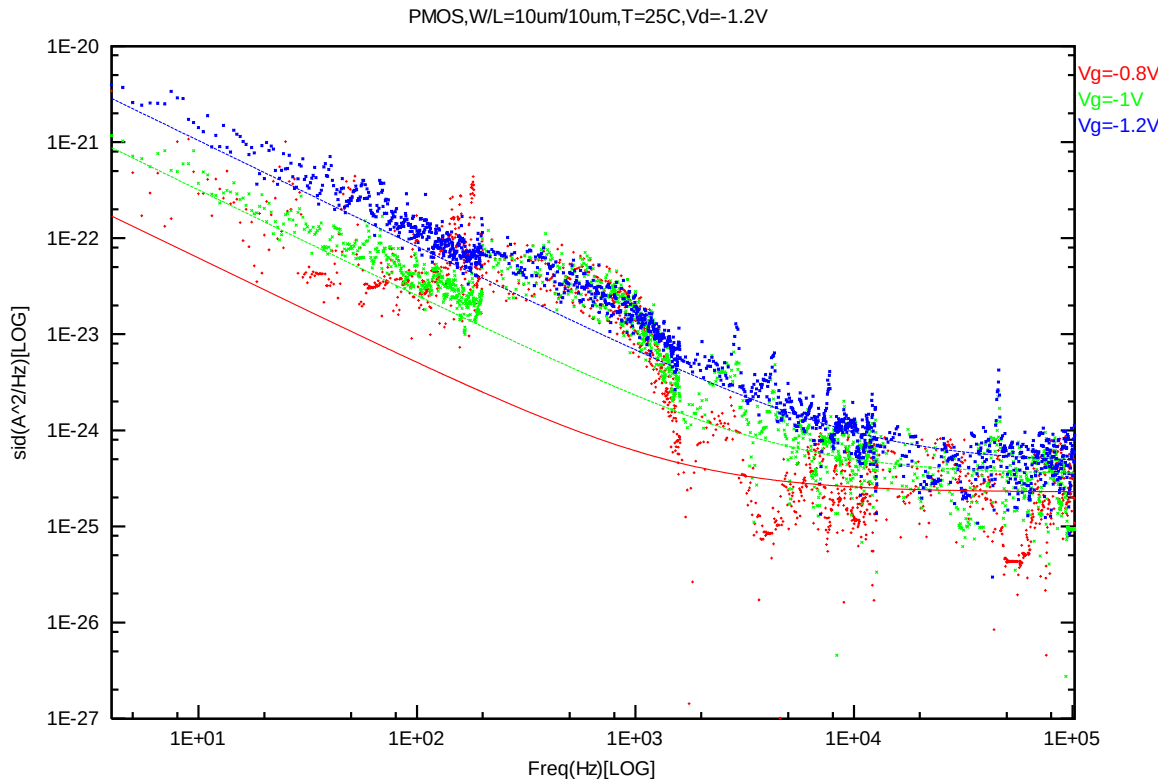


Figure 7-15 PMOS W/L=10 μm /10 μm @ Vd= -1.2V

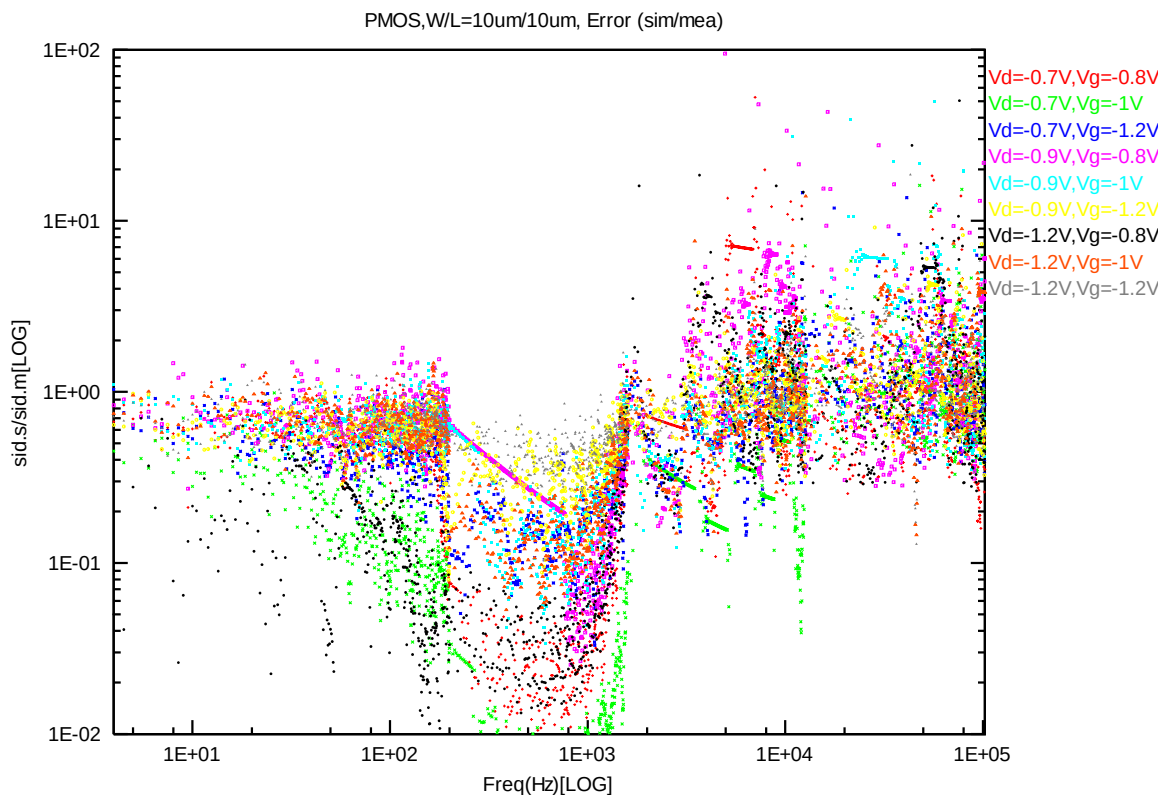


Figure 7-16 PMOS W/L=10 μm /10 μm Error Distribution Plot

7.2 Trend plots at frequency=100Hz

A. NMOS

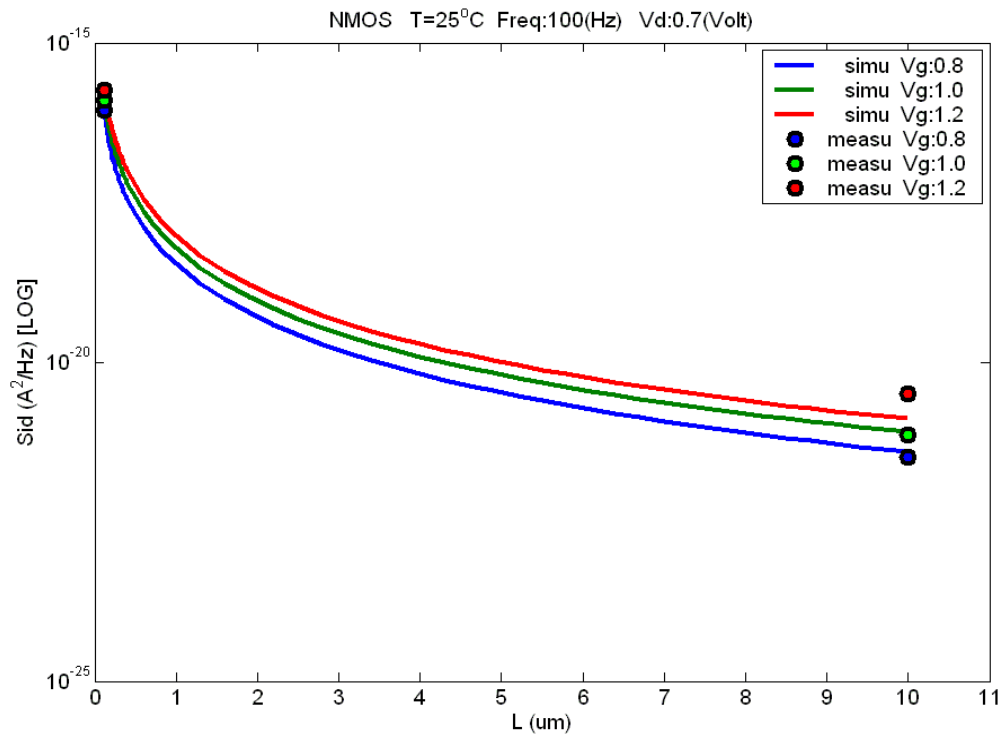


Figure 7-17 Noise vs Length for NMOS @ $V_d=0.7V$

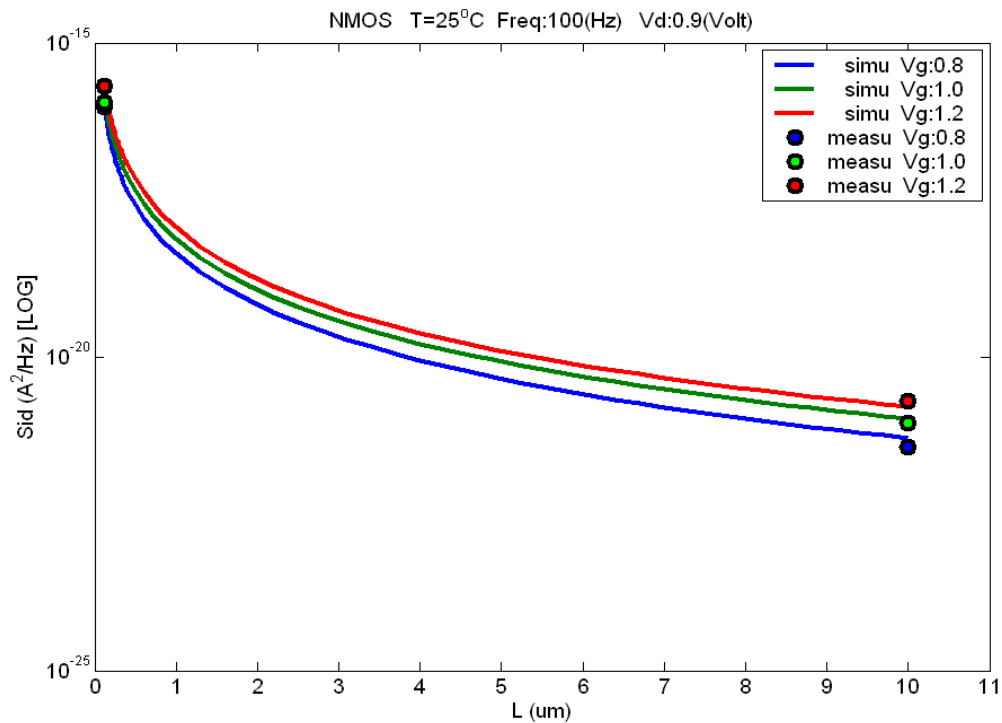


Figure 7-18 Noise vs Length for NMOS @ $V_d=0.9V$

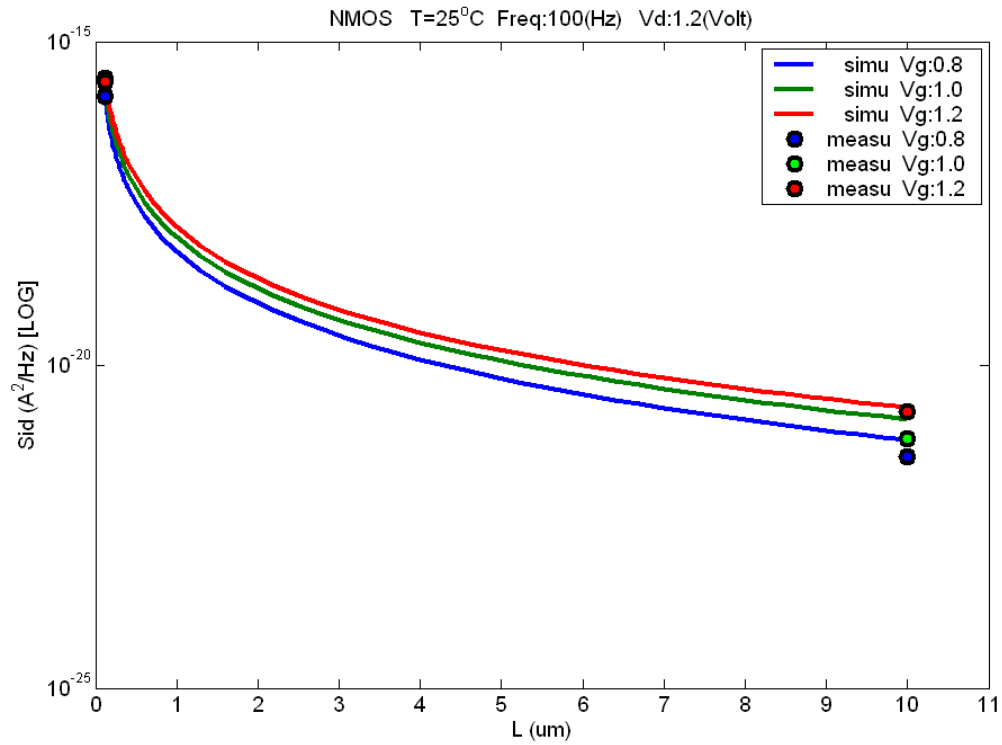


Figure 7-19 Noise vs Length for NMOS @ $V_d=1.2\text{V}$

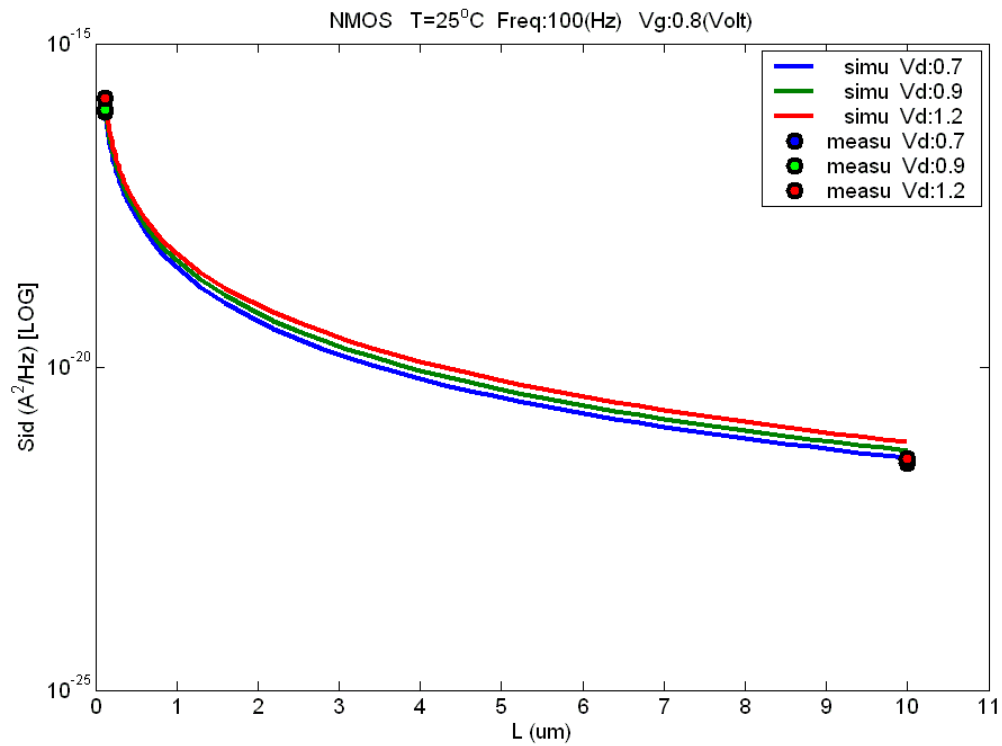


Figure 7-20 Noise vs Length for NMOS @ $V_g=0.8\text{V}$

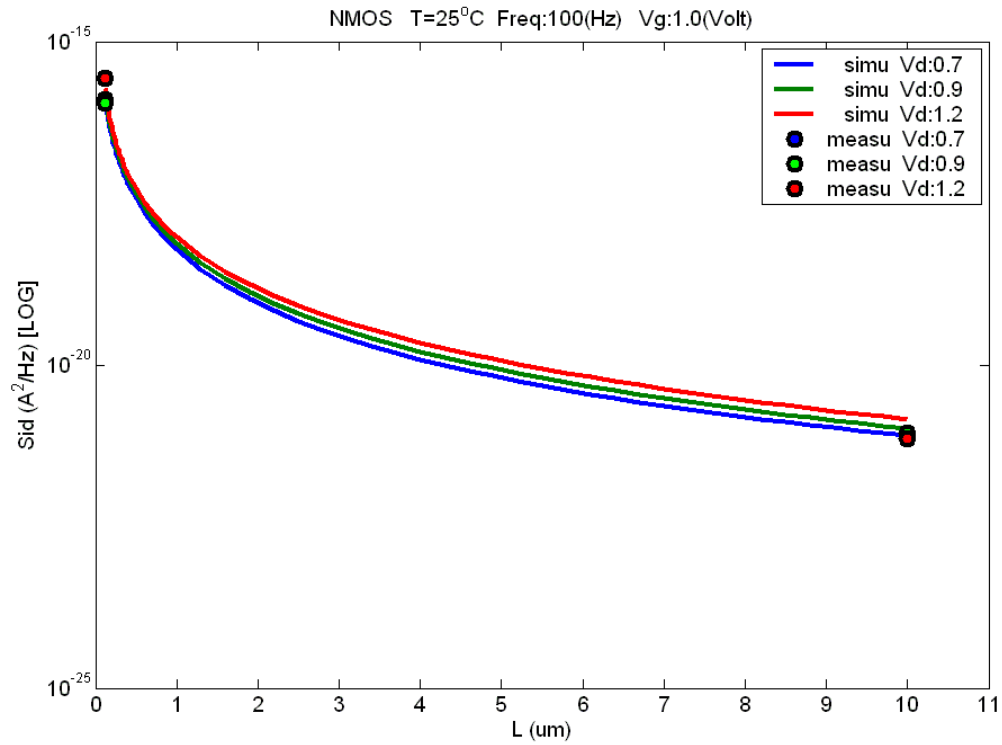


Figure 7-21 Noise vs Length for NMOS @ $V_g=1.0V$

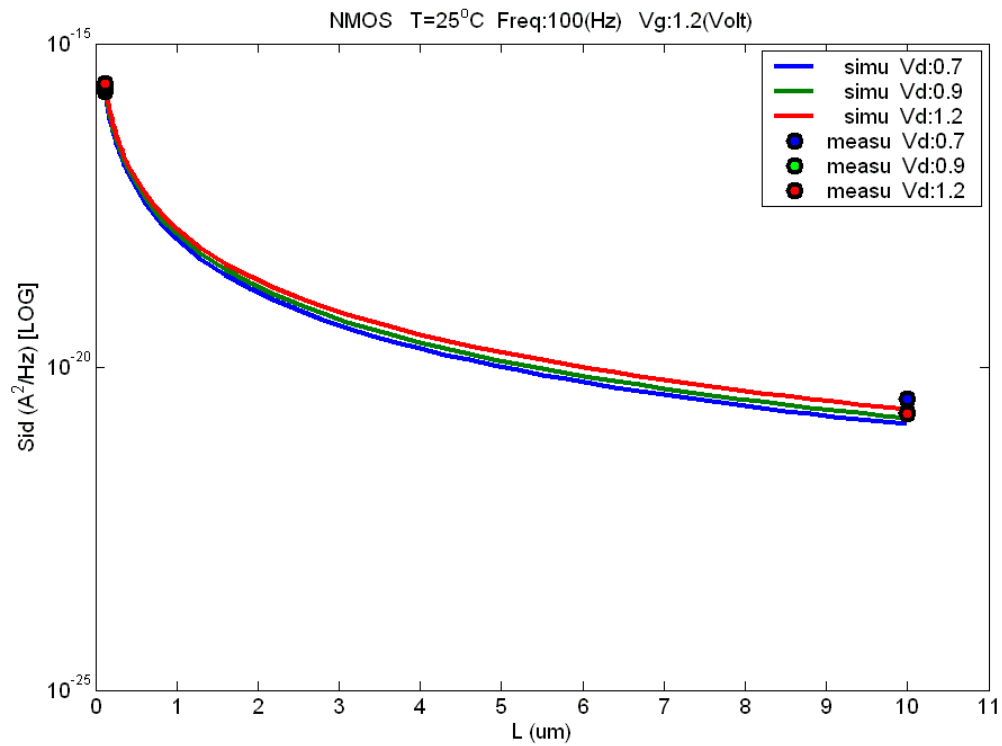


Figure 7-22 Noise vs Length for NMOS @ $V_g=1.2V$

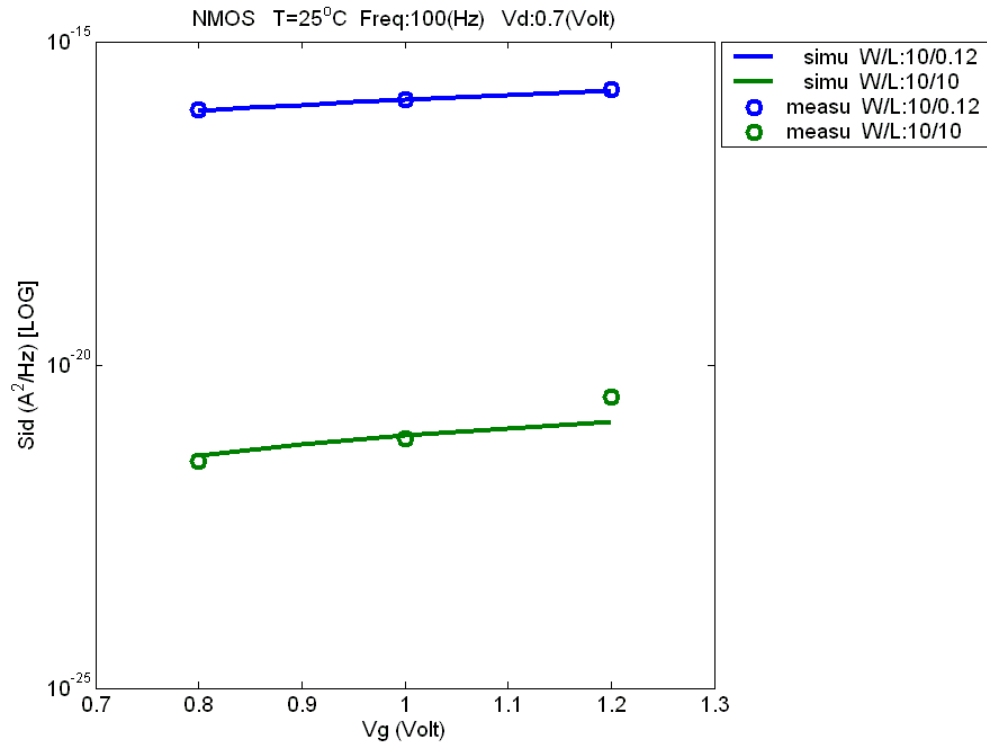


Figure 7-23 Noise vs Vg for NMOS @ Vd=0.7V

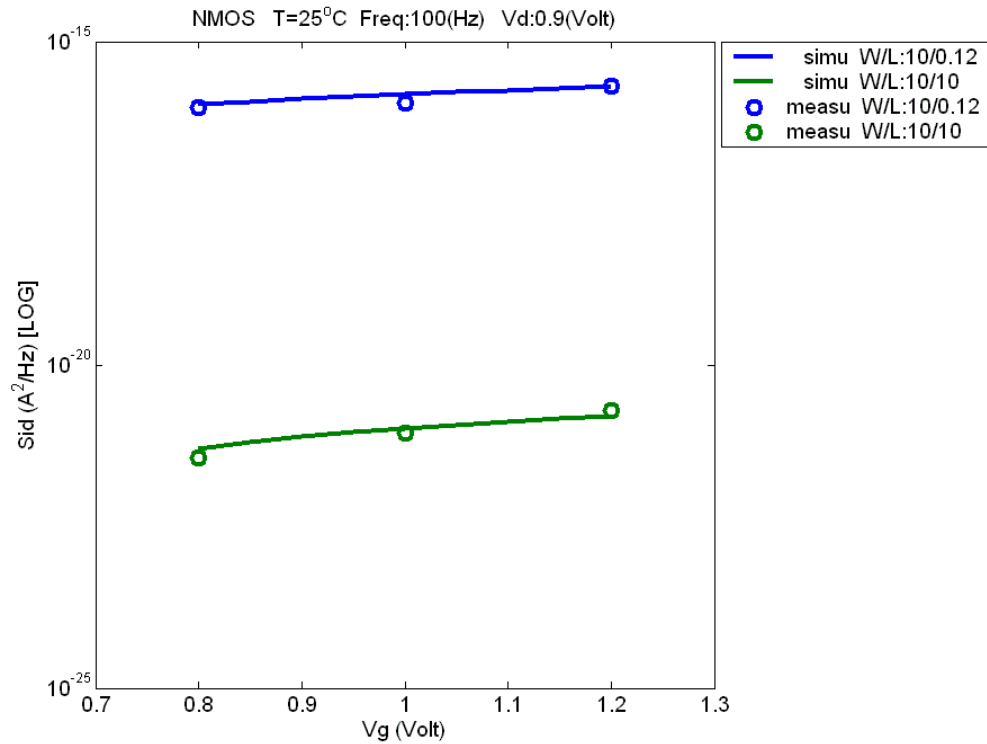


Figure 7-24 Noise vs Vg for NMOS @ Vd=0.9V

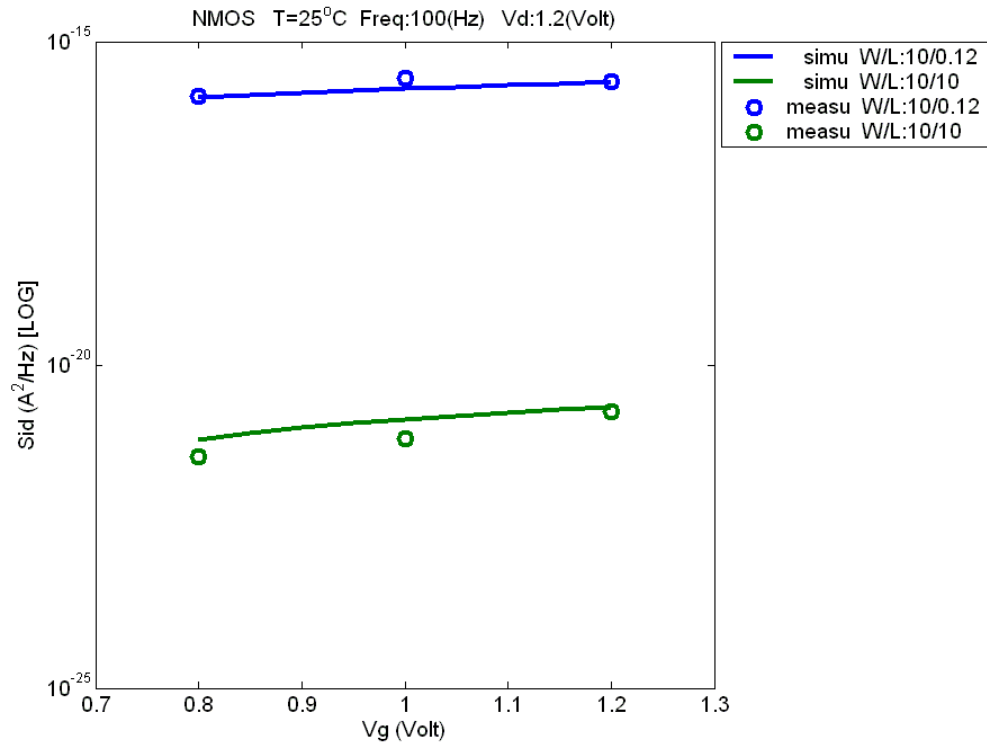


Figure 7-25 Noise vs Vg for NMOS @ Vd=1.2V

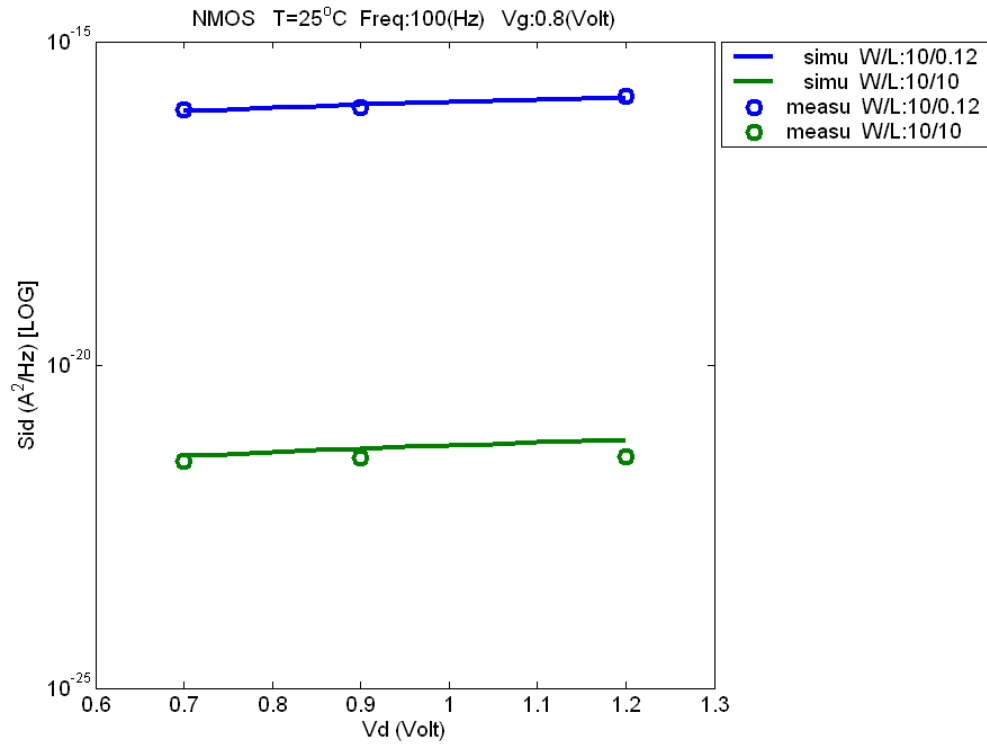


Figure 7-26 Noise vs Vd for NMOS @ Vg=0.8V

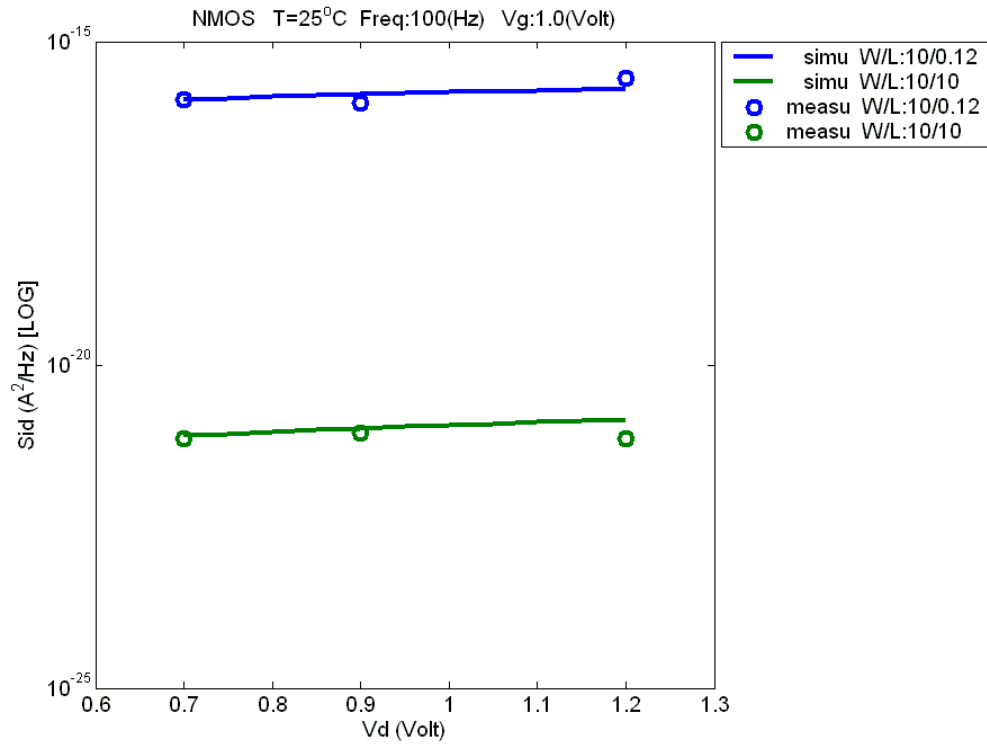


Figure 7-27 Noise vs V_d for NMOS @ $V_g=1.0V$

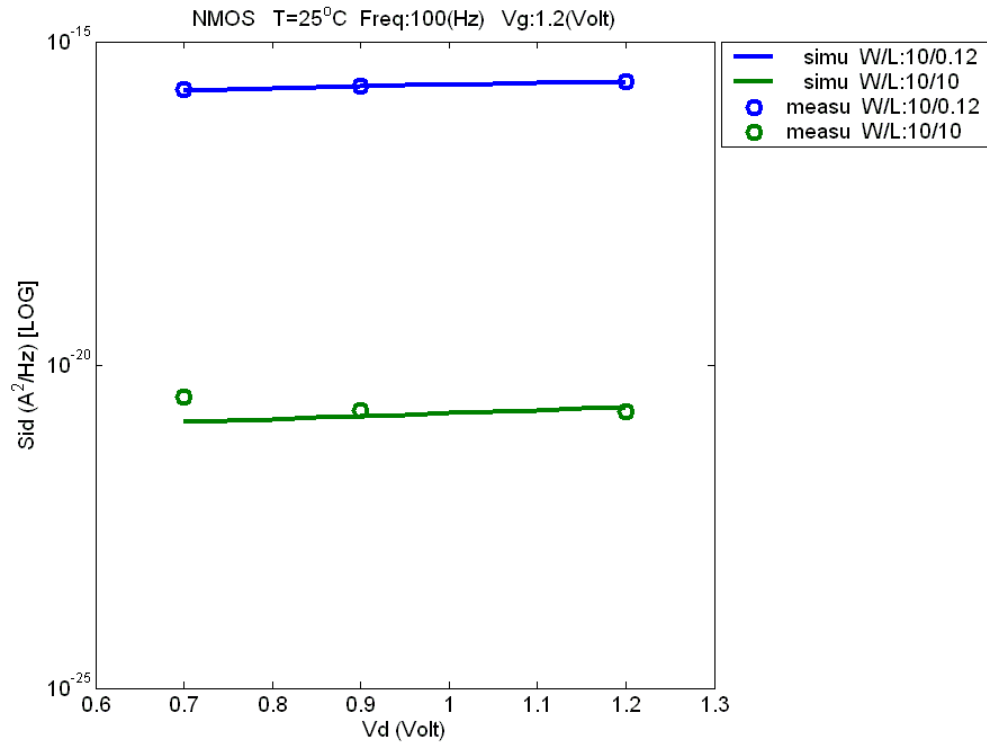


Figure 7-28 Noise vs V_d for NMOS @ $V_g=1.2V$

B. PMOS

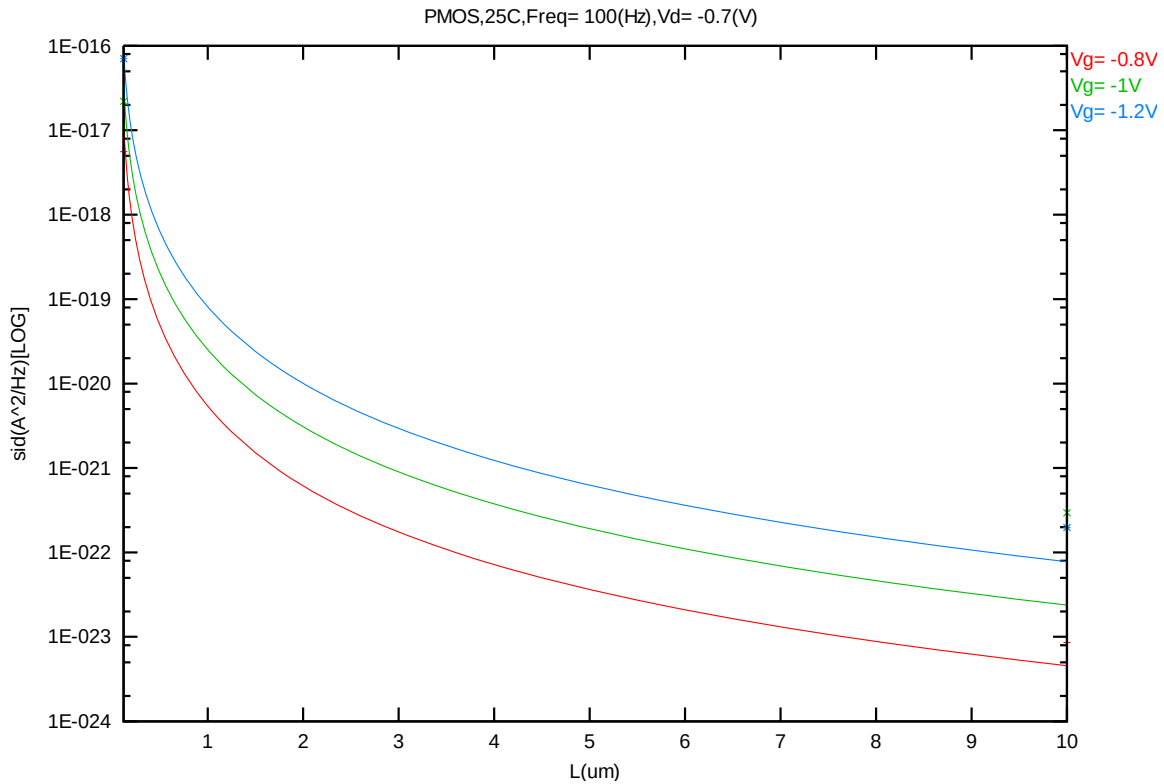


Figure 7-29 Noise vs Length for PMOS @ Vd= -0.7V

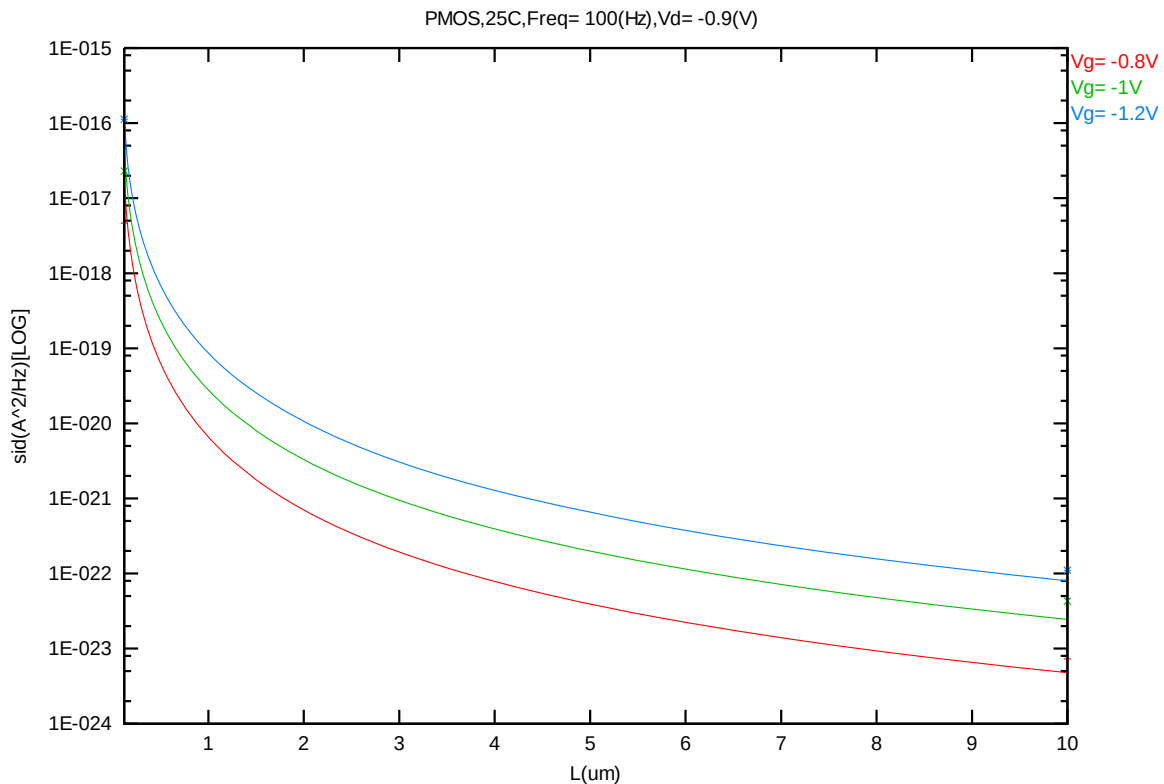


Figure 7-30 Noise vs Length for PMOS @ Vd= -0.9V

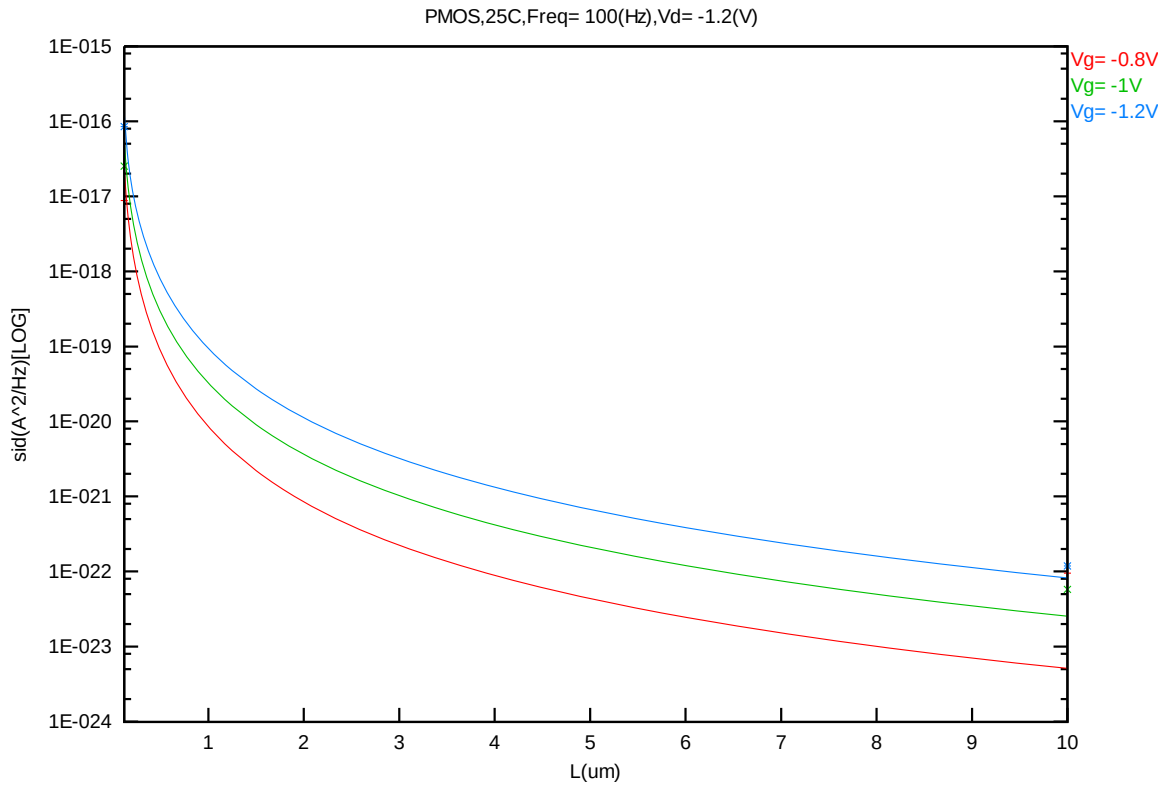


Figure 7-31 Noise vs Length for PMOS @ Vd= -1.2V

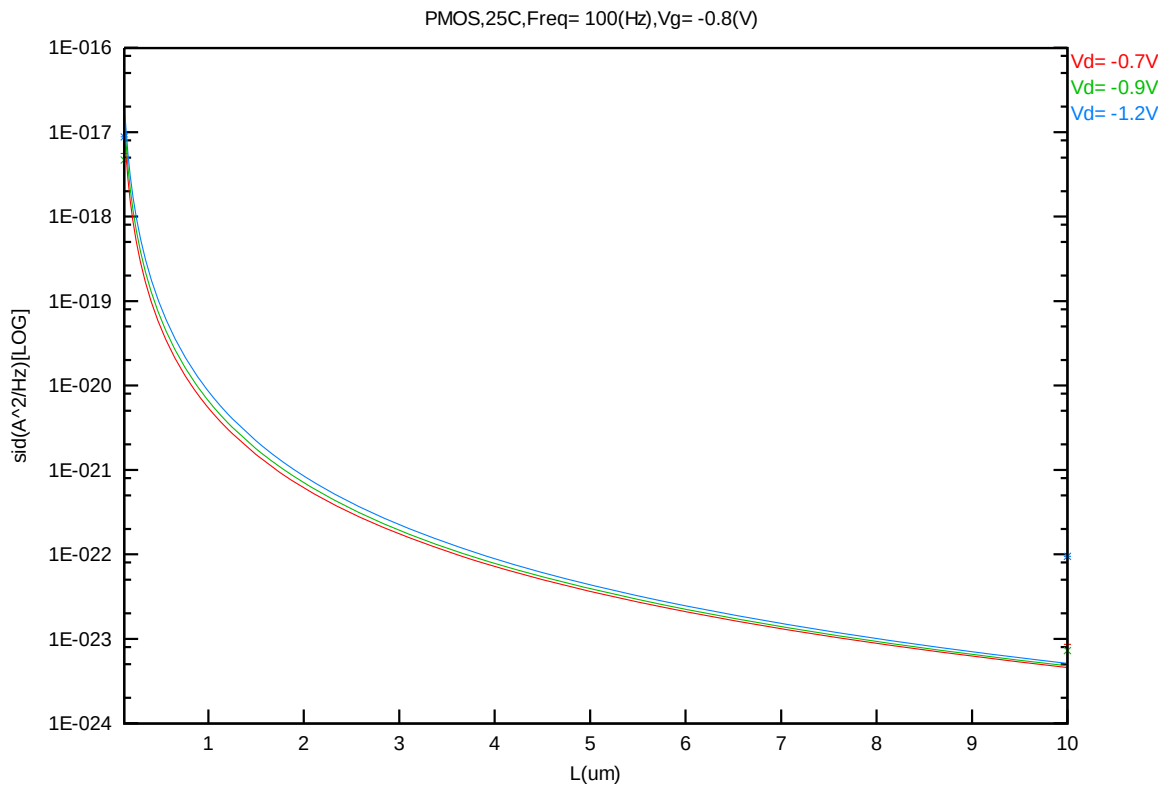


Figure 7-32 Noise vs Length for PMOS @ Vg= -0.8V

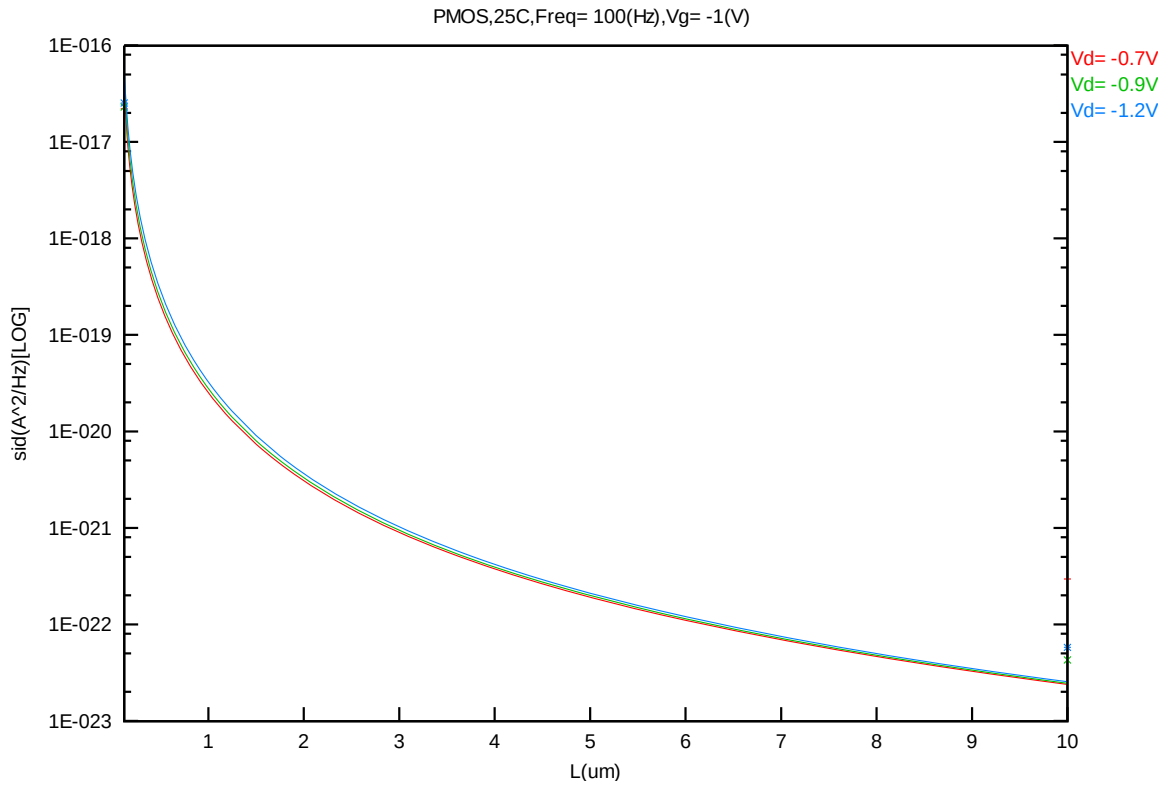


Figure 7-33 Noise vs Length for PMOS @ Vg= -1.0V

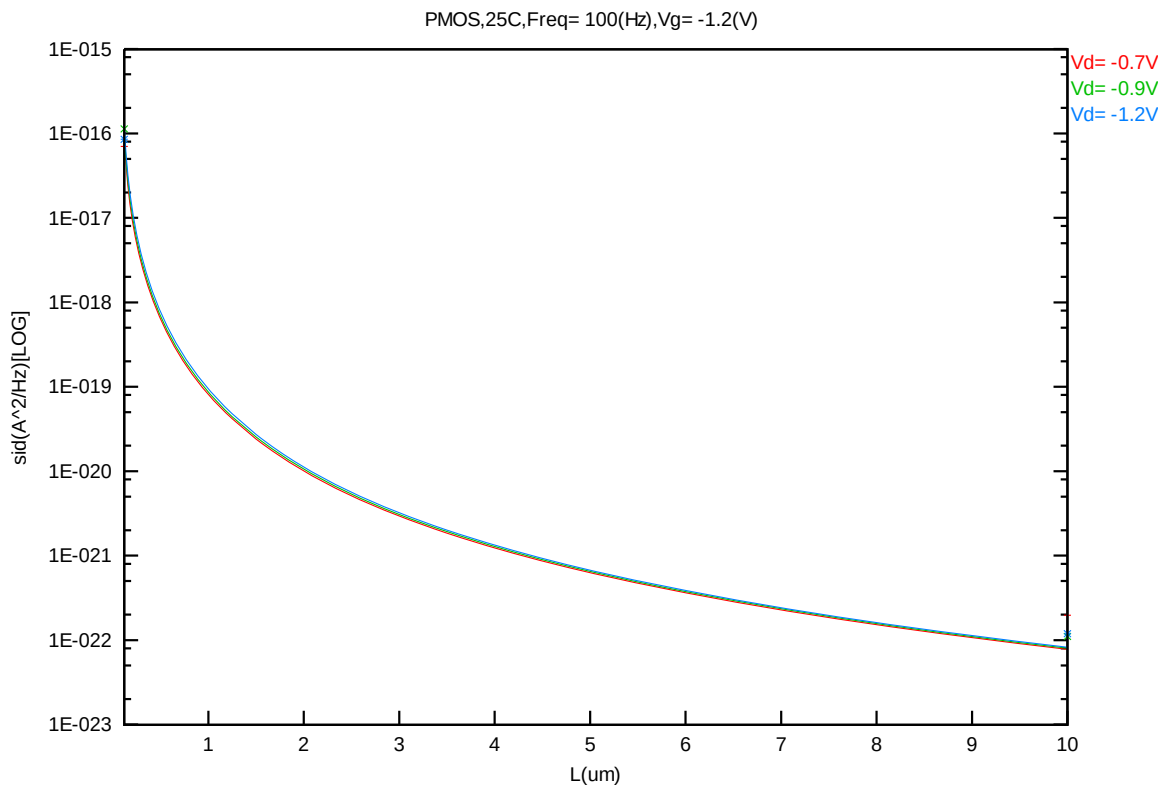


Figure 7-34 Noise vs Length for PMOS @ Vg= -1.2V

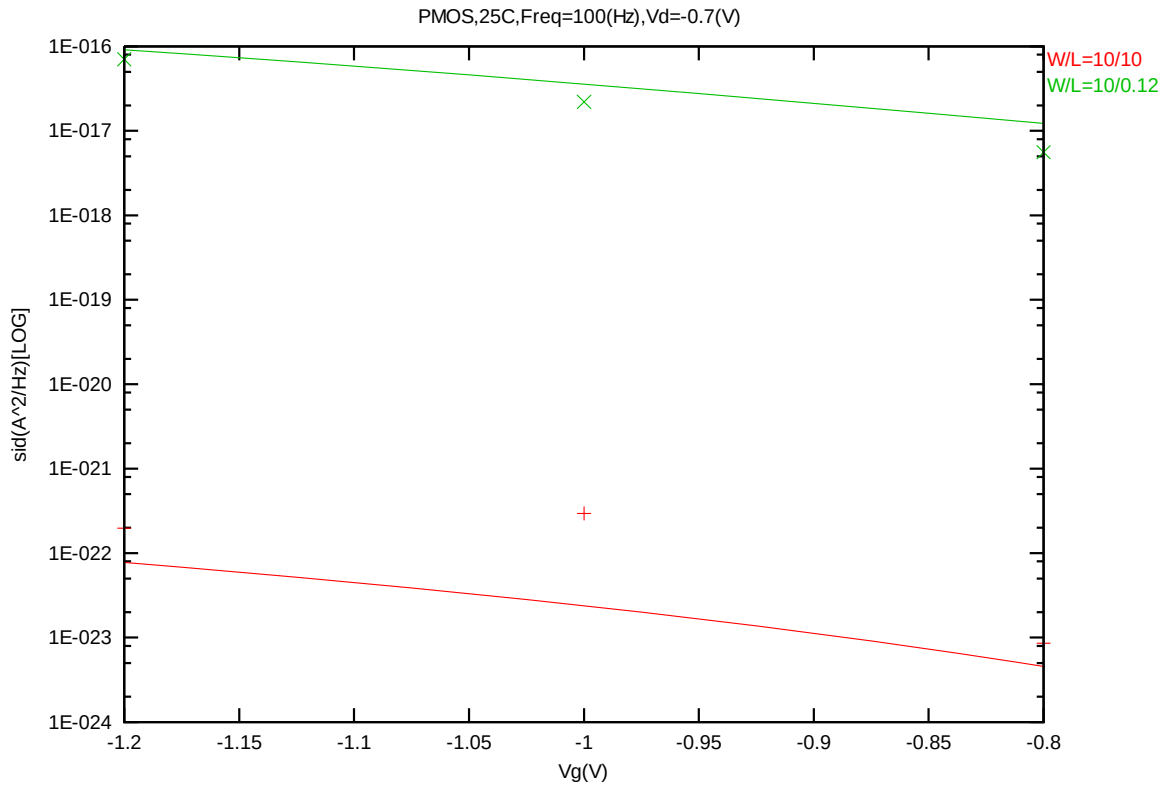


Figure 7-35 Noise vs Vg for PMOS @ Vd= -0.7V

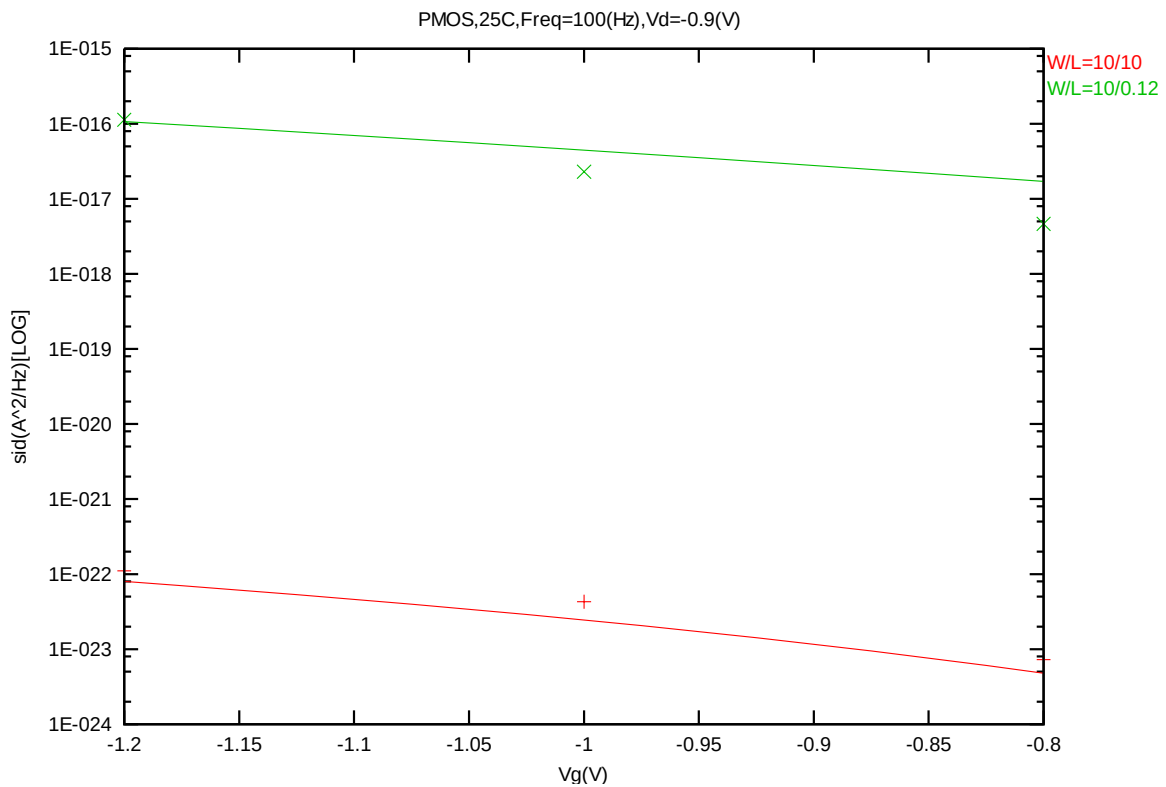


Figure 7-36 Noise vs Vg for PMOS @ Vd= -0.9V

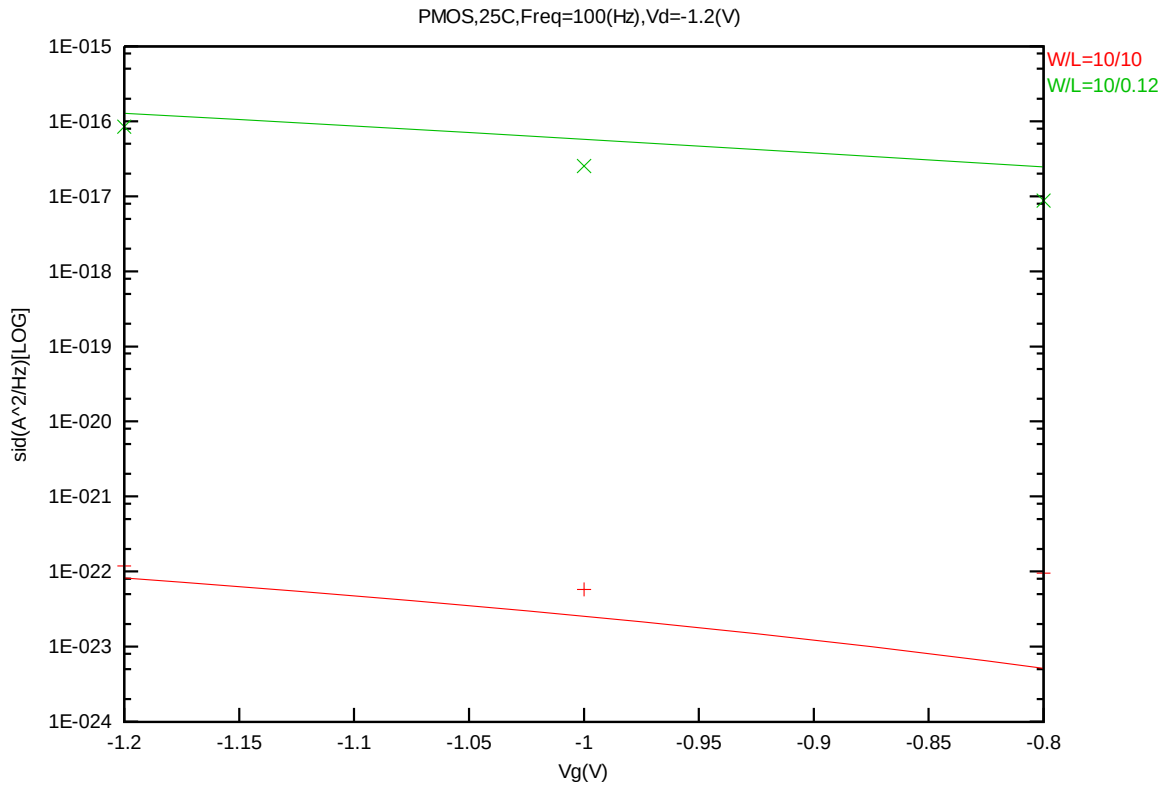


Figure 7-37 Noise vs Vg for PMOS @ Vd= -1.2V

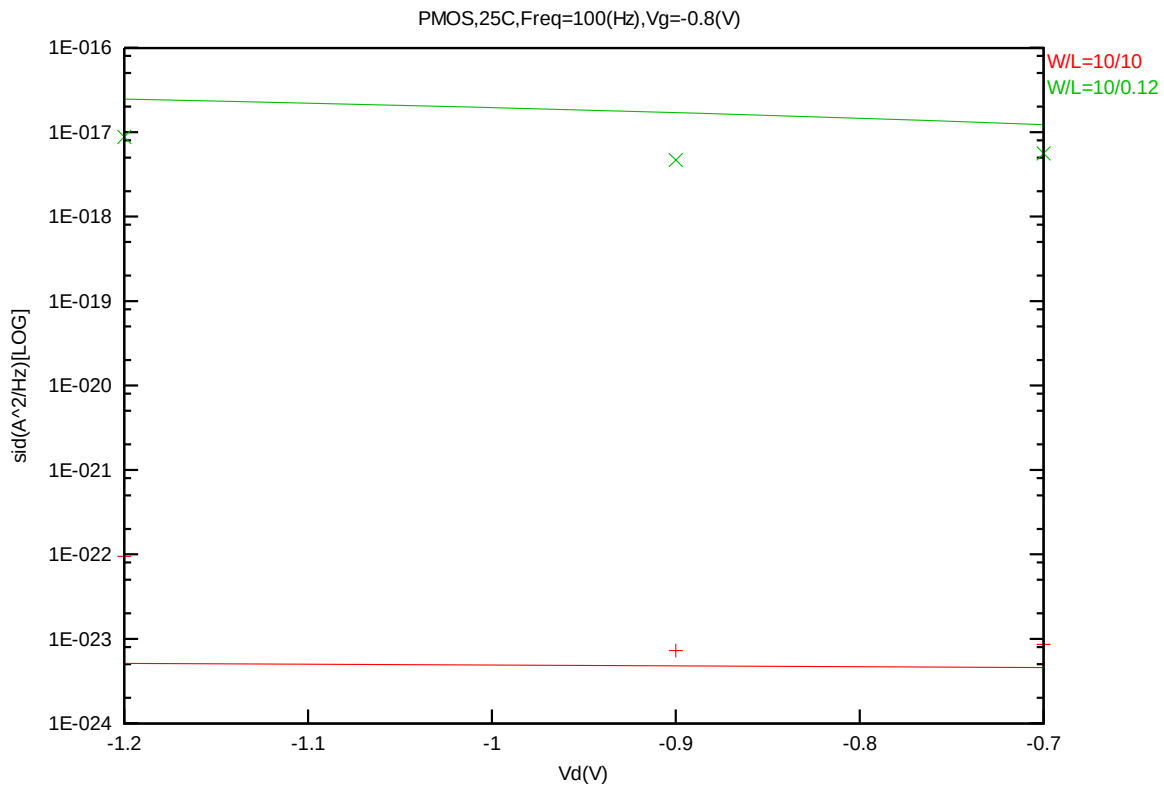
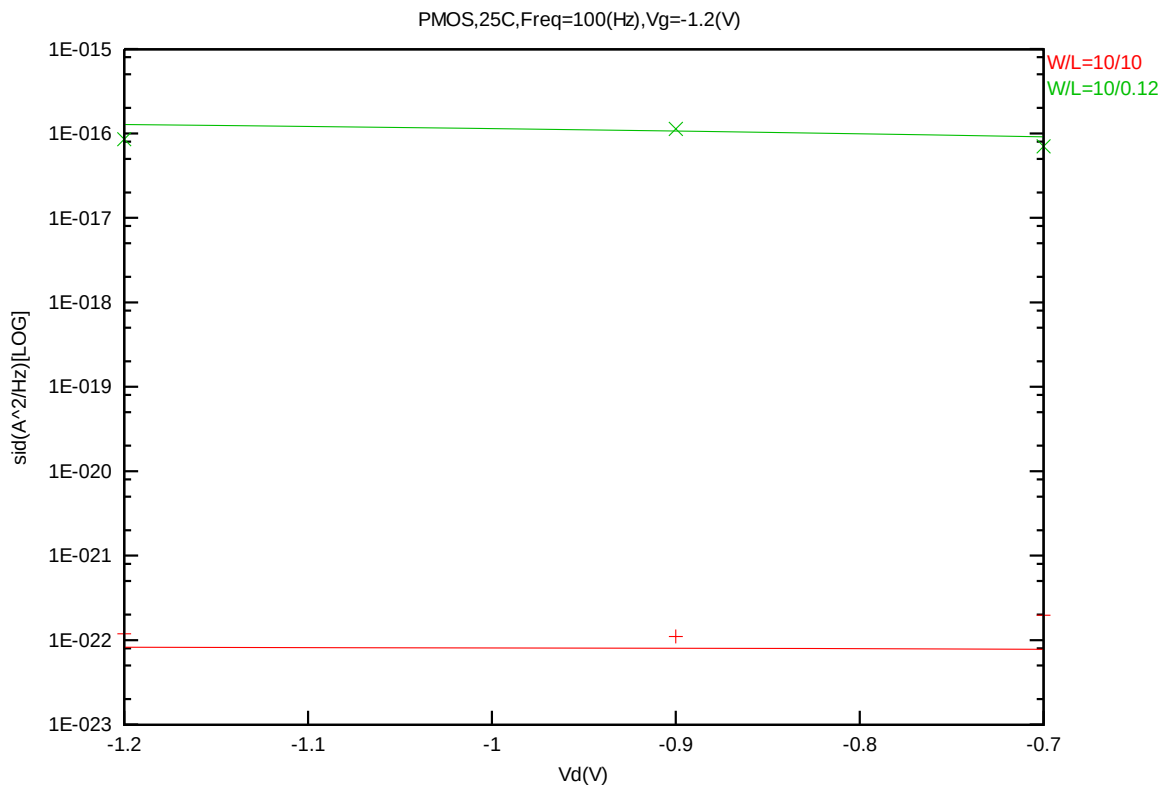
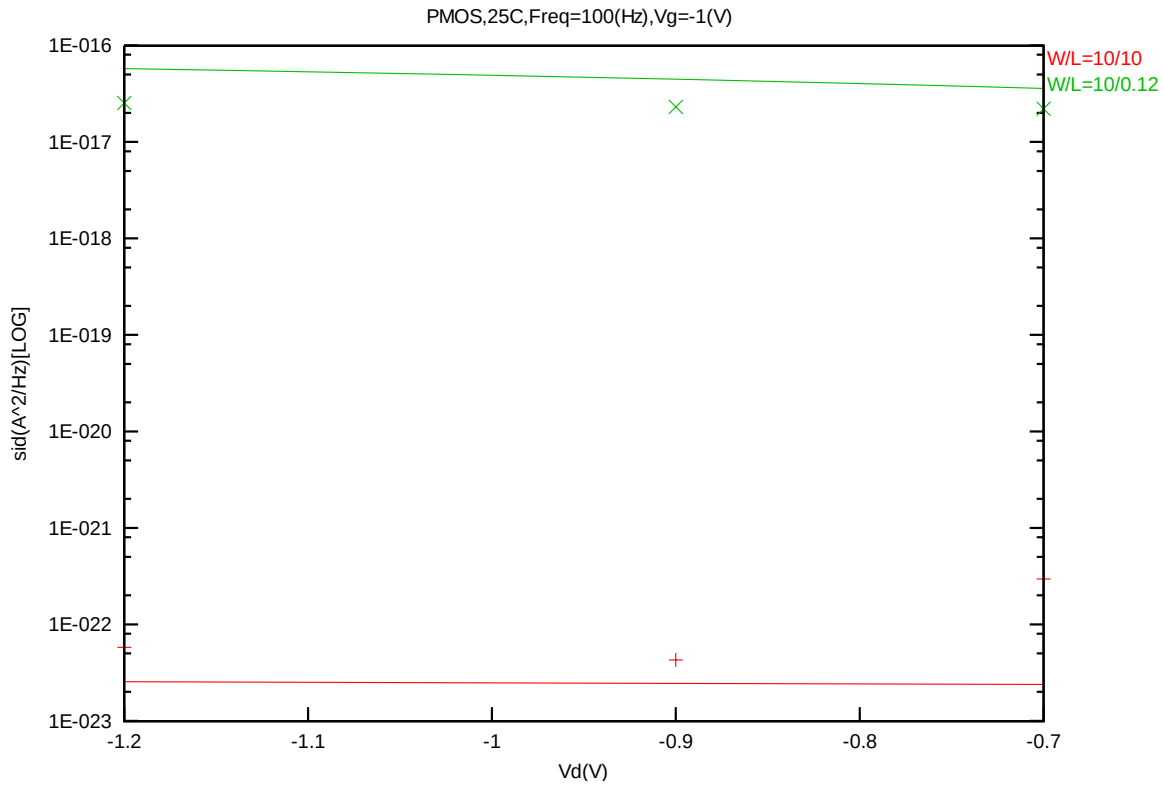


Figure 7-38 Noise vs Vd for PMOS @ Vg= -0.8V



8 Appendix

8.1 Verify corner plot.

Table 8-1 Verify wafer information

Mask ID	A11001	A11001	A11001
Lot ID	D60K6	D5N0S	D63LJ
Wafer Number	#1 and #3	#1 and #2	#8 and #9

Threshold voltage in linear region V_{TLN}

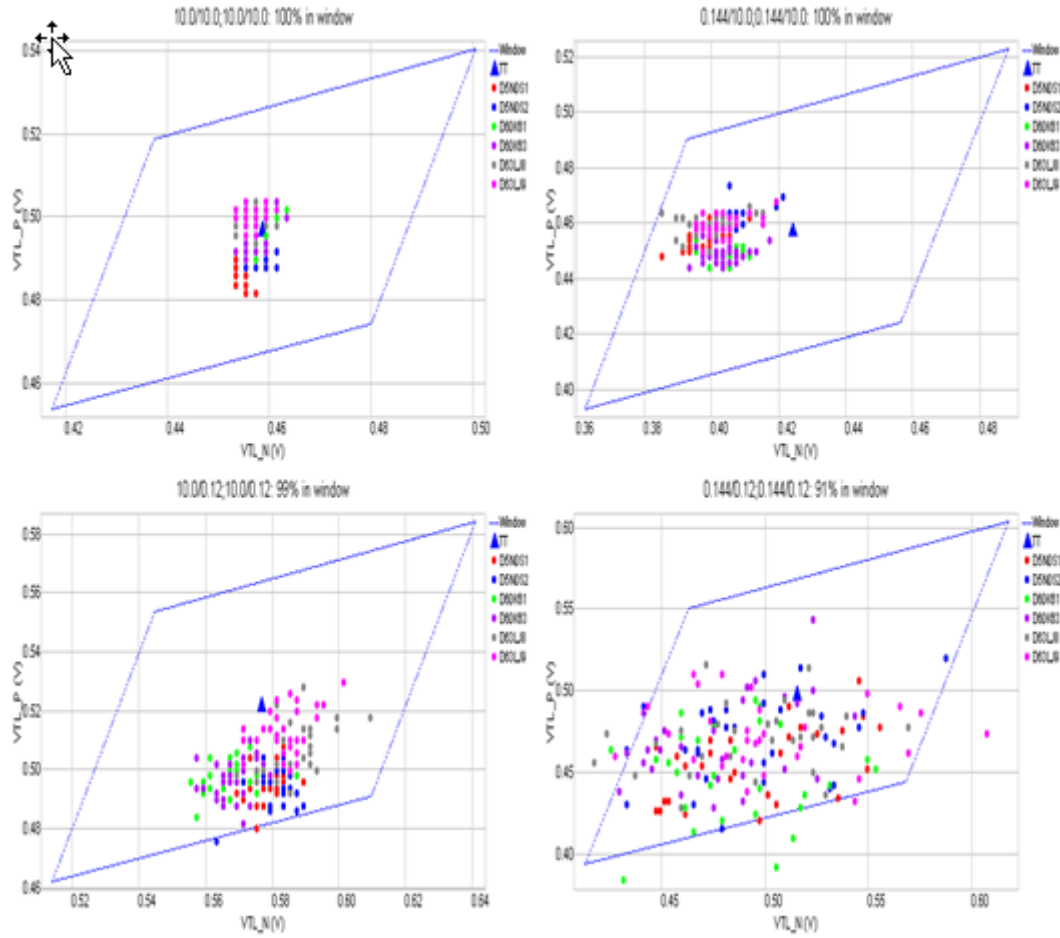


Figure 8-1 Corner graph of threshold voltage at low drain bias

Linear Current I_{LIN}

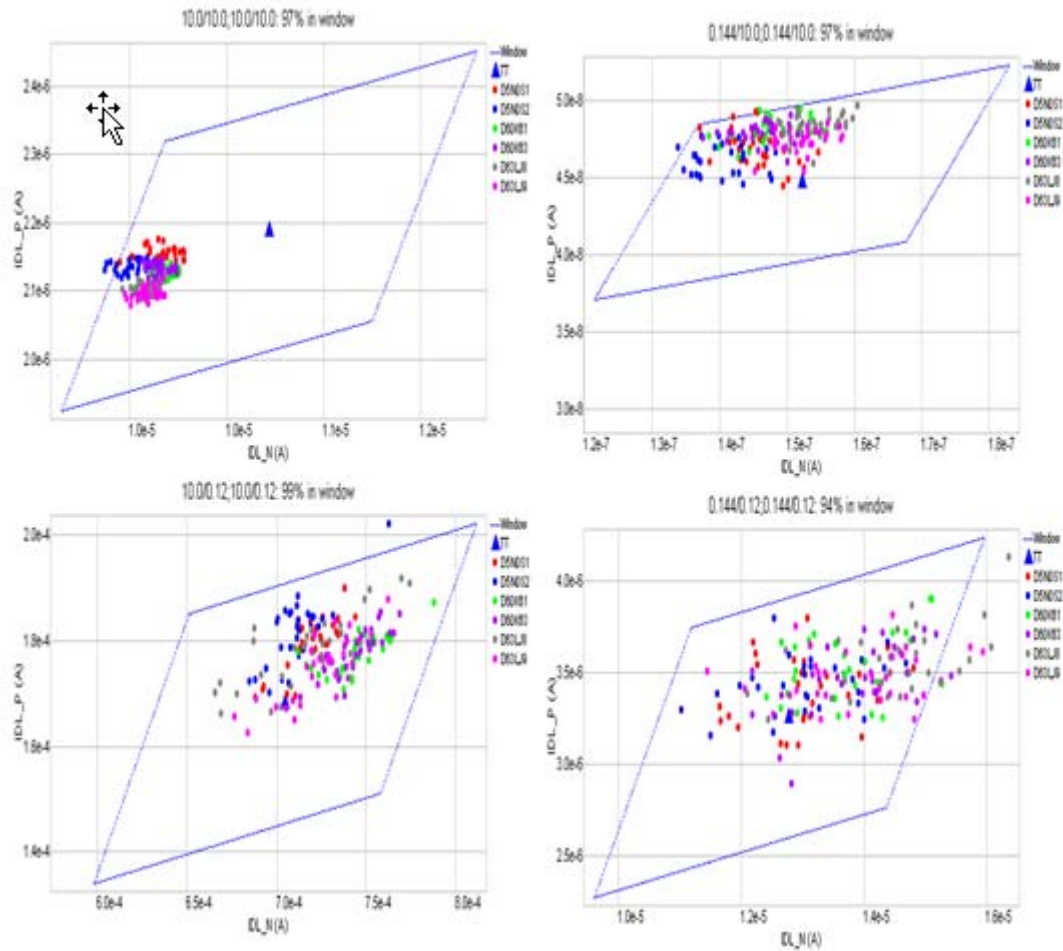


Figure 8-3 Corner graph of linear current

Saturation Current I_{ON}

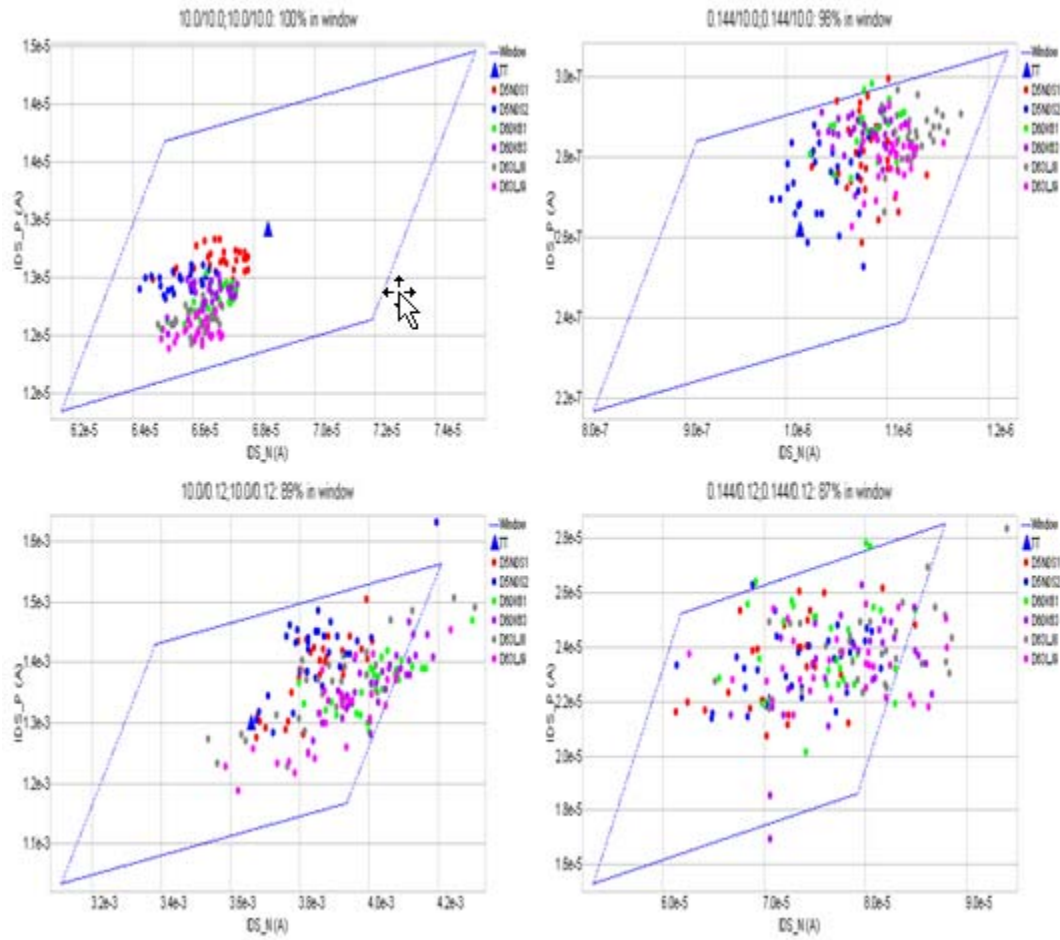


Figure 8-4 Corner graph of on-current

8.2 HSPICE Warning Message

Table 8-2 Hspice Warning Message

Item	Warning Message	Explanation
1	None	None

8.3 ELDO Warning Message

Table 8-3 Eldo Warning Message

Item	Warning Message	Explanation
1	None	None

8.4 SPECTRE Warning Message

Table 8-4 Spectre Warning Message

Item	Warning Message	Explanation
1	None	None